

COVER PAGE

**PREDICTING THE EFFECTS OF GEOMAGNETIC STORMS ON THE
IONOSPHERE UNDER INCREASING GREENHOUSE GAS
CONCENTRATIONS**

BY

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JULY, 2025

TITLE PAGE

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A PROJECT SUBMITTED

**IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
AWARD OF THE DEGREE OF BACHELOR OF SCIENCE (B.Sc) AT
THE DEPARTMENT OF PHYSICS, FACULTY OF SCIENCE,
AIR FORCE INSTITUTE OF TECHNOLOGY, KADUNA.**

DECLARATION

I BABALOLA-JACOBS OLUWASEGUN DAVID, with Matric No.: U21PH1003 hereby declare that the project work entitled “PREDICTING THE EFFECTS OF GEOMAGNETIC STORMS ON THE IONOSPHERE UNDER INCREASING GREENHOUSE GAS CONCENTRATIONS is a record of an original work done by me, as a result of my research effort carried out in the Faculty of Sciences, Air Force Institute of Technology, Kaduna under the supervision of Dr H.A. LAWAL.

All sources and references used have been duly acknowledged.

BABALOLA-JACOBS OLUWASEGUN DAVID

Date

CERTIFICATION

This is to certify that this study entitled “PREDICTING THE EFFECTS OF GEOMAGNETIC STORMS ON THE IONOSPHERE UNDER INCREASING GREENHOUSE GAS CONCENTRATIONS” was carried out by BABALOLA-JACOBS OLUWASEGUN DAVID, with matric number U21PH1003 in the Department of Physics, Faculty of Sciences, Air Force Institute of Technology, Kaduna, under my supervision.

Dr H.A. LAWAL
Supervisor

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APPROVAL PAGE

This is to certify that this study entitled “PREDICTING THE EFFECTS OF GEOMAGNETIC STORMS ON THE IONOSPHERE UNDER INCREASING GREENHOUSE GAS CONCENTRATIONS” prepared and submitted by BABALOLA-JACOBS OLUWASEGUN DAVID in partial fulfilment of the requirement for the award of degree in B.Sc. Physics at the Department of Physics, Faculty of Sciences, Air Force Institute of Technology, Kaduna, has been examined and approved. The project meets the standards and requirements set by the institution and it is hereby accepted at fulfilling the academic criteria.

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DEDICATION

This project is dedicated to God Almighty, with deepest gratitude for His endless blessings, wisdom and strength throughout my project and my entire academic journey.

ACKNOWLEDGMENT

I would like to first express my deepest gratitude to God for His endless love and guidance throughout my academic journey, especially throughout this project.

Without His support, this work would not have been possible.

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ABSTRACT

This study investigates how geomagnetic storms influence ionospheric behavior in the context of rising atmospheric greenhouse gas concentrations, with a focus on Total Electron Content (TEC), foF2, and hmF2 as ionospheric indicators. Using a data-driven modeling framework, the study begins by analyzing time series trends across solar cycles, revealing strong periodic patterns in ionospheric parameters aligned with solar activity, and a consistent upward trend in CO₂ levels that remains decoupled from solar variability.

To model the relationship between geomagnetic disturbances, greenhouse gases, and the ionosphere, the Python-based Generalized Additive Model (pyGAM) was employed due to its ability to capture nonlinear effects with high interpretability. The model was fitted using three predictors—Dst index (a measure of geomagnetic disturbance), atmospheric CO₂ concentration, and a derived solar cycle phase variable. Among the predictors, solar cycle phase was found to be the most influential, followed by CO₂, while Dst contributed the least predictive value. Surprisingly, the removal of Dst from the model led to a slight improvement in generalization performance (lower GCV), raising questions about its effectiveness as a stand-alone predictor of ionospheric variability.

These results highlight two critical insights: (1) solar activity remains the dominant driver of ionospheric changes, and (2) greenhouse gases are increasingly relevant to long-term ionospheric variability. Conversely, the traditionally relied-upon Dst index may not adequately reflect ionospheric conditions under both quiet and disturbed geomagnetic periods. The study concludes by recommending the inclusion of additional geomagnetic indices (e.g., AE, Kp, or SYM-H) and the use of higher-resolution and global datasets to better understand the coupled space-weather–climate system.

CHAPTER ONE

INTRODUCTION

The ionosphere is the ionized region of Earth's upper atmosphere, extending from approximately 48 km to 965 km above sea level ("Ionosphere", 2025). It is not a distinct atmospheric layer but rather a complex region within the thermosphere and parts of the mesosphere, where high-energy solar radiation ionizes atmospheric atoms and molecules by stripping away their electrons. This region hosts numerous satellites, including the International Space Station (ISS) (National Aeronautics and Space Administration, n.d., para. 4), and plays a critical role in modern communication and navigation systems, as radio and GPS signals propagate through it. Variations in ionospheric conditions can significantly affect satellite operations and disrupt radio signal transmission (National Aeronautics and Space Administration, n.d.). Consequently, it is essential to understand the diverse drivers and events that govern ionospheric variability. These include solar influences—such as solar flares, coronal mass ejections (CMEs), and solar wind—as well as atmospheric factors like temperature and pressure variations, atmospheric waves, thunderstorms, and turbulence. Another major contributor to ionospheric disturbances is geomagnetic storms, which can significantly reshape ionospheric structure and behavior.

A geomagnetic storm is a space weather event characterized by disturbances in Earth's magnetic field, primarily triggered by interactions between the solar wind and Earth's magnetosphere. These storms induce substantial changes in the ionosphere, including variations in electron density and total electron content (TEC), intensification of electric currents (e.g., auroral and equatorial electrojets), and dynamic expansion or contraction of ionospheric layers. Such disruptions can impair radio wave propagation, degrade GPS accuracy, interfere with navigation systems, and pose risks to satellite integrity. The intensity of geomagnetic storms is typically assessed using the Disturbance Storm-Time (Dst) index, which tracks global magnetic field depressions, and the Kp index, which measures global geomagnetic activity over three-hour intervals. Storms are categorized by severity into five levels: G1 (minor), G2 (moderate), G3 (strong), G4 (severe), and G5 (extreme). Key predictive indicators include solar wind speed and the orientation and strength of the Interplanetary Magnetic Field (IMF).

An area of growing scientific inquiry is whether the increasing concentration of greenhouse gases in the atmosphere affects ionospheric conditions, and if so, how these changes interact with or alter the ionosphere's response to geomagnetic storms. Several studies have identified correlations between the sustained rise in greenhouse gas levels—particularly carbon dioxide—and cooling trends in the thermosphere, contraction of the upper atmosphere, and changes in neutral mass density and electron concentration. These thermodynamic changes may influence ionospheric density and structure. Conversely, other research emphasizes that ionospheric variability is predominantly governed by solar activity, particularly the Sun's 11-year cycle. These contrasting findings underscore the complexity of upper atmospheric and ionospheric dynamics, and raise important questions about the interplay between long-term atmospheric composition changes and short-term space weather phenomena. Understanding

how these factors co-influence ionospheric behavior remains crucial for predicting its variability and mitigating risks to space-based technologies.

1.1 Background of the Study

Scientific understanding of the Earth's climate system dates back to the 1960s, when early climate modelers predicted that a combination of tropospheric warming and strong cooling in the upper atmosphere would result from increased carbon dioxide concentrations ([Pearce, 2023](#)). Rishbeth and Roble (1992) projected that greenhouse gas concentrations in the midlatitude middle atmosphere would double relative to early 1980s levels. Subsequent research has confirmed that rising greenhouse gas levels are strongly correlated with cooling trends in the upper atmosphere.

As part of the upper atmosphere, the ionosphere is significantly affected by this cooling. Studies have reported evidence of ionospheric contraction, accompanied by changes in its density, structure, and critical parameters such as the critical frequency (f_oF2) and the peak height of the F2 layer (h_mF2). These changes suggest that greenhouse gas-driven cooling may have long-term implications on ionospheric behavior.

In parallel, significant progress has been made in understanding the ionosphere's response to geomagnetic storms. These storms, which are primarily caused by solar activity, can substantially alter ionospheric conditions. Documented effects include increases or decreases in electron density, variations in total electron content (TEC), generation of traveling ionospheric disturbances (TIDs), decreases in the O/N_2 ratio leading to reduced ionization, and disruptions in radio wave propagation. These disturbances impact numerous technologies, including satellite communication, GPS-based systems, and high-frequency radio communication.

While both greenhouse gas concentrations and geomagnetic storms independently affect the ionosphere, limited research has explored how these two factors might interact. This project aims to investigate the relationship between geomagnetic storm effects and increasing greenhouse gas composition. Specifically, it seeks to determine whether elevated greenhouse gas levels amplify ionospheric disturbances during geomagnetic storms and to predict the nature of these compounded effects.

By deepening our understanding of this relationship, the study aims to contribute to more accurate forecasting of ionospheric behavior under future atmospheric conditions. This insight is crucial for mitigating potential disruptions to satellite-based and communication technologies, and it underscores the broader importance of reducing greenhouse gas emissions in a technologically dependent world.

1.2 Aim of the Study

This research aims to investigate and predict the effects of geomagnetic storms on the Ionosphere under increased greenhouse gas compositions.

1.3 Objectives of the Study

1. To analyze the effects of geomagnetic storms on the ionosphere, with a particular focus on their impact on electron density, total electron content (TEC), ionospheric expansion and contraction, and radio signal propagation.
2. To investigate the relationship between increasing greenhouse gas concentrations and key ionospheric parameters, such as electron density and TEC.
3. To explore the potential correlation between the effects of greenhouse gas concentrations and the impacts of geomagnetic storms on ionospheric behavior.
4. To assess how greenhouse gas concentrations may influence the severity of ionospheric disturbances caused by geomagnetic storms.
5. To compare the relative contributions of greenhouse gas composition and geomagnetic storms to ionospheric variability.
6. To predict ionospheric responses to geomagnetic storms under elevated greenhouse gas concentrations, simulating future conditions and their potential effects on radio wave propagation and technological systems.

1.4 Statement of Research Problem

Raising Earth's average surface temperature requires an immense amount of added heat energy. The approximately 2°F (1°C) increase in global average surface temperature observed since the pre-industrial era (1850–1900, based on NOAA's records) may appear minor, yet it reflects a substantial accumulation of heat within the Earth system ([NOAA Climate.gov](https://www.noaa.gov/climate), n.d.). While the lower atmosphere near the Earth's surface is warming, the upper atmosphere is cooling—an effect driven by the very same greenhouse gases that heat the lower layers ([Pearce, 2023](#)). Extending all the way to the edge of space, this cooling phenomenon was first predicted by climate models and has only recently been confirmed with precision through satellite observations ([Pearce, 2023](#)).

As the upper atmosphere cools, it contracts ([Bailey & NASA, 2021](#)), and this contraction has serious implications for the ionosphere: changing its density, altering the structure of its layers, and affecting key parameters such as electron density and neutral mass density. These changes influence satellite orbital paths and disrupt radio signal transmission. At the same time, geomagnetic storms—triggered by variations in solar wind and interplanetary magnetic fields—can dramatically disturb the ionosphere, modifying total electron content (TEC), causing traveling ionospheric disturbances (TIDs), and altering radio wave propagation. The result is a cascade of disruptions to critical technologies, including satellite communications, navigation systems like GNSS, as well as the structural integrity of satellites themselves.

With both greenhouse gas-induced cooling and geomagnetic storms significantly altering the ionosphere, an urgent and complex question emerges: could there be a connection between the two? And if so, does the rising concentration of greenhouse gases intensify the ionospheric response to geomagnetic storms—making their effects more severe, more unpredictable, or more persistent? If these phenomena are indeed interlinked, can we forecast such combined impacts more accurately? And ultimately, how much of the ionospheric variability we observe is driven by the slow, long-term shifts in atmospheric composition versus the short-term shocks of geomagnetic activity? These are the pressing questions this study seeks to explore.

1.5 Significance of the Study

Understanding how the ionosphere responds to geomagnetic storms is vital for maintaining the reliability of modern communication, navigation, and satellite-based systems. As geomagnetic storms can lead to serious disruptions in these technologies, predicting their effects accurately is a growing scientific and practical priority.

At the same time, increasing concentrations of greenhouse gases are altering the structure and behavior of the upper atmosphere. Although this phenomenon has been recognized, its specific influence on ionospheric dynamics—especially during geomagnetic storms—is not yet fully understood. This research seeks to address this gap by exploring whether and how elevated greenhouse gas composition may intensify or alter ionospheric responses to geomagnetic activity.

By combining these two areas of study, this project aims to offer new insights into the complex interactions between space weather and long-term atmospheric changes. The findings could improve our ability to forecast ionospheric disturbances, leading to more resilient communication and navigation systems in the future. Additionally, this research could contribute to the broader understanding of climate-related changes in the upper atmosphere, reinforcing the importance of monitoring greenhouse gas emissions not only for their surface-level climate impacts but also for their potential effects on space weather interactions.

1.6 Physics of Space

This study is grounded on several key physical principles and laws that govern atmospheric physics, plasma dynamics, and radiative processes.

1.6.1 Maxwell's Equations

The behavior of electric and magnetic fields in the ionosphere is described by Maxwell's equations. These equations govern the generation and interaction of electromagnetic fields, and are essential for understanding radio wave propagation and the electromagnetic impact of geomagnetic storms on the ionized layers of the upper atmosphere.

$$\nabla \cdot E = \frac{\rho}{\epsilon} - \text{Maxwell-Gauss Equation}$$

$$\nabla \cdot B = 0 - \text{Maxwell-Gauss Equation}$$

$$\nabla \times E = -\frac{\partial B}{\partial t} - \text{Maxwell - Faraday Equation}$$

$$\nabla \times B = \mu(J + \epsilon \frac{\partial E}{\partial t}) - \text{Maxwell - Ampere Equation}$$

1.6.2 Lorentz Force Law

Charged particles in the ionosphere are subjected to forces due to electric and magnetic fields, as described by the Lorentz force law. This law explains the motion of ions and electrons in the presence of geomagnetic disturbances, especially during solar storm events.

$$F = q(E + v \times B) - \text{Lorentz Force Equation}$$

1.6.3 Magnetohydrodynamics (MHD) Principles

The ionosphere is treated as a magnetized plasma, and its dynamics during geomagnetic storms are governed by the principles of magnetohydrodynamics. These principles combine fluid dynamics and electromagnetism to describe the collective motion of the ionized gas under the influence of magnetic fields.

The working principle involves the acceleration of an electrically conductive fluid (which in this case is an ionized gas called plasma) by the Lorentz force, resulting from the cross product of an electric current (motion of charge carriers i.e. ions) with a perpendicular magnetic field.

$$F = J \times B$$

J = Current Density

B = Magnetic Field

1.6.4 Radiative Transfer and Greenhouse Gas Absorption

Radiative transfer principles explain how greenhouse gases such as CO₂ and CH₄ absorb and emit infrared radiation. These gases trap heat in the lower atmosphere (the greenhouse effect) but also lead to enhanced radiative cooling in the upper atmosphere, contributing to ionospheric contraction and temperature decline.

1.6.5 The Chapman Theory of Ionization

This theory describes the formation of ionized layers in the ionosphere through solar radiation. It provides a theoretical basis for understanding how solar EUV (extreme ultraviolet) and X-ray radiation create electron density profiles in different ionospheric layers.

1.6.6 The Ideal Gas Law and Thermodynamic Principles

The behavior of the ionosphere, particularly its temperature, pressure, and density, is fundamentally governed by the Ideal Gas Law, which is expressed as:

$$PV = nRT$$

Where:

P = Pressure, V = Volume, n = number of moles of gas, R = Universal gas constant and T = Temperature

1.6.7 Wave Propagation Theory

The term wave propagation is used to describe how waves travel or move. The transmission of radio waves and the manifestation of traveling ionospheric disturbances (TIDs) are governed by wave propagation theory. Changes in electron density and the ionospheric structure during storms or atmospheric cooling affect how these waves propagate through space.

$$\frac{\partial^2 U}{\partial t^2} = c^2 \cdot \nabla^2 U - \text{Wave equation}$$

1.7 Scope and Limitation of the project

This study sets out to investigate how the ionosphere responds to geomagnetic storms under increasing greenhouse gas compositions in the upper atmosphere. It aims to analyze changes in key ionospheric parameters such as electron density, total electron content (TEC), ionospheric height (hmF2), and critical frequency (foF2) during geomagnetic storm events, and to assess whether these responses are altered or intensified by long-term greenhouse gas-induced cooling and contraction of the upper atmosphere.

The research is grounded in established physical models, including ionospheric dynamics, thermodynamic behavior, and solar-geomagnetic interactions. It also considers long-term trends in upper atmospheric temperature and composition particularly greenhouse gas driven cooling and their potential to modify the structure and variability of the ionosphere.

Data sources may include historical ionosonde data, geomagnetic indices (e.g., Kp, Dst), atmospheric composition models, and previous climate-ionosphere studies. The goal is to contribute to improved prediction of ionospheric behavior under future atmospheric conditions, particularly during geomagnetic storm activity.

CHAPTER TWO

LITERATURE REVIEW

This chapter explains in detail what the ionosphere is, factors that affect the ionosphere and its uses. It now elaborates geomagnetic storms and green house gases, what they are, the effects and how they affect the ionosphere respectively. Also, a very concise overview of the machine learning model employed to achieve result.

2.1.1 Ionosphere

The ionosphere is a region of Earth's upper atmosphere, ranging approximately from 60km to over 1000 km altitude, that is ionized primarily by solar ultraviolet (UV) and X-ray radiation. This ionization creates a layer of free electrons and ions that affects significantly radio wave propagation, making the ionosphere critical for satellite communications, GPS signals, and HF radio transmission (Ratcliffe, 1972).

Historical Perspective of the Ionosphere

The earth's ionosphere is a partially ionized gas that envelops the earth and, in some sense, forms the interface between the atmosphere and space. Since the gas is ionized, it cannot be fully be described by the equations of neutral fluid dynamics. In fact, a major revolution in ionospheric physics has occurred in the past decades as the language and concepts of plasma physics have played an increasing in the advancement of its research. The very discovery of the ionosphere came from radio wave observations and the recognition that only a reflecting layer composed of electrons and positive ions could explain the characteristics of the data. This was discovered by Edward Appleton, whose experiments proved the existence of an electrically charged layer in our atmosphere called the ionosphere (Nobel Prize, n.d.). Most of the early works in ionospheric physics was aimed at explaining the various layers and their variability with local time, latitude, season, and so forth. The ionosonde, a remote sensing device that yields electron density profiles up to but not above the altitude the altitude of the highest concentration of charged particles, was the primary research tool. Such measurements revealed wildering variety of ionospheric behavior ranging from quiet, reproducible profiles to totally chaotic ones. Furthermore, a variety of periodic and aperiodic variations were observed, with time scales ranging from the order of the solar cycle (11 years) to just a few seconds, as time passed, the emphasis of ionospheric research shifted from questions dealing with formation and loss of plasma towards the dynamics and plasma physics of ionospheric phenomena and the ionosphere's reaction to space weather effects. Ionospheric research has greatly benefitted the space physics program with the associated development of instruments for balloons, rockets, and satellites. The combination of remote ionospheric sensing and direct situ measurements made from spacecraft has accelerated the pace of ionospheric research. Equally important has been the development of plasma physics as a theoretical framework around which to organize our understanding of the ionospheric phenomena. Likewise, the continuing development of magnetospheric and atmospheric science has very much influenced our understanding of the ionosphere (Kelley, 2009, pp. 1–2).

Formation of the Ionosphere

The ionosphere is formed in the Earth's upper atmosphere by incident solar radiation interacting with, and removing electrons, from different gases. Different wavelengths of solar electromagnetic energy, in the ultraviolet and X-ray parts of the spectrum, ionize different component gases in the atmosphere. A single ionospheric layer may be regarded as being formed by a specific wavelength interacting with a single species of air molecule. We can visualize this in diagrammatic form:

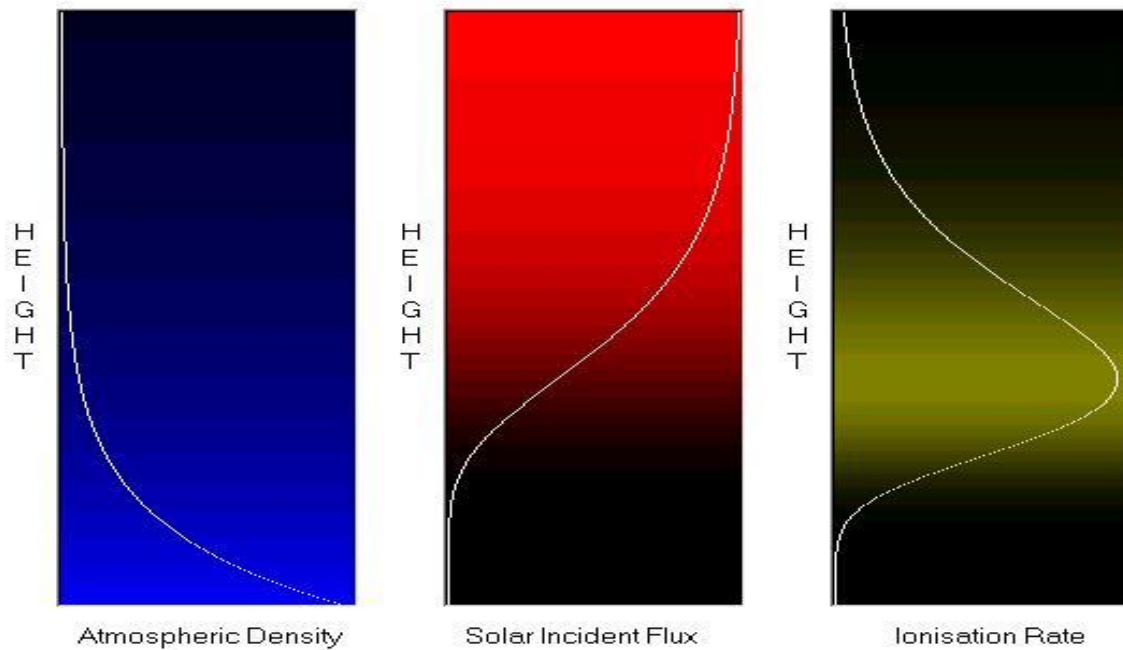


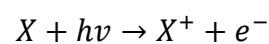
Figure 2. 2 A Visualization of the formation of the Ionosphere

As the solar radiation travels down through the atmosphere it encounters a greater density of air and the rate of ionization becomes greater. However, in doing so, it loses intensity and eventually the ionization rate decreases. This results in an ionospheric layer which shows a steep increase as altitude increases (the bottom side of the layer) and a more gradual decrease after the peak (the topside of the layer). Such a layer is called a Chapman layer after the scientist who first derived the form of this ionospheric profile, which is given by the formula:

$$N_e = N_0 e^{\{0.5[1-h-\alpha e^{-h}]\}}$$

where h is the height and α is the secant of the solar zenith angle. N_e is the electron density ([Space Academy, n.d.](#)).

The formation of the ionosphere is caused primarily by photoionization of atmospheric gases by solar ultraviolet (UV) and X-ray radiation. The fundamental reaction that describes this process is:



Where:

X = Neutral atmospheric molecule or atom (e.g., N_2 , O_2 , O).

$h\nu$ = Photon of UV or X-ray energy from the sun.

X^+ = Resulting positively charged ion.

e^- = Freed electron

This process increases electron density, which produce the ionized layers of the ionosphere (D, E, and F regions), which vary by altitude, solar activity, and time of day.

Factors that affect the Ionosphere

I. Solar Radiation (EUV/X-ray):

The density of ionization varies in response to changes in the intensity and properties of radiation from the Sun. The output of solar energy is relatively constant in the visible and near-ultraviolet portions of the spectrum. It varies appreciably, however, at shorter wavelengths, reflecting changes in the temperature of the outermost regions of the solar atmosphere. The changes are particularly large, in excess of a factor of 10, at X-ray wavelengths. Variations in the D region are correspondingly large, with smaller though still significant changes in the E and F layers (Encyclopædia Britannica, n.d., under Ionospheric variations section).

II. Solar Wind and Magnetospheric Interactions:

The solar wind is a continuous stream of protons and electrons from the sun's outermost atmosphere i.e. the corona. These charged particles flow past the solar system at speeds ranging from 400km per second to 500km per second, in a plasma state according to the National and Oceanic and Administration Space Weather prediction Center (SWPC). When this solar wind reaches earth, it compresses the sun facing side of the magnetosphere and stretches the night side into a long tail. The relationship between solar wind and the magnetosphere is important for shaping ionospheric conditions.

One of the main ways the ionosphere is affected is through magnetospheric currents and electric fields that are driven into the upper atmosphere. Intensified solar winds (such as coronal mass ejections) cause geomagnetic storms, during geomagnetic storms energy is transferred from the magnetosphere into the ionosphere via field aligned currents. These currents heat and ionize the upper atmosphere, increasing electron density in some regions while depleting it in others (Dobrijevic, 2022; Tsurutani et al., 2004; Kelley, 2009).

Additionally, enhanced particle precipitation occurs when energetic electrons and ions from the magnetosphere are funneled along magnetic field lines into the polar ionosphere. This increases ionization in the D and E regions and changes conductivity and current systems in the polar zones. The result is auroras, as well as disturbances in communication systems and satellite navigation (Zolesi & Cander, 2014).

Furthermore, solar wind-induced electric fields penetrate to lower latitudes and drive plasma transport, leading to phenomena like equatorial plasma bubbles, TEC (Total Electron Content) enhancements or depletions, and variations in the F2 layer peak

height (hmF2) and critical frequency (foF2) ([Fejer, 2002](#)). These changes depend on the strength, direction, and duration of solar wind pressure and interplanetary magnetic field (IMF) conditions.

III. Joule Heating & Neutral Winds:

Joule heating significantly impacts the ionosphere by increasing the temperature of the upper atmosphere through the dissipation of electromagnetic energy. This occurs when electric fields—often driven by solar wind–magnetosphere interactions—generate currents in the ionosphere. As these currents flow, particularly the Pedersen currents, they collide with neutral particles in the thermosphere, leading to heating ([Schunk & Nagy, 2009](#)). This heating causes thermal expansion, elevates ionospheric layers (like the F-region), and can alter the density structure of the thermosphere, affecting satellite drag and communication systems ([Hargreaves, 1992](#)).

On the other hand, neutral winds—winds composed of uncharged atmospheric particles—also play a crucial role in shaping ionospheric behavior. These winds can redistribute ionospheric plasma through ion-neutral collisions, modify electric field structures, and contribute to features like the equatorial ionization anomaly. When neutral winds move across magnetic field lines, they can generate electric fields and influence the motion of charged particles, thereby altering the ionosphere’s structure and dynamics ([Fuller-Rowell et al., 1997](#)).

IV. Geomagnetic Storms:

Geomagnetic storms significantly influence the ionosphere by affecting its electron density, structure, and dynamics. These storms are caused by enhanced solar wind conditions, especially coronal mass ejections and high-speed solar wind streams which interact with Earth’s magnetosphere and induce electric currents and magnetic field variations. The ionosphere, being an electrically charged layer of the atmosphere, responds dramatically to these disturbances ([Hargreaves, 1992](#)).

One major effect is the redistribution of ionospheric plasma. During storms, increased energy from the magnetosphere enhances Joule heating and particle precipitation, that drives thermospheric winds and causes upsurge of neutral gases. This can reduce the electron density at low and mid-latitudes and increase it in others, leading to what is known as positive and negative ionospheric storm effects ([Buonsanto, 1999](#)). These effects repeatedly give rise to Total Electron Content (TEC) fluctuations, which can damage satellite communication and GPS accuracy ([Fuller-Rowell et al., 1994](#)).

The storm-induced penetration of electric fields and the disturbance dynamo further affect ionospheric currents and plasma transport. For instance, at equatorial latitudes, storms may enhance or reverse the equatorial electrojet, modifying the Equatorial Ionization Anomaly (EIA), a key structure in the low-latitude ionosphere ([Fejer & Scherliess, 1997](#)). These processes vary with local time, season, latitude, and storm intensity, making storm-time ionospheric response complex and location-dependent.

Moreover, geomagnetic storms impact the ionospheric F2 region where critical parameters such as foF2 (critical frequency) and hmF2 (peak height) reside by altering thermospheric composition and temperature. These parameters are essential for

understanding high-frequency radio wave propagation, and their disturbance can hinder space weather prediction models ([Rishbeth & Mendillo, 2001](#)).

In conclusion, geomagnetic storms affect the ionosphere in many ways, including enhanced particle precipitation, increased thermospheric heating, and electric field penetration. These changes can affect communications, navigation systems, and satellite operations, this iterates the importance for accurate modeling and forecasting of space weather effects on the ionosphere.

V. Atmospheric Composition and Circulation:

Earth's atmospheric composition, predominantly nitrogen (N₂) and oxygen (O₂) with traces of other gases, serves as the primary reservoir for the energy and particles that eventually reach the ionosphere ([Wikipedia, 2025a](#)). Variations in these gases due to factors like greenhouse forcing alter how solar radiation penetrates and how heat and particles rise into the upper layers ([Encyclopedia of Atmospheric Sciences, n.d.](#))

Additionally, atmospheric circulation the vertical and horizontal air movement transports heat, moisture, and momentum upward. One well-known pattern is the Brewer–Dobson circulation, which moves air from the tropics to the poles in the stratosphere, affecting the global distribution of gases such as ozone and water vapor ([Wikipedia, 2025b](#)).

Lower-atmosphere systems create waves gravity, planetary, and tidal waves—that propagate upward. These waves induce fluctuations in ionospheric parameters like electron density and temperature. As summarized by *Frontiers in Space Aeronomy* (2021), “Atmospheric waves... may directly or indirectly affect winds, temperature, compositional structures...the ionospheric wind dynamo” (p. 4).

These waves break in the mesosphere/lower thermosphere region, redistributing energy and momentum that alter neutral winds and mix atmospheric constituents, causing significant impact on ionospheric structure ([Frontiers in Space Aeronomy, 2021](#)).

Neutral winds, driven by circulation and waves, generate electric fields via dynamo action at E-region altitudes. These electric fields can drift plasma, shifting the distribution of electrons vertically and horizontally. That in turn affects parameters like TEC, foF₂, and the global electron density structure ([Frontiers in Space Aeronomy, 2021](#)).

Furthermore, atmospheric circulation systems like the Hadley cell and planetary waves modulate these effects. For example, planetary waves alter the ionosphere by modulating tides, affecting upper atmospheric winds and temperatures ([Frontiers in Space Aeronomy, 2021](#)). The Brewer–Dobson circulation influences the chemical environment at ionospheric heights, potentially affecting the ionization and recombination rates ([Wikipedia, 2025b](#)).

In summary, atmospheric composition determines which gases and trace constituents are available for ionization, while circulation drives dynamic transport and wave propagation. Together, these processes shape the thermal and chemical environment in which ionospheric plasma forms and evolves. Their interplay explains why ionospheric variability cannot be fully understood by solar and geomagnetic drivers alone it is inseparably linked to the underlying atmosphere.

VI. Greenhouse gas composition:

Greenhouse gases particularly carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) play a profound role not only in surface warming but also in upper atmospheric physics. While they warm the lower atmosphere by trapping infrared radiation, in the thin upper layers, they actually cool the atmosphere by efficiently emitting infrared energy to space (Roble & Dickinson, 1989; Añel et al., 2025). This cooling shrinks the atmosphere, affecting dynamics, composition, and crucial ionospheric parameters.

Firstly, increasing CO₂ in the thermosphere, which acts as a major radiative coolant, leads to significant cooling and contraction of the upper atmosphere (Roble & Dickinson, 1989; Laštovička et al., 2008). As Añel et al. (2025) explain, “Greenhouse gas emissions ... cause cooling and thermal shrinking and affect the atmospheric structure” (p. 1). This contraction causes the ionosphere’s F-region peak (hmF₂) to drop lower in altitude, altering its density structure (Añel et al., 2025; Laštovička et al., 2008).

Numerical modeling supports these observations. Kogure et al. (2022) show that increased CO₂ leads to a cooler thermosphere and lower hmF₂: “The ionosphere is cooling ... and the peak altitude of the electron density (i.e., hmF₂) descends and the density increases (decreases) below (above)” ~230–300 km (p. 1). Similarly, Liu et al. (2020) demonstrate that zonal wind and meridional circulation strengthen due to CO₂ cooling, further influencing ionospheric dynamics: “Thermosphere cooling is stronger around solstice ... semidiurnal tides are strongly reduced” (p. 1).

In contrast to the troposphere, where GHGs cause warming, the upper atmosphere cools a phenomenon referred to as “greenhouse cooling” (Añel et al., 2025; Roble & Dickinson, 1989). Añel et al. (2025) describe that in the thin upper atmosphere, infrared emissions escape quickly, causing net cooling:

“Greenhouse gases in the much lower density upper atmosphere are optically thin ... resulting in infrared radiative cooling” (p. 1).

This cooling effect decreases thermospheric density (confirmed by satellite drag data) and alters ionospheric layer heights and densities (Añel et al., 2025; Laštovička et al., 2008).

Lastly, changes in atmospheric composition and circulation due to greenhouse gas trends can modify ionospheric currents and electrodynamics, influencing geomagnetic activity’s coupling with the ionosphere (Cnossen, 2014). Cnossen (2014) found that CO₂ increases contributed as much as geomagnetic field changes to changes in peak electron density and currents, showing how GHG-driven structural and thermal changes propagate into ionospheric behavior.

In summary, rising GHG concentrations cool and contract the upper atmosphere, lowering ionospheric layer heights (e.g., hmF₂), altering electron densities, and shifting circulation patterns. These effects impact radio propagation, satellite drag, and our ability to model long-term ionospheric trends making consideration of GHG composition essential in any ionospheric study.

VII. Diurnal and Seasonal Cycles:

The Earth's ionosphere exhibits strong patterns of variability tied to diurnal (daily) and seasonal cycles, which significantly influence its density structure, critical frequencies, Total Electron Content (TEC), and ionospheric dynamics. These cyclic variations shape the overall conditions of the ionosphere and play a central role in satellite communications, navigation, and geomagnetic storm responses.

Every day, solar extreme ultraviolet (EUV) and X-ray radiation ionizes the upper atmosphere, creating the ionosphere on the sunlit side, while nighttime recombination reduces ionization levels (Chapman, 1931, as cited in Ogwala et al., 2019). This results in a pronounced diurnal cycle. For example, TEC values typically rise sharply after sunrise, peak around midday, and decline exponentially after sunset. An observational study in Nigeria noted, “day-to-day variation in OBS-TEC is higher during the daytime than nighttime for all the years” (Ogwala et al., 2019, p. 1).

These daily fluctuations are driven by solar-driven processes such as enhanced neutral winds and electric fields that modulate electron production and dynamics (Gorney, 1990; Forbes et al., 2006, as cited in Liu et al., 2020).

On seasonal timescales, the ionosphere also demonstrates systematic behavior. Seasonal studies of global TEC and critical frequencies indicate that equinox periods tend to exhibit the highest diurnal peaks, while solstices generally show lower maxima. For instance, Teunissen and Sagi (1967) found that “the largest diurnal peaks in daytime content tend to occur during the equinoxes and the smallest diurnal peaks tend to occur during solstices” (as cited in Yizengaw et al., 2014).

Such patterns have been observed globally. Studies using COSMIC satellite data from 2006–2019 show clear seasonal trends in NmF₂ and hmF₂ (critical frequency and peak height) associated with variations in thermospheric winds and neutral composition (Liu et al., 2020).

Atmospheric tides—driven by solar heating—propagate upward, modulating ionospheric winds, density, and temperature in both diurnal and seasonal sequences (Wikipedia, 2023).

The diurnal and seasonal variations are further complicated by geographic and geomagnetic differences. Research on TEC in Birnin Kebbi (northern Nigeria) from 2011 to 2014 revealed distinct seasonal variation: higher TEC during equinoxes than solstices, though specific patterns varied yearly (Ogwala et al., 2019).

Equatorial regions in particular show unique patterns related to the Equatorial Ionization Anomaly, where crests of enhanced electron density form away from the magnetic equator due to electrojet-driven fountain effects—variations that exhibit both diurnal and seasonal asymmetry (Zhao et al., 2020).

In summary, diurnal and seasonal cycles impose predictable structure on the ionosphere: daily peaks tied to sunlight, and seasonal modulation linked to solar geometry, neutral atmospheric dynamics, and geomagnetic context. Any comprehensive ionospheric study or model must therefore account for these temporal cycles to accurately reflect natural ionospheric variability.

VIII. 27-Day Solar Rotation Effects:

The Sun's 27-day synodic rotation brings active regions like sunspots to repeatedly face Earth, causing cyclical variation in solar ultraviolet (EUV) radiation and solar wind streams. These cycles drive predictable changes in the ionosphere's electron density and related parameters.

A seminal study by Ma et al. (2012) and further supported by Schmölter et al. (2022) emphasizes that “the ~27 - day variations in solar radiation and geomagnetic activity, caused by solar rotation, are the main drivers of the ionospheric ~27-day signatures” (Ma et al., 2012; Schmölter & von Savigny, 2022). This connection means that as the Sun rotates, increased EUV input enhances ionization in the upper atmosphere, visibly altering ionospheric characteristics.

Indeed, Schmölter & von Savigny (2022) describe the mechanism:

“Photoionization of O to O⁺ ... defines the so-called F region ... The 27-day signatures ... show height-dependent variations based on the contribution of the different processes” (p. 1).

Crucially, these ionospheric changes are not instantaneous. Schmölter et al. (2021) investigated the delay and found that “a delay of a few hours to 2 days occurs” in the ionospheric response to solar rotation-driven changes in EUV and solar wind (Schmölter et al., 2021). They elaborate that variations in electron density at the F-region peak lag behind the solar EUV flux by about one day, confirming a cascade effect between radiation and atmospheric response.

Further explaining this effect, Schmölter et al. (2021) state:

“The absorption of solar extreme ultraviolet (EUV) radiation in the upper atmosphere defines the ionospheric plasma ... The 27-day solar rotation period is a significant variation of the solar activity that is known to impact the ionospheric plasma” (p. 1).

These cyclical dynamics also extend to molecular oxygen densities and temperature structures. For example, Schmölter & von Savigny (2022) found the delayed appearance of signatures in O₂ densities at lower heights, with larger delays in higher altitudes attributed to delayed photodissociation and recombination effects.

Moreover, the impact isn't limited to radiation-induced changes. Quasi-27-day solar wind streams often emanating from coronal holes generate geomagnetic oscillations. As noted by Scherliess et al. (2013), “Solar wind high-speed streams ... when [they] reach Earth ... induce 27-day variations in the ionosphere.” These recurrent geomagnetic drivers reinforce the radiation-based cycles, creating a compound effect.

IX. Ionospheric Temperature:

The ionospheric temperature encompassing both electron and ion temperatures plays a pivotal role in determining plasma density, dynamics, and energy balance in the Earth's upper atmosphere. Since the ionosphere is a partially ionized plasma, its temperature directly influences collisional processes, heat conduction, and chemical reactions.

At lower altitudes (150–300 km), electrons gain most of their heat from solar extreme ultraviolet radiation and energetic particle precipitation, creating a temperature significantly higher than the surrounding neutral gases. As noted by Brace et al. (1962):

“The absorption of EUV should cause the electron temperature to exceed the gas temperature between approximately 150 and 350 km in the quiet midday ionosphere” (Brace et al., 1962, as cited in *Extreme two-phase change...*, 2025).

This electron–neutral temperature difference fuels a variety of dynamical responses. For instance, when electron temperature (T_e) increases, the collision frequency drops, reducing energy transfer from electrons to ions, and altering plasma thermal structure. As reported by Köhnlein et al. (2015):

“... electron density and electron temperature are related via ... electron critical angular frequency and Debye wavelength” (Köhnlein et al., 2015, para. 4).

At high latitudes, this relationship becomes pronounced during geomagnetic storms, where electron precipitation raises both electron temperature and electron density:

“Particles precipitating into the polar ionosphere ... heat the electrons. ... cause a strong increase in the electron density and temperature in the F region” (Borries et al., 2024, para. 2).

As electron temperature increases, Joule heating and thermal conduction become significant. The ionospheric plasma dissipates energy along magnetic field lines into neighboring regions, affecting ion and neutral temperatures. Hanlon demonstrates that ion thermal conduction can maintain a stratified ionospheric temperature, influencing ion peak and plateau levels (Hanlon, n.d.).

Furthermore, during geomagnetic storms, a “two-phase change of ionospheric electron temperature” is observed—first a rapid rise due to prompt penetration electric fields, then a fall as disturbance dynamo electric fields take over (Smirnov et al., 2025). This dynamic thermospheric shift alters vertical plasma drifts, reshaping electron density and cooling rates in the F-region (Smirnov et al., 2025).

Uses of the Ionosphere

The ionosphere, a region of Earth’s upper atmosphere (approximately 60–1,000 km), plays a critical role in modern life by enabling long-distance communication and navigation, supporting space weather research, and protecting aerospace operations. One of its most essential functions is its ability to refract and reflect radio waves—particularly in the high-frequency (HF) band—which allows signals to “bounce” beyond the horizon for transcontinental and maritime communications (Bora, 2017; Frontiers Editorial Office, 2021). This phenomenon—skywave propagation—not only connects remote regions but ensures emergency and military communications remain viable when other systems fail (Wikipedia, n.d.-a; Shukla, 2011).

Beyond HF communications, the ionosphere can also degrade or delay signals traversing its layers, impacting satellite-based systems like GPS. As signals travel through the ionized plasma, they experience phase shifts, group delays, and scintillation—effects that introduce errors into navigation and timing systems (GFZ German Research Centre for Geosciences, 2023; Royal Belgian Institute for Space Aeronomy, n.d.; SciTechDaily, 2008). To maintain

high accuracy, GNSS receivers and other systems apply ionospheric models or dual-frequency corrections to compensate for these disturbances (GFZ German Research Centre for Geosciences, 2023; Stationary Ionization Model Team, 2016).

The ionosphere's sensitivity to solar and geomagnetic conditions makes it an invaluable probe of space weather. Instruments like ionosondes, incoherent scatter radars, and satellite missions such as ICON capture real-time changes in electron density, composition, and dynamics—essential data for understanding solar storms, geomagnetic substorms, and their effects on Earth's upper atmosphere (Eos Editorial Board, 2022; Sadiq, 2008; Gaia Space Mission Team, 2023). Programs like the IRI model owe their success to decades of global ionospheric monitoring, which inform not just science but practical forecasting and alerting systems (Eos Editorial Board, 2022).

In scientific and engineering contexts, the ionosphere is also deliberately manipulated or excited to study its response. Projects like HAARP and SuperDARN use HF transmissions to create plasma waves and reveal underlying ionospheric dynamics and conductivity structures critical for understanding fundamental geospace physics (Wikipedia, 2024; Wikipedia, 1995).

In summary, the ionosphere is far more than a passive atmospheric layer: it enables global communication, shapes navigation accuracy, preserves space-borne infrastructure, and advances geospace science through both observation and active experimentation. Its complex behavior under solar and terrestrial influences makes it indispensable to both society's technological backbone and scientific research.

As enumerated earlier, the ionosphere is affected by a lot of factors, including time factors but in this project the focus will be on two major factors, geomagnetic storms and greenhouse gases.

2.1.2 Geomagnetic Storm

Geomagnetic storms are large-scale disturbances of Earth's magnetosphere, driven primarily by solar wind structures such as coronal mass ejections (CMEs) or corotating interaction regions (CIRs), which transfer energy into the magnetospheric system and ignite currents and plasma flows across the magnetosphere and ionosphere (NASA, 2023; National Oceanic and Atmospheric Administration [NOAA], 2023). The Dst index, indicating hourly changes in the horizontal geomagnetic field near the equator, is used to define storms—with Dst dropping below -50 nT marking the threshold—more negative values signifying intense or super-storm events (NOAA National Centers for Environmental Information [NCEI], 2023; NASA, 2023). Concurrently, the Kp index, derived from standardized 3-hour local K indices across 13 mid-latitude observatories, categorizes storm intensity into NOAA's G-scale (G1–G5), where G1 signals minor activity and G5 designates extreme conditions (NOAA, 2023; South African National Space Agency [SANSI], 2023; NASA, 2023).

During a geomagnetic storm, the injection of energy into the magnetosphere feeds potent currents, notably the ring current and auroral electrojets. These currents, together with energetic particle precipitation, heat and distort the thermosphere and ionosphere, altering atmospheric density profiles and interfering with satellite operations and radio-communication systems (NOAA, 2023). Notable historical storms such as the Carrington Event and the 1989 Quebec blackout underscore the potential for severe disruption to power grids, communication networks, and navigation systems (Cliver & Svalgaard, 2004; NASA, 2023).

Formation of Geomagnetic Storms

Geomagnetic storms are large-scale disturbances in Earth's magnetosphere, typically triggered by interactions between solar emissions and the Earth's magnetic field. The primary drivers of these storms are Coronal Mass Ejections (CMEs) and high-speed solar wind streams often emanating from coronal holes on the Sun's surface (Gonzalez et al., 1994).

When a CME or fast solar wind stream reaches Earth, it carries with it a magnetic field—known as the Interplanetary Magnetic Field (IMF). If the IMF has a strong southward component ($B_z < 0$), it can reconnect with Earth's northward-pointing magnetic field lines (Dungey, 1961). This magnetic reconnection process injects solar energy and plasma into the magnetosphere, severely disrupting its equilibrium (Tsurutani & Gonzalez, 1997).

The injected energy compresses the dayside magnetosphere and stretches the nightside into a long magnetotail. This configuration intensifies currents, such as the ring current, which weakens Earth's magnetic field at the surface—measured as a drop in the Dst index (Gonzalez et al., 1994; Pulkkinen, 2007).

If the storm includes substorms, auroral electrojets and ionospheric disturbances increase, fueling further instability and energy redistribution throughout the geospace environment (Kamide & Kusano, 2015). The magnitude of the storm depends on the speed, density, and magnetic orientation of the solar wind structures.

Effects on the Ionosphere and Ionospheric Parameters

Geomagnetic storms profoundly disturb the ionosphere, altering both its composition and structure. They commonly cause positive and negative storm phases, impacting total electron content (TEC), electron density, critical frequency (f_oF_2), peak height (hF_2), and ionospheric temperature. These changes vary in magnitude and timing across geographical regions and storm intensity.

1. Total Electron Content (TEC) and Electron Density

During geomagnetic storms, TEC can either increase or decrease depending on storm phase and region. As summarized by Chukwuma et al.:

“A geomagnetic storm may lead to an increase or decrease in the electron density as compared to quiet days. A positive ionospheric storm is seen during the main phase, while in the recovery phase, negative storms are pronounced” (Chukwuma et al., 2021, p. 9).

Observational studies confirm this behavior: “TEC and NmF_2 were increased during a minor geomagnetic disturbance, but hmF_2 did not change” (Liu et al., 2016, p. 7).

Storm-induced structural changes in TEC, such as Storm-Enhanced Density (SED), often reflect intense storm phases. For example, during the March 2015 storm:

“TEC increased by about 10 TECU (50%) during the storm” (Liu et al., 2016, p. 11).

2. Critical Frequency (f_oF_2) and Peak Height (hmF_2)

The critical frequency f_oF_2 is tightly linked to maximum electron density:

“The response of the ionospheric F₂ layer consists of both a depletion and increase of the electron concentration called ‘positive’ and ‘negative’ phases” (Proëlss, 1999, p. 112).

Storms drive foF₂ variations regionally. In Europe during the November 2012 storm, researchers found midday jumps in O/N₂ ratios coincident with abrupt changes in foF₂ suggesting thermospheric composition and heating as influences.

Furthermore:

“During the main phase extremely depleted electron density was observed in foF₂ at all stations” (Tammy et al., 2023, p. 4).

Peak height hmF₂ also shifts significantly courtesy of storm-time electric fields and winds—which elevate the F₂-layer altitude, prolonging its plasma lifetime.

3. Ionospheric Temperature

Geomagnetic storms also heat the ionosphere:

- Electron temperature (T_e) rises due to particle precipitation and electric field penetration:

“Particles precipitating heat the electrons cause a strong increase in the electron density and temperature in the F region” (Unno et al., 2024, p.3).

During top-side storm events, T_e increases are often coupled with density rises but sometimes anti-correlated due to reduced collisional cooling.

- Ion temperature is also affected heated via friction and energetic inputs—modifying thermal structure and altering plasma scale height.

4. Thermosphere Composition & Neutral Winds

Geomagnetic storms alter thermospheric composition and circulation:

- Auroral heating enhances neutral winds and lowers the O/N₂ ratio: “The value of the O/N₂ ratio decreased in all high-latitude regions” during storm events (Doe & Lee, 2023, p.6).
- These winds and compositional shifts propagate equatorward, causing “negative ionospheric storms” depletions in electron density by enhancing recombination rates.
- Conversely, positive storm effects arise when electric fields uplift plasma, decreasing recombination via travel time delays.

Effects on Space Research and Exploration

Geomagnetic storms significantly influence space research and exploration, affecting satellites, spacecraft, and mission planning. The incursion of these storms initiates complex interactions that have profound implications for modern space endeavors.

One of the foremost impacts of geomagnetic storms is on satellite operations in low Earth orbit (LEO). Abrupt heating and expansion of the thermosphere increase atmospheric density, leading to heightened drag forces. As noted by Parker and Linares (2024), “total mass density enhancements in the thermosphere are identified by tracking satellite drag decay characteristics” (para. 7). This drag surge can shift satellite trajectories by several kilometers

within days, which is critical for mission-critical precision tasks, such as collision avoidance and formation flying (Parker & Linares, 2024).

Historically, this effect has even led to satellite loss events. For example, in February 2022, a geomagnetic storm prompted the premature re-entry of approximately 40 Starlink satellites. Ahmad et al. (2025) explained that “magnetic substorms significantly affected species densities... leading to an increase in atmospheric drag” (p. 1), demonstrating how storms jeopardize satellite fleet health.

Geomagnetic storms also pose risks to spacecraft electronics and instrumentation. Charged particles from the solar wind can induce spacecraft charging and single-event upsets, flipping bits in satellite computer systems (NASA, 2003, as cited in Space Weather, n.d.) a threat that increases equipment malfunction rates during intense geomagnetic events (Space Weather, n.d.).

Moreover, there are increased radiation hazards for astronauts and high-altitude crew. Geomagnetic storms can elevate radiation exposure levels aboard the International Space Station or polar-flight aircraft. NASA highlighted that “astronauts receive more of a radiation dose than earthbound people... especially during a solar storm” (Time, 2024, sec. Getting ready for the next one).

In addition, navigation and communication systems suffer notably. GNSS signals traveling through the ionosphere become disturbed, affecting precision and reliability. The heightened ionospheric electron content and scintillation induce errors in positioning systems at critical times (Ahmad et al., 2025; Parker & Linares, 2024).

Finally, these storms interrupt telemetry and ground-satellite communications. Storm-induced auroral phenomena and magnetic fluctuations can distort HF and VHF signal propagation, requiring ground operators to rely on backup protocols during storm events (NASA Jet Propulsion Laboratory [JPL], n.d.; NASA Science, n.d.).

2.1.3 Greenhouse Gases

Greenhouse gases (GHGs) are atmospheric gases that trap heat within Earth’s atmosphere by absorbing and re-emitting infrared radiation, thereby contributing to the greenhouse effect and influencing global climate patterns. The primary greenhouse gases include carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), ozone (O₃), and water vapor (H₂O) (NASA, 2022; IPCC, 2021).

Carbon dioxide, largely produced by fossil fuel combustion and deforestation, is the most significant anthropogenic GHG due to its long atmospheric lifetime. Methane, though less abundant, has a far greater heat-trapping capability per molecule and is emitted by agriculture, waste management, and fossil fuel extraction (EPA, 2023). Nitrous oxide arises primarily from agricultural soil management and industrial activities (UNEP, 2021).

Water vapor is unique among GHGs because its concentration is controlled by temperature: it amplifies warming through a positive feedback loop rather than acting as a direct driver (Lindsey, 2023). While ozone is a short-lived gas in the troposphere, it also contributes to localized warming and air quality degradation (WMO, 2018).

Collectively, these gases trap outgoing longwave radiation and cause Earth's average surface temperature to rise, a process fundamental to understanding global warming and climate change (IPCC, 2021).

How are Greenhouse gases related to the Ionosphere

Greenhouse gases (GHGs), including carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), and water vapor, are widely recognized for their role in trapping heat in the lower atmosphere. However, their influence extends beyond surface-level climate change and into the upper atmosphere particularly the ionosphere. While the ionosphere is primarily shaped by solar radiation and geomagnetic activity, increasing concentrations of greenhouse gases have been shown to alter its thermal structure, electron density, and general dynamics over long timescales (Roble & Dickinson, 1989; Qian et al., 2023).

In the thermosphere and ionosphere, CO₂ acts differently than it does near Earth's surface. Instead of warming the upper atmosphere, it serves as an efficient radiator, emitting infrared radiation into space and thus contributing to significant cooling in these higher layers (Emmert, 2015). This cooling causes the thermosphere to contract, which leads to a reduction in atmospheric density at satellite altitudes. NASA satellites have confirmed this contraction, attributing it to the increasing levels of anthropogenic greenhouse gases in the atmosphere (NASA, 2021). As a result, the background density and temperature of the ionosphere are altered, which can affect satellite drag, orbital decay rates, and ionospheric dynamics.

Moreover, this cooling impacts the production and loss rates of ionospheric plasma. The contraction of the thermosphere changes the distribution of key neutral species such as atomic oxygen and molecular nitrogen, which in turn influences the electron density and height of the F2 layer (Pedatella et al., 2022). A lower neutral density reduces the efficiency of ionization processes, leading to long-term declines in parameters such as Total Electron Content (TEC) and peak electron density (NmF2) (Qian et al., 2023).

These changes are particularly evident during geomagnetic storms. Normally, such storms inject energy into the upper atmosphere, heating it and increasing TEC and ionospheric disturbances. However, in a cooler, contracted thermosphere, the storm-time response is muted. Studies have shown that while relative TEC variability may increase, the absolute changes in electron density and altitude shift are reduced under high-CO₂ conditions (Pedatella et al., 2022). This implies that greenhouse gas-driven background changes modulate not only daily ionospheric behavior but also its response to extreme space weather events.

In addition, the positive feedback loop involving water vapor where warming leads to more evaporation and more greenhouse effect can indirectly affect the ionosphere by modifying the underlying thermodynamic structure of the atmosphere (Lindsey, 2023). This further complicates the chain of interactions between greenhouse gases and ionospheric parameters.

Effects of greenhouse gases on the ionosphere and ionospheric parameters

The increasing concentrations of greenhouse gases (GHGs) such as carbon dioxide (CO₂) have long been known to cool the upper atmosphere—a process often termed “greenhouse cooling”—because these gases efficiently radiate thermal energy to space at high altitudes (Cicerone, 1990; Roble & Dickinson, 1989). This cooling effect leads to contraction of the thermosphere and ionosphere, resulting in lower neutral and electron densities at key altitudes (Roble & Dickinson, 1989; Cnossen et al., 2024).

This contraction has significant consequences for ionospheric parameters. Thermospheric shrinkage reduces neutral density at fixed altitudes, which in turn affects the total electron content (TEC) by lowering electron production and recombination rates (Pedatella et al., 2025; Rishbeth & Roble, 1992). Model simulations reveal that with rising CO₂ levels, storm-time TEC responses become weaker in absolute terms, while relative variability increases due to lower base densities (Pedatella et al., 2025; Pedatella et al., 2025).

Electron density specifically has declined by roughly 16.6% from the 1920s to the 2010s at mid-latitudes, attributed mainly to this GHG-induced cooling of the ionosphere (Cnossen et al., 2024). Similarly, peak parameters such as foF₂ (critical frequency) and hmF₂ (peak height) show long-term trends linked to GHG increases and reduced solar activity. FoF₂ has exhibited decreasing trends at many stations, while hmF₂ tends to lower as cooler, contracted thermospheric layers rise (Jakowski et al., 2024).

Ionospheric temperature patterns are likewise influenced. Cooler thermospheric conditions lead to lower ion and electron temperatures overall, particularly on the nightside, due to reduced collisional heating. However, warming may still occur during geomagnetic storms as GHGs alter neutral winds and energy input pathways (Pedatella et al., 2025; Cnossen et al., 2024).

Neutral mass density also decreases in the upper atmosphere with rising CO₂. Model outputs indicate a ~20–25% reduction in absolute thermospheric neutral density changes during storms when CO₂ rises from ~400 to ~900 ppmv (Pedatella et al., 2025). Observationally, model-data comparisons note drops of about -1.85×10^4 electrons/cm³ in electron density over decades (Cnossen et al., 2024).

In summary, greenhouse gas increases cool and contract the upper atmosphere, leading to lower TEC, electron density, foF₂, hmF₂, and neutral and electron temperatures. These effects interact with solar and geomagnetic forcing, modulating ionospheric responses to both long-term trends and individual storm events (Pedatella et al., 2025; Cnossen et al., 2024; Jakowski et al., 2024).

Effects on Space Research and Exploration

Rising concentrations of greenhouse gases (GHGs) such as carbon dioxide, methane, and nitrous oxide are not only altering Earth's climate but also significantly impacting space operations. Research has found that as GHG emissions continue to increase, they cause the upper atmosphere particularly the thermosphere to cool and contract, reducing atmospheric drag at altitudes where satellites and debris orbit (Hatfield, 2021; Parker & Linares, 2025). Consequently, space debris and satellites remain aloft longer, increasing the probability of orbital collisions and presenting new challenges for space sustainability.

Parker and Linares (2025) estimate that by 2100, satellite carrying capacity in low Earth orbit (200–1,000 km) could decline by 50–66% under high-emission scenarios, resulting from diminished thermospheric drag (Parker & Linares, 2025). This 'shrinking' thermosphere was directly observed by NASA mission data showing sustained contraction over decades because of GHG-induced cooling (Hatfield, 2021).

Compounding this, analyses from the Naval Research Laboratory indicate a 10% decrease in mean thermospheric density over the past 35 years, reinforcing the thermosphere's sensitivity to elevated CO₂ (Emmert et al., 2004).

These atmospheric changes have multiple implications for space research:

1. Orbit management: Reduced drag prolongs satellites' orbital lifetimes, but also increases the persistence of debris, heightening Kessler syndrome risks. Consequently, satellite operators must revise station-keeping strategies to maintain orbital accuracy (University of Birmingham, 2025; MIT News, 2025).
2. Collision risk and sustainability: With more objects occupying crowded orbits and a diminished atmospheric cleaning mechanism, collision avoidance maneuvers are becoming more frequent and essential for mission safety (The Verge, 2025; The Guardian, 2025).
3. Mission architecture and design: Satellites may require enhanced collision resilience, onboard propulsion, and advanced traffic monitoring to contend with reduced drag and a debris-rich environment—driven partly by GHG-induced thermospheric changes (MIT News, 2025).

2.1.4 Relationship between the Ionosphere, Geomagnetic Storms and Greenhouse gases

Geomagnetic storms and greenhouse gas (GHG) concentrations both significantly influence the ionosphere, albeit through distinct mechanisms that interact to shape its structure, dynamics, and variability. Geomagnetic storms inject energy into the magnetosphere–ionosphere system via solar wind drivers such as coronal mass ejections (CMEs) and corotating interaction regions (CIRs), triggering complex electromagnetic and plasma processes that lead to enhanced ionization, heating, and circulation changes in the ionosphere (Ratovsky et al., 2022; NOAA SWPC, 2025). These disturbances manifest as increases or decreases in Total Electron Content (TEC), electron density, critical frequency (f_oF_2), peak height (h_mF_2), ionospheric temperature, and neutral winds (Ratovsky et al., 2022; Gao et al., 2023; NOAA SWPC, 2025). For example, intense storms can elevate TEC and electron density in some regions (positive storm effects), while causing depletions elsewhere due to storm-enhanced recombination and altered thermospheric circulation (negative storms) (Ratovsky et al., 2022; Gao et al., 2023).

In contrast, greenhouse gases influence the ionosphere indirectly by altering the neutral thermosphere's thermal structure and composition. Increasing CO₂ concentrations enhance radiative cooling at higher altitudes, causing thermospheric contraction, lower neutral densities, and cooler temperatures (Pedatella et al., 2025; Cnossen et al., 2013; Wright et al., 2014). These changes reduce satellite drag and alter the background state against which geomagnetic disturbances occur. Critically, they also enhance relative storm-time responses: although absolute TEC and neutral density perturbations during storms may weaken, the relative changes compared to the lowered background mean become more pronounced in high-CO₂ conditions (Pedatella et al., 2025; Cnossen et al., 2013).

When combined, geomagnetic storms and elevated greenhouse gases yield a nuanced interaction. Storm-driven enhancements in TEC, f_oF_2 , and electron density may be less strong in absolute terms due to the cooler, contracted thermosphere under high-CO₂ conditions, yet the percentage variability (relative to baseline) often increases. Additionally, background cooling influences storm-driven neutral winds and O/N₂ ratios, further modulating ionospheric responses such as F₂ layer peak height shifts and recombination rates (Pedatella et al., 2025; Cnossen et al., 2013).

In essence, geomagnetic storms act as transient energy injections creating ionospheric disturbances, while greenhouse gases enact long-term baseline modifications. These two influences intersect: a cooler, thinner thermosphere shaped by GHGs alters how the ionosphere reacts to geomagnetic storms, producing storm responses that may be smaller in absolute magnitude but more irregular and pronounced relative to background conditions. As GHG concentrations continue to rise, accurate modeling and forecasting of ionospheric behavior—especially during space weather events—must incorporate both storm drivers and evolving thermospheric backgrounds (Pedatella et al., 2025; Cnossen et al., 2013).

2.1.5 PyGAM Model

Definition and Unique Features of pyGAM

The Python Generalized Additive Model (pyGAM) is a flexible, open-source statistical modeling library designed to model complex, nonlinear relationships between a response variable and multiple predictors. pyGAM extends the classical Generalized Additive Model (GAM) framework by incorporating advanced spline-based smoothing techniques, allowing for adaptive model fitting and robust interpretability (Servén & Brummitt, 2020).

Applications and Strengths

pyGAM is particularly well-suited for scientific, environmental, and atmospheric modeling tasks where relationships between variables are not linear or obvious, but interpretability remains essential. It has found applications in areas such as climate science, biostatistics, ecology, and economics due to its ability to provide interpretable models while maintaining flexibility. This makes pyGAM an attractive alternative to traditional linear regression and more opaque machine learning models like neural networks or random forests.

What Makes pyGAM Unique

Several core features distinguish pyGAM from other modeling tools:

- **Smooth term fitting using penalized splines:** Unlike many machine learning models, pyGAM explicitly models the shape of each predictor's effect through splines. These smooth terms are adjusted by the user-defined or automatically tuned regularization (λ), which balances model complexity and overfitting (Servén & Brummitt, 2020).
- **Interpretability:** Each predictor's individual influence on the response variable is visualized using partial dependence plots, enabling researchers to identify trends and patterns without black-box abstraction.
- **Automatic hyperparameter tuning:** pyGAM supports grid search for selecting the optimal number of splines and λ value to minimize model error, specifically through Generalized Cross Validation (GCV), which is used as a key performance measure.
- **Support for non-normal distributions:** The framework handles a wide range of link functions and distributions, making it extensible for various types of dependent variables (e.g., Gaussian, Poisson, Binomial).

Due to these capabilities, pyGAM has become a powerful tool for uncovering subtle, nonlinear dependencies in atmospheric and geophysical data—making it ideal for ionospheric studies

where variables like Dst index, CO₂, and solar activity influence each other in complex, indirect ways.

Theoretical Overview of pyGAM

PyGAM (Python Generalized Additive Models) is based on the theory of Generalized Additive Models (GAMs), which extend linear regression by modeling non-linear relationships between the response and predictor variables. Rather than assuming a fixed linear contribution from each predictor, GAMs assume the response is a sum of smooth functions of the predictors:

$$y = \beta_0 + f_1(x_1) + f_2(x_2) + \dots + f_n(x_n) + \varepsilon$$

Where:

- y is the dependent (response) variable (e.g., TEC, foF2, hmF2),
- $f_i(x_i)$ is a smooth, non-linear function of predictor x_i (e.g., Dst, CO₂, solar cycle phase),
- β_0 is the intercept,
- ε is the error term.

Each smooth function $f_i(x_i)$ is estimated using regression splines.

Spline Basis and Function Construction

pyGAM typically uses cubic B-splines to construct smooth functions. These splines divide the predictor's domain into a series of intervals separated by knots, fitting polynomial segments between the knots and ensuring continuity up to the second derivative.

Each function $f_i(x_i)$ is expressed as a linear combination of basis functions:

$$f_i(x_i) = \beta_1 \cdot B_1(x_i) + \beta_2 \cdot B_2(x_i) + \dots + \beta_k \cdot B_k(x_i)$$

Where:

- $B_j(x_i)$ are the spline basis functions,
- β_j are coefficients estimated from the data,
- k is the number of basis functions (determined by the number of knots and spline degree).

This flexible representation allows the model to adapt to non-linear trends in the data.

Regularization and Overfitting Control

To prevent overfitting, pyGAM adds a smoothness penalty to the loss function. The penalized residual sum of squares is minimized as follows:

$$\text{Loss} = \sum (y_i - \hat{y}_i)^2 + \lambda \cdot \int [f''(x)]^2 dx$$

Where:

- The first term is the sum of squared residuals,
- The second term is a penalty on the second derivative of the smooth function (to reduce excessive curvature),
- λ is the smoothing parameter selected via generalized cross-validation (GCV).

Higher values of λ result in smoother fits; lower values allow more flexibility

Breakdown of all the performance parameters in PyGAM

1. Log-Likelihood

The log-likelihood measures the probability of the observed data under the model. A higher value indicates a better fit of the model to the data (Wood, 2017).

Better model: Higher log-likelihood implies better model adequacy.

2. Scale

The Scale is an estimate of the residual variance; for Gaussian models, it is equivalent to the Mean Squared Error (Hastie, Tibshirani, & Friedman, 2009).

Better model: Lower scale values suggest more accurate predictions with smaller residuals.

3. Deviance

The deviance quantifies how far the model's predictions deviate from a perfect fit. It is analogous to the residual sum of squares and is minimized for better-fitting models (Wood, 2017).

Better model: Lower deviance signals improved model accuracy.

4. Total Variance Explained (R^2 Analog)

This measures the proportion of variance in the dependent variable explained by the model. Like R^2 in linear regression, it indicates goodness-of-fit (Hastie et al., 2009).

Better model: Higher values (closer to 1) indicate stronger explanatory power.

5. Effective Degrees of Freedom (EDF)

EDF represents the flexibility of the fitted spline. Higher EDF implies more model complexity. However, excessively high EDF may cause overfitting (Simonoff, 1996).

Better model: A balanced EDF prevents both overfitting and underfitting.

6. Generalized Cross Validation (GCV) Score

GCV approximates the model's out-of-sample prediction error while penalizing complexity. It is automatically minimized during model tuning in pyGAM (Serban, 2019).

Better model: Lower GCV reflects better generalization and appropriate smoothness.

7. Akaike Information Criterion (AIC)

AIC evaluates the trade-off between model fit and complexity. It penalizes the number of parameters to avoid overfitting (Akaike, 1974).

Better model: Lower AIC suggests a model that balances fit and parsimony.

Variable Importance Assessment in pyGAM

Variable assessment in pyGAM (Python implementation of Generalized Additive Models) centers on interpreting how much each predictor contributes to the response variable while accounting for the complexity and flexibility inherent in non-linear relationships. One of the defining features of pyGAM is its interpretability through Partial Dependence Plots (PDPs) and the impact on model performance when a variable is removed, both of which are central to variable assessment.

The most commonly used method to assess variable importance in pyGAM involves comparing the full model's metrics to those from a series of reduced models—each missing one predictor. Key indicators include changes in Total Variance Explained, AIC, GCV Score, and Deviance.

If omitting a variable significantly worsens these metrics, the variable is considered influential (Zschech et al., 2022; Khorrami Chokami & Rabitti, 2024).

For instance, Khorrami Chokami and Rabitti (2024) proposed a formal game-theoretic approach to quantify variable contributions within GAMs. Their method calculates exact Shapley values, offering a theoretically grounded importance index that aligns closely with empirical degradation methods like those used in pyGAM. Similarly, Lai et al. (2024) applied variable contribution scoring via the `gam.hp` package in R and confirmed its alignment with traditional metrics-based evaluation in pyGAM.

Variable importance is further explored by monitoring changes in partial deviance—the difference in deviance between the full model and the reduced model. A high partial deviance implies that the excluded variable plays a key role in explaining non-linear variability in the outcome (Yang et al., 2023).

Moreover, Zschech et al. (2022) emphasized that pyGAM provides better local feature attribution than many black-box models, as it visualizes how changes in predictor values affect the outcome directly, a feature crucial for applications where domain interpretability is as important as performance.

Thus, variable assessment in pyGAM combines theoretical rigor, empirical metrics, and interpretability—all of which support nuanced, statistically grounded insights about feature relevance in complex models.

PyGAM vs Deep learning models

When comparing pyGAM (Python Generalized Additive Models) to deep learning models, especially in the context of scientific domains like space weather–ionosphere interactions or climate–atmosphere studies, several critical distinctions emerge—chiefly around interpretability, flexibility, and model selection rationale.

1. Interpretability: A Core Strength of pyGAM

pyGAM is fundamentally designed around interpretable machine learning, where each predictor contributes additively through smooth functions. This enables users to visually and analytically isolate the influence of each variable, making it invaluable in scientific fields that demand explainability (Siems et al., 2023; Zschech et al., 2022). In contrast, deep learning models—such as fully connected neural networks or convolutional neural networks—are often black boxes. While they may achieve high predictive accuracy, they offer little insight into why predictions are made, unless complex post hoc interpretation tools are applied, which can still yield ambiguous results (Chang, Caruana & Goldenberg, 2021).

2. Capturing Nonlinearity Without Sacrificing Transparency

Both pyGAM and deep learning can capture nonlinear relationships, but pyGAM does so with clear structure and regularization controls, such as penalized splines, which prevent overfitting while maintaining interpretability (Voss et al., 2021). Deep learning models can model highly nonlinear functions but tend to be data-hungry, often requiring large, diverse training sets to generalize well—and their complexity often leads to overfitting in small-to-medium datasets, a typical setting in geophysical and ionospheric research.

3. Model Transparency and Scientific Rigor

pyGAM's ability to produce partial dependence plots and report effective degrees of freedom, deviance, AIC, and GCV scores helps scientists directly link model behavior to domain-specific theory. This is essential when assessing the individual contribution of CO₂, geomagnetic indices like Dst, or solar cycle phase to ionospheric variability. Deep learning models typically offer few avenues to perform formal statistical assessments, making them less suited for exploratory or confirmatory scientific analysis ([Mueller et al., 2024](#)).

4. Balance of Performance and Practical Utility

While neural networks can outperform GAMs in predictive tasks involving image or sequence data, in domains involving tabular, structured data—like solar–terrestrial physics—GAMs often match or exceed neural networks in utility and interpretive clarity ([Kruschel et al., 2025](#)). In climate modeling, for example, GAMs offer robust performance and clearer feature contributions than ML models like Random Forest or MLPs ([Bonsoms & Ninyerola, 2024](#)).

2.2 Review of Literature

In recent years, there has been a lot of research on how greenhouse gas compositions affect the upper atmosphere, there has been investigations on how the effects of greenhouse gases on the ionosphere affect the electron density of the ionosphere, satellite communications, and radio wave propagation. There are also several studies on how geomagnetic storms affect the ionosphere. Studies have been made to investigate whether the variations we see in the ionosphere are related to greenhouse gas compositions and also if the variations we observe in the ionosphere are majorly just caused by solar activity. This will be a review of few of these studies. A dedicated section will also be included to analyze studies that have identified specific geomagnetic parameters as reliable predictors of ionospheric impacts.

2.2.1 Study by Rishbeth and Roble (1992)

The study titled Cooling of the Upper atmosphere by enhanced greenhouse gases – modelling of thermospheric and ionospheric effects predicted that the concentrations of greenhouse gases particularly carbon dioxide and methane are expected to double by mid-21st century in the middle atmosphere (Rishbeth & Roble, 1992). They used the NCAR Thermosphere/Ionosphere General Circulation Model (TIGCM) to predict the effects of doubled greenhouse gas compositions on the ionosphere and thermosphere. They predicted that the thermospheric temperatures will be lowered by 30 – 40k and air density at heights of 200 – 300 km to reduce by 20 – 40 percent, which will increase the orbital lifetime of satellites. The height of the ionospheric F2-layer peak will drop on average by about 15km, with some negative effect on radio propagation. The F2-layer critical frequency will be affected.

This study provides the foundation of how increasing greenhouse gas composition has effect on the upper atmosphere particularly the ionosphere. This research provides a crucial context on how greenhouse gas induced cooling can influence ionospheric dynamics, which would be essential for understanding and predicting the behavior of the ionosphere during geomagnetic storms under increasing greenhouse gas composition.

The limitation of this research is there is no relationship between the effects caused by greenhouse gas compositions and those resulting from solar and geomagnetic activity. The

ionosphere is influenced by a lot of overlapping factors like, 11-year solar cycle, solar flux variations, geomagnetic storms and greenhouse gas compositions. As such, we cannot isolate the predicted changes in ionospheric parameters to greenhouse gas – induced cooling alone. Even with long term data and trend analysis, the dynamic and interconnected nature of the upper atmosphere makes it difficult to isolate the independent contribution of rising greenhouse gas levels without other external influence from external factors.

2.2.2 Study by Bremer (1992)

This study by Bremer (1992) titled Ionospheric trends in mid-latitudes as a possible indicator of the atmospheric greenhouse effect. Long term ionosonde measurements in mid-latitudes (i.e. in the German town - Juliusruh: 54.6°N, 13.4°E;) from 1957 – 1990 were used. The study identified that the peak height of the F2 layer appears to be decreasing, but in contrast the change in electron densities in the F2-layer and E-layer was not significant. He further added that the results agreed with the predictions of Rishbeth et al who expected the lowering of the E and F2 layer caused by the cooling of the upper atmosphere due to increased greenhouse gas effect.

This study is significant because it establishes that there are long term trends in the ionosphere which are related to greenhouse induced cooling of the upper atmosphere, also the study took data from midlatitude regions (i.e. a region between the equator and the poles.) This is important because in this region the effects of geomagnetic storms and solar activity on the ionosphere are not as strong as at the equator or at the poles.

The study is still limited as it does not isolate the trends caused by greenhouse gases to those caused by solar activity and geomagnetic storms. So, it is difficult to attribute the trends observed to increase in greenhouse gases alone.

2.2.3 Study by Danilov (2001)

This study Danilov (2001) analyzed long term ionospheric data to find trends in the ionosphere that can be attributed to greenhouse gas induced cooling. The study revealed that data on radio absorption in the D-region indicate some systematic trends which may be considered as a manifestation of a temperature decrease around the boundary between the mesosphere and thermosphere (Danilov, 2001).

The study also revealed strong positive trends of electron concentration in the upper D-region, positive trends of E-layer critical frequency, foE and negative trends of the ratio of two principal ions in the E-region (i.e. nitric oxide and oxygen). These changes may be as a result of dynamical processes in the ionosphere.

This study also considered the trends of F2-layer parameters studied in many publications, that is trends in the critical frequency foF2 and maximum height hmF2. And it found that trends in critical frequency depended strongly on geomagnetic latitude and geomagnetic activity.

This made Danilov conclude these geomagnetic variations might account for the observed ionospheric trends. But he still gave room that there might be other factors contributing to the trends, such as greenhouse gases or other human activities.

This study is significant for this research because it provided valuable insight of how data can show strong correlation between increasing greenhouse gas composition and ionospheric variability trends and also show that the ionospheric variability is majorly caused by geomagnetic activity. And this is a crucial foundation why there is a need to understand the proportion of the effects of greenhouse gas composition on the ionosphere and geomagnetic storms. And also why we must discover the relationship between the effects of the geomagnetic storms and greenhouse gas cooling on the ionosphere.

The limitation of this study is that it could not isolate the specific impact of greenhouse gases on the ionosphere amidst other complex factors.

2.2.4 Study by Upadhyay and Mahajan (1998)

The study Atmospheric greenhouse effect and ionospheric trends by Upadhyay and Mahajan (1998) was aimed to analyze long term trends in the F2-layer of the ionosphere particularly the peak height (hmF2) and critical frequency (foF2) and to explore whether those trends could be attributed to greenhouse gas effect. In this study ionosode data for some 31 stations were analyzed and the solar activity effect trends were removed and the remaining trends for peak height anomaly (AhmF2) and critical frequency anomaly (AfoF2) were studied for these stations. And the results were negative for some stations and positive for others. These mixed results made Upadhyay and Mahajan (1998) to conclude that the ionospheric data did not provide any definitive evidence of a global long - term trend.

This study is significant because it removes the solar cycle effects from the data to be able to achieve its result. This is important because the ionosphere is dependent on many factors which makes it important to isolate the effects of greenhouse gases to the data in order to approximately attribute the effects observed in the ionosphere to greenhouse gas effect alone.

This study is limited even though it attempts to remove solar activity effects using regression of the study parameters (hmF2 and foF2) and solar radio flux as this may not fully account for all solar and geomagnetic influences on the ionosphere. Also, the studies analysis is limited to the F2 layer only and no analysis on how other ionospheric layers respond to ionospheric effects. And lastly the study is inconclusive because the reasons for different trends remained unexplained.

2.2.5 Study by Blagoveshchensky et al (2018)

This research Impacts of magnetic storms on the global TEC distribution by Blagoveshchensky et al (2018) analyses the variations in total electron content (TEC) during six geomagnetic storms of different intensities from $DST_{min} = -46nT$ to $DST_{min} = -223nT$. The change in TEC compared to the normal value from the past 27 days was calculated, the difference called – delta TEC (δTEC). This difference was calculated for three meridians: American, Euro-African and Asian-Australian. The observed the following results:

- The largest δTEC values (peak changes) happened almost simultaneous along all three meridians at the beginning of the storm.
- A transition from weak storm to a superstorm does not affect the intensity of the δTEC maximum (Blagoveshchensky et al, 2018). This means that storm strength doesn't always mean bigger TEC effect.
- The seasonal effect was most pronounced in Asia and Australia meridian, less often along the Euro-African meridian and not revealed along the American meridian.
- There are some common features of TEC variation revealed during each storm phase, but there is no clear dependence of TEC responses on storm phases. And the effects during each storm was different at different locations, meaning location matters.
- The correlation coefficients (R) of TEC changes (δTEC values) along the three meridians were analyzed for each storm, yielding results were $R > 0.5$ general, this shows that TEC changes across regions are strongly related. Also, the possible reason for some exceptions were: the complexity of phenomena during the intense storms and discordance in local time of the geomagnetic storm beginning along different meridians (Blagoveshchensky et al, 2018).
- The intensity of R decreased with growth of storm intensity.

This paper shows that geomagnetic storms cause immediate and measurable increase in TEC. It also warns me of the kind of relations geomagnetic strength has with TEC, as the relationship is not linear and simple. It shows me that there are a lot of other factors that affect TEC like season, latitude, and pre-existing ionospheric conditions. This paper however does not study geomagnetic storm effects on other ionospheric parameters like critical frequency (foF2), electron density and peak height (hmF2). It also does not include greenhouse gas effects or their impact on the ionosphere. Therefore, this paper does not help me to understand the combined effects of greenhouse gases and geomagnetic storms.

2.2.6 Study by Mikhailov and Marin (2001)

The study titled “An interpretation of the foF2 and hmF2 long – term trends in the framework of the geomagnetic control concept” explains the patterns in how foF2 (critical frequency of F2 layer) and hmF2 (peak height of F2 layer) have changed over the years using the idea that geomagnetic activity controls these trends, based on contemporary f2-layer storm mechanisms. Mikhailov & Marin (2001) tells us that the way foF2 and hmF2 change over time is strongly dependent on long-term varying geomagnetic activity whose effects cannot be removed from the trends using conventional indices of geomagnetic activity. Therefore, any interpretation of the foF2 and hmF2 trends should consider the geomagnetic effects as an alienable part of the trend analysis. Mikhailov & Marin (2001) explained that periods with negative and positive foF2 and hmF2 trends correspond to the periods of increasing or decreasing geomagnetic activity with the turning points around 1955, and end of 1960s and 1980s, where foF2 and hmF2 trends change their signs, also that such changes can be explained by neutral composition, temperature and thermospheric winds, all of which are related to geomagnetic activity variations. It further goes on to suggest that natural causes, especially geomagnetic storms are the main drivers of change in F2-layer rather than an artificial one, related to the

thermosphere cooling due to greenhouse gas effect. Therefore, one should expect the thermosphere heating for the periods of increasing geomagnetic activity rather than cooling.

This study is important as it argues that long term trends in ionospheric parameters is majorly caused by geomagnetic activity, not greenhouse gas effects, it also shows that periods of increasing greenhouse gas activity correspond with changes in foF2 and hmF2 trends. This helps confirm without any doubt that geomagnetic disturbances are meaningful predictors of ionospheric conditions. But although they link geomagnetic activity and ionospheric trends, they don't quantify how much greenhouse gas trends contribute. This is something this project will fix.

2.2.7 Study by Danilov (2013)

This study titled, Ionospheric F-region response to geomagnetic disturbances in *Advances in Space Research*, Danilov (2013) provides a comprehensive review of the ionospheric F2-region's reaction to geomagnetic storms, commonly referred to as ionospheric storms. The paper classifies the storm-time ionospheric responses into two major phases: the *positive phase*, where electron density increases, and the *negative phase*, where it decreases. These phases display intricate spatial and temporal variability, making their behavior difficult to model or generalize across different geophysical contexts. The study outlines the primary morphological features of ionospheric storms and the associated physical mechanisms—such as thermospheric winds, changes in composition (e.g., O/N₂ ratio), and electric fields—that control the F-region's response.

Danilov's work is anchored in empirical and morphological observation, summarizing decades of prior literature rather than conducting new modeling or data analysis. He highlights that despite significant advances, understanding the dynamics of positive and negative storm phases remains incomplete, especially regarding their predictability and dependency on local time, latitude, and storm intensity (Danilov, 2013).

The review by Danilov is significant as it provides a consolidated reference on the known patterns and processes governing F2-region responses to geomagnetic storms. It helps establish the foundational context for newer studies in ionospheric modeling by listing known dependencies and persistent challenges in understanding storm effects. His synthesis also emphasizes the necessity of more physically interpretable and spatially aware models to handle the complex, nonlinear behavior of the ionosphere during disturbed periods.

According to Danilov (2013), a critical theoretical concern in space weather research is the overreliance on the Dst index as a universal predictor of ionospheric response. Danilov argues that the Dst index does not always correlate clearly or consistently with ionospheric variability, especially for the F2-layer parameters. He states that the ionospheric reaction to geomagnetic storms is highly variable and often exhibits regional and temporal dependence, suggesting that Dst alone cannot capture the complexity of these interactions.

Danilov emphasizes that although Dst is widely used as a global proxy for geomagnetic disturbance intensity, it does not directly describe the physical mechanisms impacting the ionosphere, such as thermospheric winds, electric fields, or composition changes induced by

solar radiation or particle precipitation. Therefore, using Dst as the sole geomagnetic parameter can mask significant ionospheric dynamics.

The review by Danilov is significant as it provides a consolidated reference on the known patterns and processes governing F2-region responses to geomagnetic storms. It helps establish the foundational context for newer studies in ionospheric modeling by listing known dependencies and persistent challenges in understanding storm effects. His synthesis also emphasizes the necessity of more physically interpretable and spatially aware models to handle the complex, nonlinear behavior of the ionosphere during disturbed periods.

Despite its thorough review of storm morphology, Danilov's study is primarily descriptive and lacks quantitative assessment or predictive modeling. It does not utilize modern statistical or machine learning techniques to assess or model the underlying mechanisms. Moreover, it does not evaluate the role of longer-term climate drivers, such as greenhouse gases or account for how these interact with storm-time ionospheric variability. Nor does the study disaggregate the contribution of geomagnetic indices (e.g., Dst) at different latitudes or timescales using regression or modeling frameworks.

2.3 Studies Identifying Geomagnetic Parameters as Predictors of Ionospheric Variability

2.3.1 Study by Maltseva

This study investigated two key ionospheric parameters, total electron content (TEC) and F2-layer critical frequency (foF2). It studied how they relate to each other during space weather events in the Southern Hemisphere, to see if TEC can be used to predict foF2 using a mathematical relationship that depends on a factor slab thickness (τ). It found that during times that Dst (Geomagnetic Disturbance) indicates a storm is happening the relationship between TEC and foF2 changed (i.e. the relationship became more complex, and slab thickness (τ) values from IRI models are no longer accurate). This makes Dst a useful variable when modeling or interpreting ionospheric measurements like foF2 and TEC ([Maltseva, 2021](#)).

The problem with this study is that it only considers short term impacts of geomagnetic disturbances on the ionosphere. The study is comparative and does not use any machine learning, predictive or statistical modelling. It does not also consider other factors that could be affecting the ionosphere apart from geomagnetic disturbances like greenhouse gas effects, local time and season effects, solar radio flux (F10.7), Atmospheric waves, and other geomagnetic indices.

2.3.2 Study by Li et al (2021)

This study tries to predict foF2 (f2-layer critical frequency) using a deep learning model called LSTM (Long Short-Term Memory) which is good at handling the time series data. The model was trained using foF2 data from 10 stations in China and Australia covering the years 2006 to 2019, to check the model accuracy they tested it on 2015 (when solar activity was high) and 2019 (when solar activity was low). The model used inputs local time, solar zenith angle, Sunspot number (SSN), solar radio flux (F10.7) index, Ap-Index, Kp-Index, geographical

coordinates, and neutral winds. They compared their model with a backpropagation neural network (BPNN), a genetic algorithm backpropagation Neural Network (GABP), and IRI 2016. They discovered their LSTM model was the best at predicting foF2 accurately, even during geomagnetic storms, while IRI2016 model was the worst among them (Li et al., 2021). The importance of this work is it used geomagnetic parameters and solar activity parameters as inputs in predicting a very important ionospheric parameter foF2. But the training and test data were taken from only 10 ionospheric stations in China and Australia, which restricts the model's generalizability, also the model was only evaluated during two solar conditions (i.e. two years), this will not help to capture long term variability, the study only focused on forecasting foF2, and it cannot be used to model other ionospheric parameters like TEC, hmF2, etc. The model also uses a lot of parameters which are not readily available or accurately measured in real time forecasting or in live space weather monitoring systems. Also, the LSTM model incorporates a wide range of input variables and it does not assess the statistical relevance or importance of each input feature, this raises the possibility of feature redundancy, where non-informative inputs may increase model complexity without improving predictive accuracy, and deep learning models like LSTM are known to be highly expressive, which also makes them susceptible to overfitting, especially on time series data.

In this study, the training data spans 2006 to 2019, while the validation is performed using 2015 and 2019, which fall within the training period, even if data from different stations were used, temporal overlap or similar geomagnetic conditions may result in information leakage, potentially inflating model performance. A better evaluation approach would involve using strictly out of sample years or cross-validation schemes that ensure no temporal bias.

2.3.3 Study by Tshisaphungo et al (2023)

This study tries to predict how the ionosphere changes during geomagnetic storms using Neural Networks (NN) and Linear Regression (LR), to model the changes in the ionospheric f2-layer critical frequency (foF2) which is important for radio communication. The data came from Grahamstown ionosonde station in South Africa (Location: 33.3°S, 26°E) from 1996 to 2014 only for training the models, six storm events were set aside to use for testing. The model was trained based on only storm time events during the time period. The input parameters for the model were SYM-H which is the strength of the ring current around earth, Auroral Electrojet (i.e. AE), and lastly local A index which measures the average geomagnetic disturbances in a specific area. It discovered that both models did well and was able to explain how the ionosphere changed when a storm hit. They also discovered that the two models did well in all selected storms which fell within the date period (1996-2014). But when the models were evaluated on storm periods beyond 1996-2014, the NN model gives a better performance compared to the LR model for a storm that reached a minimum Dst index of -155nT during 19-23 December 2015. In both models the parameter SYM-H is the variable that accounts for most of the variance. It accounts for 41% in NN and 54% in LR (Tshisaphungo et al., 2023).

This study is important as it is one of the studies that models the ionosphere with geomagnetic parameters, using both complex and linear analytic methods. Also, it tells how important model parameters are to the model. But the study has limitations, it was trained exclusively on storm time data, without including quiet-time conditions, this improves storm focused predictions,

but it limits the model's ability to distinguish between storm and non-storm behaviour, which makes the model unsuitable for real-time detection of ionospheric changes. Also, the study uses data from only one ionosonde station, making the ionospheric responses location dependent. The model also does not include solar indices or greenhouse gas parameters. The study showed that SYM-H is the most significant input but it does not check as to whether the data overfit to SYM-H. NN models are black box models by nature, and the study does not include explainability tools like (SHAP or LIME) to understand how predictions are made and to confirm whether the NN model learned physically meaningful patterns. Only two storms outside the training range were used for extrapolation testing, this is too small a validation data and it may not reflect extreme storm scenarios or variability across different solar cycles. So these are the many limitations, but a significant limitation of the study lies in the assumption or reliance on linear relationships between geomagnetic indices and ionospheric parameters. In reality, the ionospheric response to geomagnetic storms is often non-linear and highly variable, with the same storm intensity producing positive effects in some regions and negative effects in others, depending on local time, season, storm phase, latitude, and solar activity. Linear models (such as LR) may fail to capture this complexity, and even neural networks, if not properly tuned or interpreted, can obscure rather than clarify these asymmetric behaviors.

CHAPTER THREE

DATA COLLECTION AND METHODOLOGY

In this chapter the description of data collection and methods employed in the study will be discussed.

It details data sources for climate, geomagnetic and ionospheric parameter data from 1960 till date.

It also describes the data analysis techniques, statistical methods and modeling methods used, with a comprehensive description of methodology used to obtain results.

3.1 Data Sources

Here the description of the data sources used to get the data, and then detailing the type of data collected from those sources, the length (number of years) and the coverage region for the data sourced. There was difficulty accessing global data for all the variables in consideration for this project, therefore some regional alternatives were chosen with details below.

3.1.1 GIRO - Global Ionosphere Radio Observatory

Brief Description:

GIRO, that is Global Ionosphere Radio Observatory is a global network of ionosodes, high frequency radar systems dedicated to monitoring the earth's ionosphere, especially the vertical electron density in the upper atmosphere (Reinisch & Galkin, 2011). GIRO is coordinated by the Lowell Digisonde International Center (LDIC) at the University of Massachusetts Lowell, and serves as a central hub for the collection, archiving, and dissemination of ionospheric data from a worldwide network of ionosonde stations (Galkin et al., 2012).

As part of the Digital Ionogram Database (DIDBase), GIRO collects real-time and historical ionograms, which are used to extract key ionospheric parameters. These include the critical frequency of the F2 layer (foF2), the peak electron density (NmF2), and the peak height of the F2 layer (hmF2) (Galkin & Reinisch, 2008). These parameters are crucial for applications in radio communication, satellite navigation, and space weather research.

GIRO also contributes to space weather forecasting by providing data that help track ionospheric disturbances triggered by solar activity and geomagnetic storms. It supports both operational needs (such as HF communication reliability) and scientific studies aimed at understanding ionospheric variability in response to external drivers, including solar radiation, geomagnetic activity, and greenhouse gas-induced upper atmospheric cooling (Reinisch et al., 2004; Qian et al., 2011).

It provides regional data of ionospheric conditions, i.e. its data is grouped into different regions.

About Data:

This is the source for all the daily average Total Electron Content (TEC), F₂ Critical Frequency (foF₂) and F₂ Peak Height data (hmF₂) data used in this project. This data represents the data

of the ionospheric condition at a given time and it serves as our dependent variables for this project.

For this project, data was gotten from the Digital Ionogram Database (DIDBase) from the region Roquetes, a town in Catalonia region of northeastern Spain and was selected because it had observations for the largest number of years, i.e. from 1988 till Date.

3.1.2 World Data Center for Geomagnetism, Kyoto

Brief Description:

The World Data Center for Geomagnetism, Kyoto (WDC Kyoto) is one of the leading global repositories for geomagnetic data and indices. It is hosted by the Data Analysis Center for Geomagnetism and Space Magnetism at Kyoto University, Japan. The center serves as a key component of the international data infrastructure supporting research in space weather, solar-terrestrial interactions, and geophysical phenomena (WDC for Geomagnetism, Kyoto, n.d.-a). Established as part of the World Data Center system created during the International Geophysical Year (IGY) in 1957–1958, WDC Kyoto has since become a central hub for the long-term archiving and dissemination of geomagnetic data. It provides access to a wide range of parameters, including hourly, daily, and monthly geomagnetic field values, and indices such as Dst (Disturbance Storm Time), AE (Auroral Electrojet), ASYS/SYM, and K indices, which are widely used for characterizing geomagnetic storms and substorms (WDC for Geomagnetism, Kyoto, n.d.-b; Love, 2008). WDC Kyoto has played a pivotal role in maintaining and updating this index since the 1950s (Sugiura & Kamei, 1991). The Dst index is produced by the World Data Center for Geomagnetism, Kyoto using data from four low-latitude observatories: Hermanus (South Africa), Kakioka (Japan), Honolulu (Hawaii, USA), and San Juan (Puerto Rico, USA). These stations are positioned near the magnetic equator, allowing for accurate measurement of the global ring current's effect on the Earth's magnetic field (Sugiura & Kamei, 1991; WDC Kyoto, n.d.-a). The observatories' data are corrected and averaged to create a global index of geomagnetic storm strength, which has been maintained since 1957 (Love & Gannon, 2009; WDC Kyoto, n.d.-b).

One of the key strengths of WDC Kyoto is its commitment to free and open access to data, which ensures that researchers, operational agencies, and policymakers worldwide can retrieve standardized, high-quality geomagnetic measurements. This supports global collaboration in areas such as satellite mission planning, power grid protection, aviation safety, and climate studies involving solar-terrestrial coupling (WDC for Geomagnetism, Kyoto, n.d.-a).

About Data:

Two types of daily-averaged geomagnetic data was obtained from WDC, Kyoto, they are listed as follows:

I. Geomagnetic Disturbance Index (Dst – Index)

The DST daily average data was sourced from here from 1960 till date. Dst index which measures strength of the globally symmetric ring current in the Earth's magnetosphere, particularly during geomagnetic storms. It basically measures how much the earth's magnetic field is disturbed or weakened by geomagnetic storms due to the ring current. The data is the

global representative for global disturbance to the earth's magnetic field, i.e. it is not regional data.

II. Ap Index

The Ap index is a planetary geomagnetic index derived from measurements of geomagnetic activity at multiple observatories worldwide. It quantifies the level of geomagnetic disturbance and is essential for space weather modeling, particularly in evaluating magnetospheric energy input (Kurazhkovskaya & Kurazhkovskii, 2023).

The data were sourced from WDC Kyoto, spanning from 1960 to 2024. The Kyoto WDC aggregates observations from a global network of magnetometers and applies standardized methods for index generation.

Despite its compilation from multiple locations, the Ap index is a regional aggregate with global applicability, making it a quasi-global metric. While not a direct satellite-based measure, its global classification is justified by its derivation from stations evenly distributed in latitude and longitude, providing a representative snapshot of global geomagnetic conditions (Kim & Oh, 2025).

3.1.3 NOAA Global Monitoring Laboratory

Brief Description:

The National Oceanic and Atmospheric Administration (NOAA) is a U.S. federal scientific agency focused on the conditions of the oceans, atmosphere, and space. It plays a central role in monitoring climate, weather, and environmental changes (NOAA, n.d.-a). One of its key divisions, the Global Monitoring Laboratory (GML), specializes in long-term, high-precision observations of atmospheric greenhouse gases, ozone-depleting substances, and other climate-relevant atmospheric compounds (NOAA GML, n.d.-b). GML manages global networks of observation sites for long – term atmospheric observatories at Barrow, Alaska; American Samoa; and the South Pole, including the renowned Mauna Loa Observatory, Hawaii. To support climate research, model validation, and policy assessments on climate change and air quality (NOAA GML, n.d.-b).

Mauna Loa Observatory, Hawaii, USA

The Mauna Loa Observatory (MLO), located at an altitude of approximately 3,397 m on the northern flank of Mauna Loa Volcano in Hawaii, is a premier atmospheric baseline station operated by NOAA's Global Monitoring Laboratory (GML). Its remote, elevated location minimizes local contamination, making it ideal for measuring free-tropospheric background air (NOAA GML, n.d.-a). Continuously active since the 1950s, MLO is famously known for the landmark Keeling Curve tracking the rise of atmospheric CO₂ since 1958 and continues to monitor key gases such as CO₂, CH₄, N₂O, ozone-depleting substances, aerosols, and solar radiation (Keeling Curve, 2025; NOAA GML, n.d.-a; Wikipedia, 2024a).

MLO is part of multiple global networks, including the Greenhouse Gas Reference Network and the Network for the Detection of Atmospheric Composition Change (NDACC), serving both calibration and scientific measurement needs (NOAA GML, n.d.-a; Wikipedia, 2024a).

After disruptions due to the 2022 volcanic eruption, the observatory resumed critical measurements via solar-power systems in 2023 (NOAA GML, 2025).

About Data:

This is the source of the daily CO₂ concentration data in the atmosphere measured in parts per million by volume (1ppmv). Explicitly, 1ppmv signifies that for every 1 million air molecules, 1 molecule is CO₂. The data was gotten from may 1974 till date. The data was gotten from a single location, Mauna Loa Observatory, Hawaii. But even though this is a regional data it is used for global analysis to show global CO₂ trends because the observation is remote with little to no industrial activity around and it is located in a high – altitude environment, which makes the CO₂ trends in this region a very good representative of global trends.

3.1.4 NASA OMNIWeb

NASA OMNIWeb is a curated, multi-mission database maintained by the Space Physics Data Facility (SPDF) at NASA's Goddard Space Flight Center. It integrates solar wind, interplanetary magnetic field (IMF), plasma, energetic particle, and geophysical index data from numerous spacecraft including ACE, WIND, IMP-8, and ISEE-3. The datasets have been time-shifted to the Earth's bow shock and standardized into hourly, daily, and high-resolution formats, allowing for consistent intercomparison and long-term studies (Vokhmyanin, Stepanov, & Sergeev, 2019).

OMNIWeb archives span from 1963 to the present, with some parameters going back to 1960. It includes:

- IMF components (B_x, B_y, B_z)
- Solar wind speed, density, and temperature
- Plasma beta and dynamic pressure
- F10.7 cm solar radio flux
- Geomagnetic indices such as AE, K_p, and Dst

The richness and continuity of this dataset make OMNIWeb an indispensable tool for space weather research, long-term solar cycle analysis, and machine learning-based prediction models (Topliff, Cohen, & Bristow, 2010). Its quality-controlled, merged format from both in situ spacecraft and ground-based sources ensures broad applicability in global geophysical and heliophysical studies.

OMNIWeb is considered a global data source because its primary measurements are taken from spacecraft located in interplanetary space—primarily at or near the L1 Lagrange point—and then referenced to Earth's bow shock. These data are **free** from local geomagnetic noise, thus capturing large-scale heliospheric and magnetospheric conditions relevant across all geographic regions (Papitashvili & Papitashvili, 2000)

About Data:

From here, there were a lot of datasets that were downloaded from this source. They are explained as follows:

I. Solar Radio Flux (F10.7)

The F10.7 solar radio flux represents the intensity of solar radio emissions at a wavelength of 10.7 cm (2800 MHz), which correlates strongly with solar activity, particularly ultraviolet and X-ray emissions. This parameter serves as a key proxy for monitoring solar EUV output and ionospheric variability, making it vital for modeling space weather and ionospheric disturbances (Henney, Toussaint, & White, 2012).

The data for F10.7 flux from NASA OMNIweb spans the period 1960–2024, offering a long-term continuous dataset. OMNIweb integrates solar-terrestrial observations from multiple satellites and ground-based sources, centralized by NASA's Space Physics Data Facility, which enhances reliability and global accessibility.

F10.7 data are classified as global because they represent whole-sun emissions measured from ground-based observatories such as those in Ottawa and Penticton, which capture integrated solar disk output observable from Earth (Binod et al., 2019). This makes the dataset suitable for universal applications across geophysical, atmospheric, and space weather disciplines.

II. Interplanetary Magnetic Field (IMF) B_x

The IMF B_x component represents the projection of the interplanetary magnetic field vector along the Sun–Earth line. It plays a role in shaping the structure of the heliospheric current sheet and is significant for interpreting Parker Spiral configurations in the solar wind (Kutiev et al., 2013).

NASA OMNIweb provides IMF B_x measurements from 1960–2024 by compiling solar wind data from interplanetary missions like ACE, WIND, and IMP. These datasets are aggregated at Earth's bow shock, making them highly representative of interplanetary conditions.

IMF B_x data are global in classification because they are derived from satellite observations in interplanetary space rather than localized measurements. This ensures that the data reflect large-scale heliospheric magnetic structures affecting all near-Earth space environments (Kim & Oh, 2025).

III. Interplanetary Magnetic Field (IMF) B_y

The IMF B_y component indicates the magnetic field's direction perpendicular to the Sun–Earth line and along the ecliptic plane. It affects the development of magnetospheric convection patterns and is critical in assessing asymmetries in geomagnetic disturbances (El-Borie et al., 2020).

Like B_x , IMF B_y data from 1960–2024 are obtained from OMNIweb, where spacecraft measurements near Earth's magnetosphere are compiled and transformed into a geocentric solar ecliptic (GSE) coordinate system.

This parameter is also considered global, as the interplanetary field it describes is not restricted to one region but influences planetary magnetospheres on a heliospheric scale. Global monitoring of B_y is essential for understanding large-scale solar wind–magnetosphere interactions (Heybatli, Arikan, & Arikan, 2025).

IV. Interplanetary Magnetic Field (IMF) B_z

IMF B_z is the north-south component of the interplanetary magnetic field and is one of the most critical parameters for space weather forecasting. When B_z turns southward (negative), it allows enhanced coupling with Earth's magnetosphere, leading to geomagnetic storms (Kutiev et al., 2013).

OMNIweb provides IMF B_z values from 1960–2024 using standardized measurements from near-Earth satellite missions. The consistent and cross-mission calibrated values allow for uninterrupted long-term analysis of geoeffective interplanetary conditions.

B_z is considered global, similar to other IMF components, due to its derivation from space-based sensors monitoring the interplanetary environment. Its global significance lies in its direct modulation of the entire magnetosphere, rather than localized effects (Pandey & Dubey, 2017).

3.2 Data Analysis

Prior to model development, a comprehensive data analysis will be conducted to gain insights into the structure, behavior, and interrelationships among the variables under investigation. This process is crucial for detecting patterns, trends, correlations, and potential issues such as missing values or multicollinearity, all of which can significantly influence model performance and interpretation.

Therefore, in this section I will be outlining all the data analytical processes that I intend to carry out on my data and the reason why.

Outline of the Data Analysis processes:

To begin, multiple datasets sourced from different institutions will be integrated into a unified time framework using Microsoft Excel. These datasets include atmospheric CO_2 concentrations obtained from NOAA's Global Monitoring Laboratory (GML), geomagnetic Dst index values from the World Data Center for Geomagnetism, Kyoto, and ionospheric parameters, total electron content (TEC), critical frequency of the F2 layer (foF_2), and F2 peak height (hmF_2) acquired from the Global Ionosphere Radio Observatory (GIRO).

Following the merging process, the main analysis will be performed in Python. The dataset will first be cleaned by removing all rows containing missing values to ensure consistency. Subsequently, time series plots for each variable (Dst, CO_2 , TEC, foF_2 , and hmF_2) will be generated to visually assess their temporal behavior across the study period. Stationarity tests,

such as the Augmented Dickey-Fuller test, will be applied to each variable to determine the extent to which their statistical properties remain constant over time.

To investigate relationships between variables, scatter plots will be created between the independent variables (CO_2 and Dst) and the dependent ionospheric parameters (TEC, foF2, and hmF2). Correlation coefficients will also be computed to quantify the strength and direction of these associations. To further examine temporal variation in these relationships, the dataset will be divided by calendar year. For each year, individual time series and scatter plots will be generated, and correlation analyses will be conducted to evaluate how inter-variable relationships evolve annually.

Additionally, scatter plots and correlation coefficients will be employed to assess potential multicollinearity among the dependent variables themselves. This step is essential for identifying any strong interdependencies that could bias the model's estimations. Finally, given the known influence of solar activity on the ionosphere, the dataset will be segmented according to established solar cycles. For each cycle, time series plots and scatter plots will be constructed to analyze how relationships between geomagnetic storms, greenhouse gas concentrations, and ionospheric parameters differ across phases of solar activity.

Collectively, these analytical steps will provide a robust understanding of the underlying dynamics in the dataset and inform the subsequent modeling phase with a solid empirical foundation.

3.3 Tools and Python Libraries

Python is the primary language used for data manipulation, analysis and modelling. Here, the tools and python libraries that are used to analyze, understand and interpret the data collected are discussed.

3.3.1 Microsoft Excel

Microsoft excel is a spreadsheet application developed by Microsoft; it is used to organize data, preprocess, and perform preliminary analysis on the data collected. Data was collected from various sources for this project and the different data, had different time formats and measurement intervals. Microsoft excel has a well-structured and user-friendly environment for merging, sorting, and synchronizing the data. Excel will be used to put the data in the form that can easily be read for analysis with python. Thus, Microsoft Excel serves as a foundational step in ensuring data consistency and readiness for subsequent statistical and machine learning modelling efforts.

3.3.2 Pandas

Pandas is an open-source data analysis and manipulation library in python; it is used in scientific and engineering fields for its efficient handling of structured data. In this project, pandas prepare the merged datasets originally aligned using Microsoft excel for advanced analysis. Pandas data frame structure, provides a spreadsheet-like representation of data, enabling easy indexing, slicing, filtering, and reshaping of multidimensional datasets. It is

crucial for reading Excel and CSV files and removing irrelevant features and cleaning the data collected for analysis.

3.3.3 Matplotlib

Matplotlib is a python library used for creating static, animated, and interactive visualizations in data analysis. Matplotlib plots are essential for generating clear and informative plots that show the relationship between the independent (Dst, CO₂) and dependent variables (TEC, foF2, hmF2). These plots being easy to understand enhance the exploratory data analysis process, allowing effective communication of findings in the study.

3.3.4 NumPy

NumPy is a python library for numerical computing, it is used for numerical computing, and manipulation of large arrays and matrices. It performs mathematical operations, reshaping of datasets, and slicing of arrays to prepare the data for modelling.

3.3.5 Scikit-learn

Scikit-learn is a machine learning library in python, it facilitates model development, data preprocessing, and performance evaluation. It provides tools for splitting the data set into training and testing subsets using functions such as `train_test_split`, which helps to ensure that the data is trained on a representative portion of the data while preserving the integrity of the test set for unbiased evaluation. It supports tasks needed to verify model assumptions, like feature selection, normalization, and statistical assessments. It is also very compatible with other python libraries, like pandas and NumPy.

3.3.6 PyGAM

PyGAM (Python Generalized Additive Models) is a python library and will serve as the primary modeling framework due to its flexibility and interpretability in handling non-linear relationship between variables. PyGAM is well-suited for this for this research as it can handle both linear and non-linear relationships and is very interpretable. Unlike traditional linear models, PyGAM allows the fitting of smooth splines to each predictor independently, enabling the model to capture subtle trends and variations over time. PyGAM offers tools for visualizing the partial dependence plots and calculating the deviance explained by each feature, which is important for interpreting the role of each independent variable. PyGAM can model complex patterns without requiring extensive assumptions.

3.3.7 Jupyter Notebook:

Jupyter Notebook is an open-source web-based environment that allows for the creation and sharing of documents that contain live code, equations, visualizations, and narrative text (Beg

et al., 2021). It supports interactive computing across dozens of programming languages, making it a widely adopted tool in research and education.

The use of Jupyter Notebook in this project is justified by several technical and pedagogical advantages. It facilitates an interactive and iterative programming approach, allowing for code execution and result visualization in a single interface (Johnson, 2020). This enables real-time testing and debugging of code, which enhances efficiency and reduces development time.

Jupyter supports reproducibility and transparency in research. The integration of code, output, and documentation within the same document ensures that analyses can be re-run and verified by other researchers, fulfilling a core requirement in scientific workflows (Pimentel, Murta, & Braganholo, 2019; Beg et al., 2021). The format is particularly effective for data-driven tasks, allowing for step-by-step exploration and explanation of processes (Pimentel, Murta, Braganholo, & Freire, 2021).

Its educational benefits include promoting better understanding through visualization and markdown-based explanations. Jupyter has been found effective in improving engagement and learning outcomes, especially in science and engineering contexts (Cardoso, Leitão, & Teixeira, 2018; Ruiz-Sarmiento, Baltanas, & Gonzalez-Jimenez, 2021). Moreover, it simplifies collaboration, versioning, and sharing, which are critical for academic projects and peer reviews (Reades, 2020).

3.4 Methodology

This study aims to predict the effect of geomagnetic storms on the ionosphere under increasing greenhouse gas (GHG) conditions. To achieve this, a data driven modelling approach is adopted, combining atmospheric, geomagnetic, and ionospheric measurements across multiple decades and solar cycles. The research is tailored to capture how rising greenhouse gas concentrations especially carbon (iv) oxide (CO_2) influence the way geomagnetic storms affect the ionosphere, particularly the ionosphere's Total Electron Content (TEC), F_2 -layer critical frequency (f_oF_2), and peak height (h_mF_2).

To start data is collected and preprocessed from various reputable sources. These include CO_2 data from the NOAA Global Monitoring Laboratory (GML), geomagnetic storm data (Dst index) from the World Data center for Geomagnetism, Kyoto, and ionospheric parameters from the Global Ionospheric Radio Observatory (GIRO). All datasets are merged and aligned on a common time scale using Microsoft Excel, followed by careful analysis and modeling in Python using tools like Pandas, Matplotlib, NumPy, and Scikit-learn.

To build the predictive model, the pyGAM (Python Generalized Additive Model) framework is used. The model is ideal for uncovering non-linear relationships between predictor and response variables. The predictor variables include CO_2 concentration, the Dst index, and a numerically encoded solar solar cycle phase to reflect the cyclical nature of solar activity. The response variables are ionospheric parameters (TEC, f_oF_2 , h_mF_2).

The model is trained using 70% of the dataset, and its performance is evaluated through residual analysis, partial dependence plots, and explained deviance statistics. A key component of the analysis involves calculating the partial deviance explained by each predictor variable during different solar cycles. This is done to evaluate how the influence of geomagnetic storms (Dst)

on the ionosphere may be modulated by rising greenhouse gases. By comparing these effects across solar cycles 23, 24, and 25, the study seeks to reveal whether increasing CO₂ levels amplify, suppress, or otherwise alter ionospheric responses to geomagnetic disturbances.

CHAPTER FOUR

RESULTS AND DISCUSSION

This chapter presents the results derived from a comprehensive statistical and machine learning analysis conducted to investigate the effects of geomagnetic storms on the ionosphere in the context of increasing greenhouse gas concentrations. By examining the interrelationships among geomagnetic parameters, greenhouse gas variables, and ionospheric responses, this study aims to identify key predictive parameters for ionospheric variations over time. Furthermore, the analysis explores how fluctuations in greenhouse gas concentrations may modulate the influence of geomagnetic activity on ionospheric dynamics. The findings detailed in this chapter offer critical insights into the complex interactions governing space weather phenomena and atmospheric composition.

4.1 Data Points

The data utilized in this study span the period from 1974 to 2025. The temporal coverage of each parameter, however, is not uniform. This variation arises from the fact that the data were sourced from different repositories, each with its own archival range and data availability. Despite this, the dataset remains sufficiently robust, encompassing five solar cycles specifically cycles 21 through 25. Each solar cycle spans approximately 11 years; however, full coverage was not achieved for solar cycle 21, as the available data begin mid-cycle, and cycle 25 is ongoing.

Following data acquisition, a significant number of missing and uncertain values were observed across several parameters. Nonetheless, the dataset was deemed adequate for preliminary analysis. To evaluate the completeness and annual distribution of each parameter, bar plots were generated to visualize the number of valid data points recorded per year. These visualizations provide insight into the temporal density and reliability of the data across the examined period.

The distribution plots for the five principal parameters are presented below.

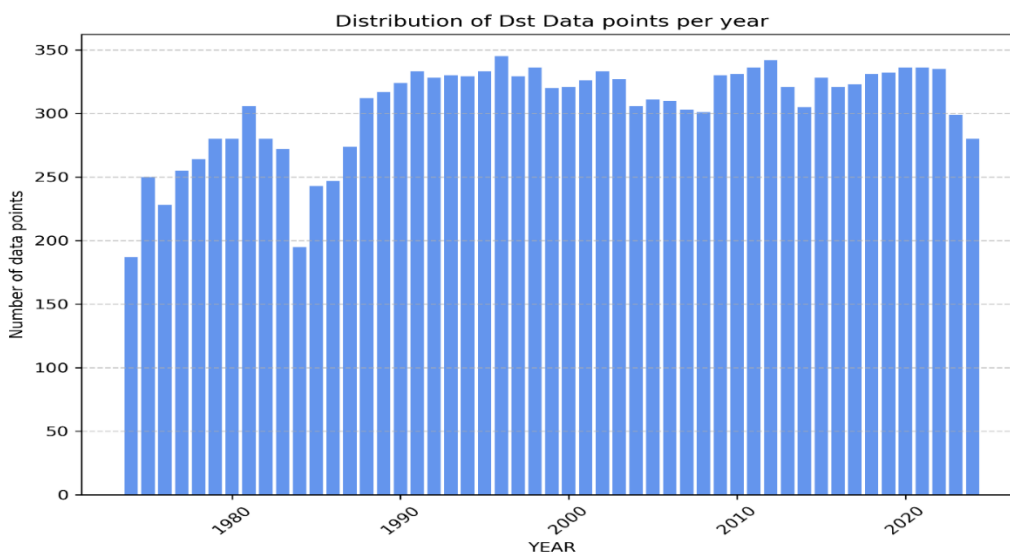


Figure 4. 2 A Bar chart showing the yearly distribution of data points of geomagnetic disturbances.

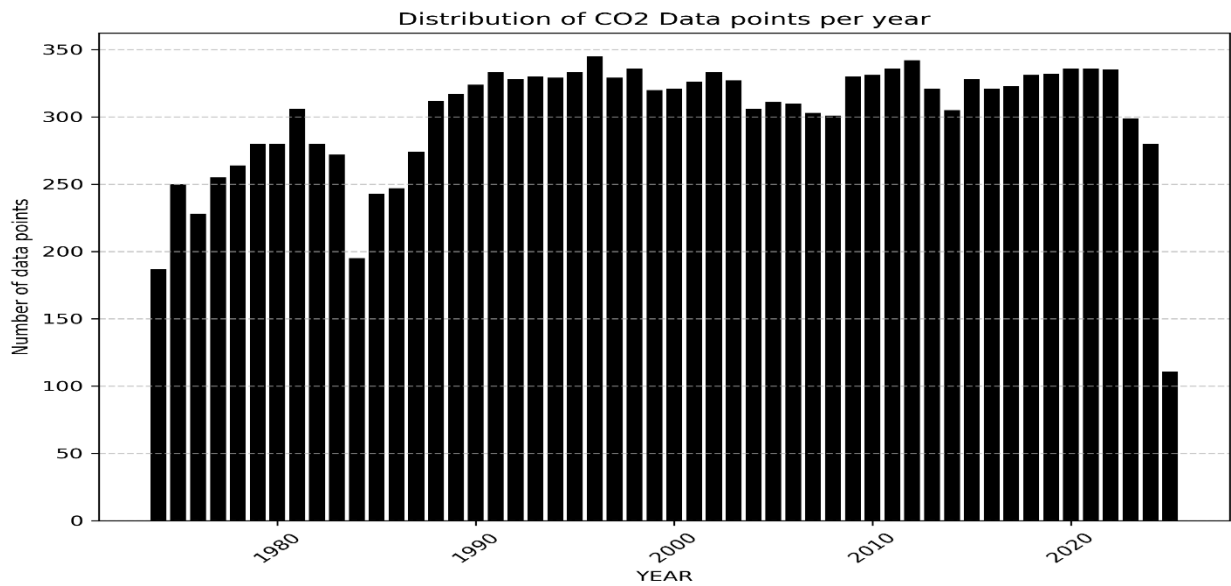


Fig.4.2 A Bar chart showing the distribution of daily averaged data points of co2 concentration yearly.

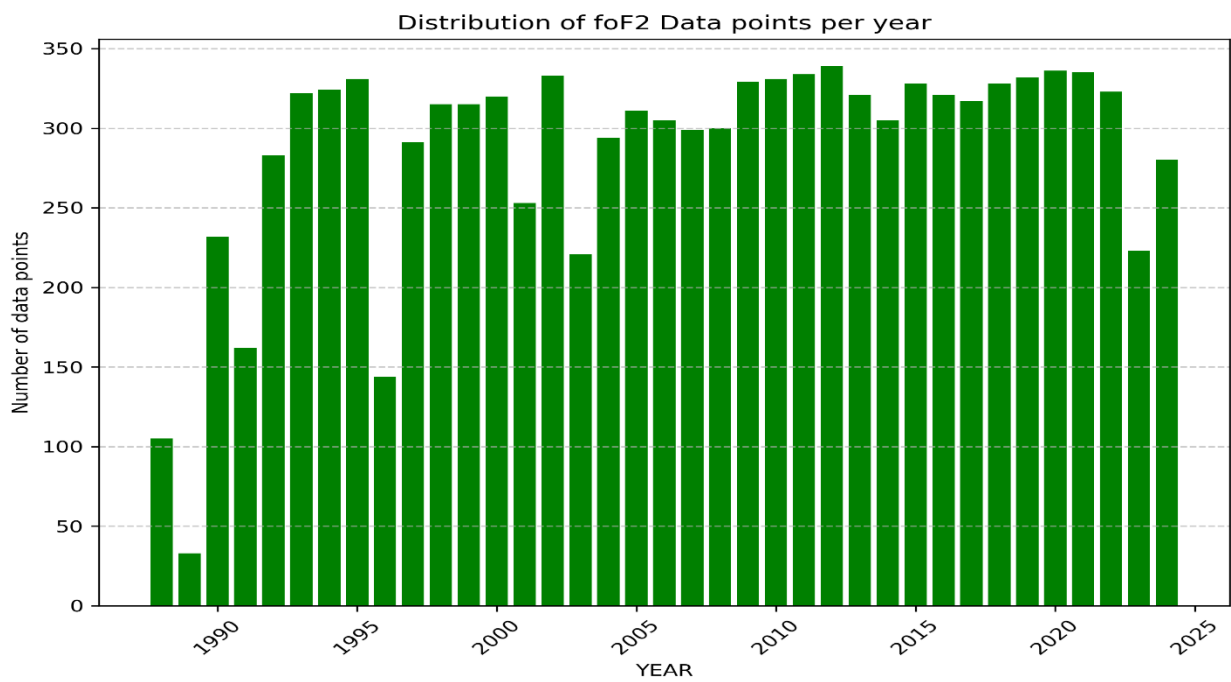


Fig.4.3 A Bar chart showing the yearly distribution of daily averaged data points of F2-layer critical frequency.

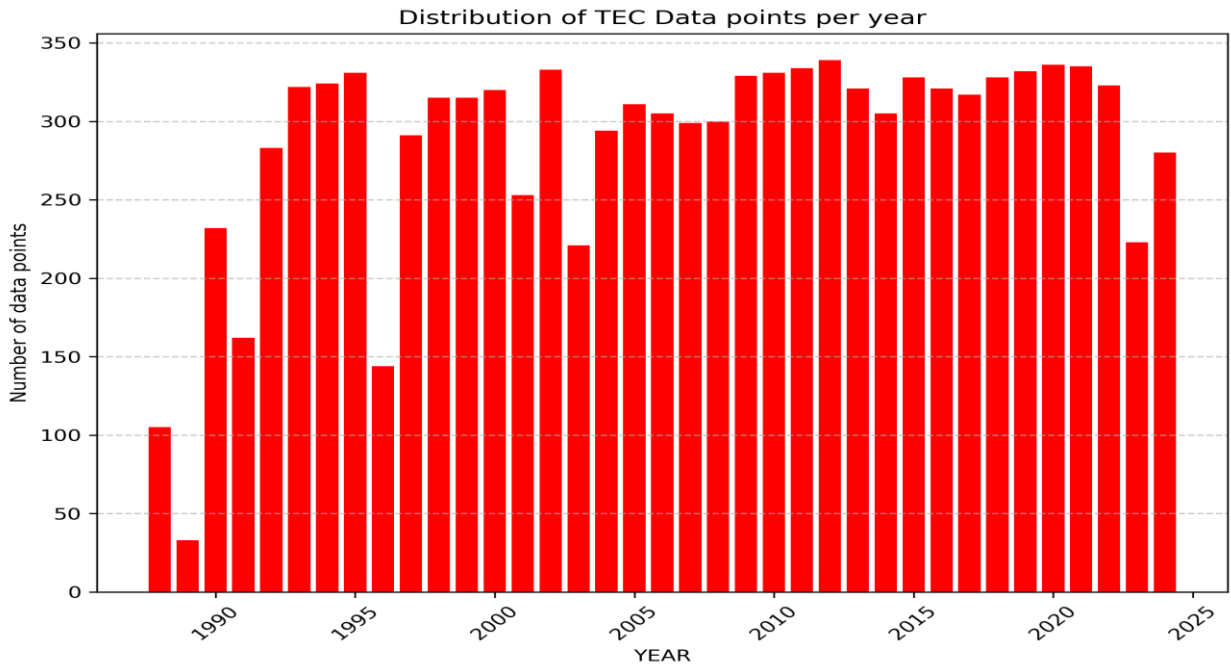


Fig.4.4 A Bar chart distribution of daily averaged data points of total electron content yearly

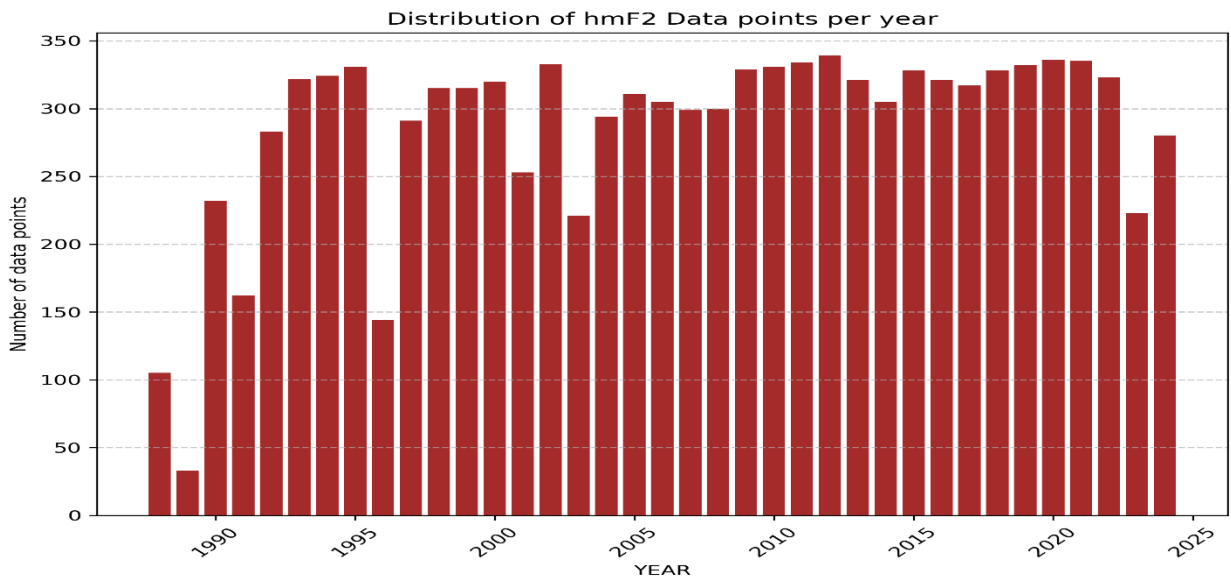


Fig.4.5 A Bar chart distribution of daily averaged data points of F2-layer peak height yearly

Analysis of the distribution plots revealed that the initial years of the dataset (particularly the first five years) contain relatively fewer data points compared to subsequent periods. This discrepancy may be attributed to the limitations in measurement technologies and observational infrastructure available during the early years of data acquisition. Additionally, for parameters that include data for the year 2025, the number of recorded observations is notably lower. This is expected, as the dataset was downloaded in May 2025, thus only representing partial-year coverage.

Furthermore, it was observed that certain intermediate years within the dataset also exhibit sparse or uneven data coverage. This inconsistency in annual data density may be linked to the effects of intense geomagnetic storms, during which satellite instruments used for monitoring often experience malfunctions, degraded performance, or complete data loss. In such

conditions, the recorded values may be flagged as unreliable, or in many cases, no measurements are available at all. These operational disruptions contribute to the uneven distribution of data points across the full temporal range.

4.2 Timeseries Analysis

There five primary parameters, comprising two independent and three dependent variables. The Independent variables are Geomagnetic Disturbance (Dst) and carbon (IV) oxide atmospheric concentration (CO₂), while the dependent variables are the ionospheric parameters Total electron content (TEC), F2-layer critical frequency (foF₂), and F2-layer peak height (hmF₂). These parameters exhibited notable variability over time, highlighting the importance of conducting a thorough temporal analysis. This variability necessitated an examination of the patterns and trends inherent in the data to provide insight into the potential relationships among the parameters. Understanding these dynamics was deemed essential for anticipating how the variables might interact within the modeling framework and for informing expectations regarding their correlative and functional behavior.

4.2.1 Timeseries plots of Dst-Index (Geomagnetic Disturbances)

Time series plots of geomagnetic disturbances based on the Dst index exhibited characteristics of stationarity, and it was observed that majority of the Dst values are above -60 nT, corresponding primarily to quiet or weak geomagnetic storm conditions. This observation aligns with expectations, as strong geomagnetic storms do not occur on a daily basis. The stationary nature of the series is further supported by the extended temporal coverage of the dataset, which includes extensive quiet-time intervals. Given that most Dst values fall within the range of 0 to -60 nT, it was anticipated that, in subsequent scatter plot analyses, relationships between the Dst index and other parameters may exhibit clusters concentrated within this range. This expectation is based on the data density observed in the time series. The Dst time series plot is displayed below.

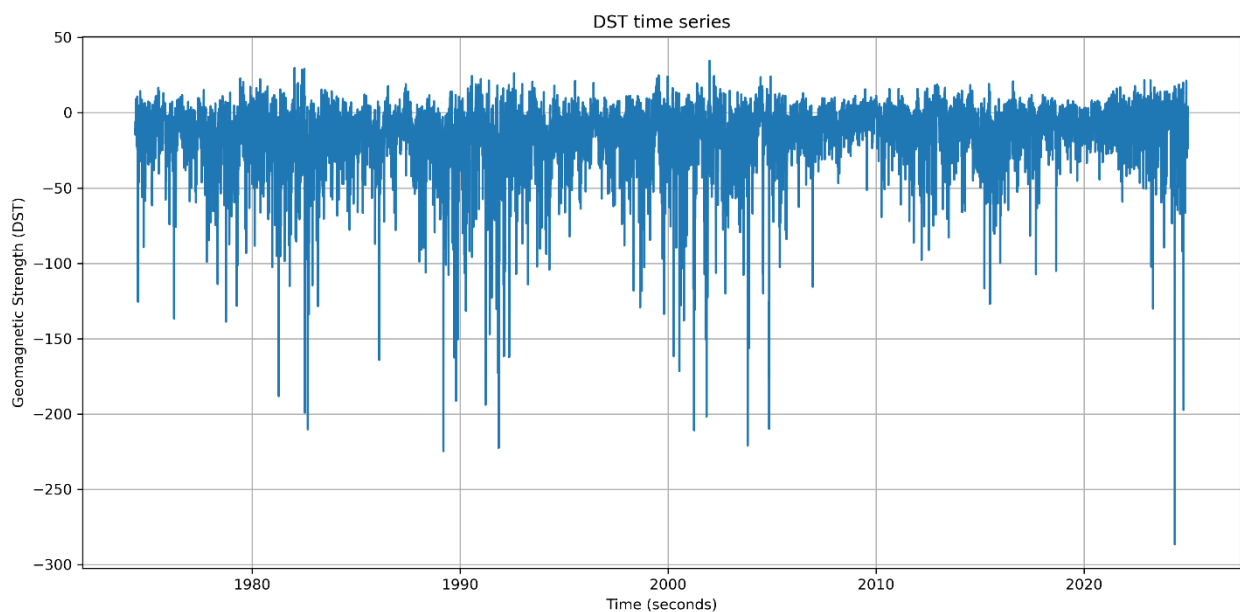


Fig4.6 A timeseries plot of Geomagnetic disturbance (Dst).

4.2.2 Timeseries plots of Carbon (IV) Oxide (CO₂)

The time series plot of atmospheric carbon dioxide (CO₂) concentration was found to be non-stationary, exhibiting a progressive upward trend over time. The pattern of increase was not smooth but followed a distinct and recurring zigzag structure, indicating periodic fluctuations superimposed on the overall rise. Upon closer inspection, it was observed that each increase in concentration was typically preceded by a temporary decline, after which the concentration rose to a level higher than the previous peak. This recurring behavior suggested that CO₂ levels oscillated in a repeating cycle, with each cycle resulting in a net increase over time. Based on the visible structure of the time series, it was inferred that the periodicity of this pattern was likely around one to two years. The corresponding time series plot of CO₂ concentration is presented below.

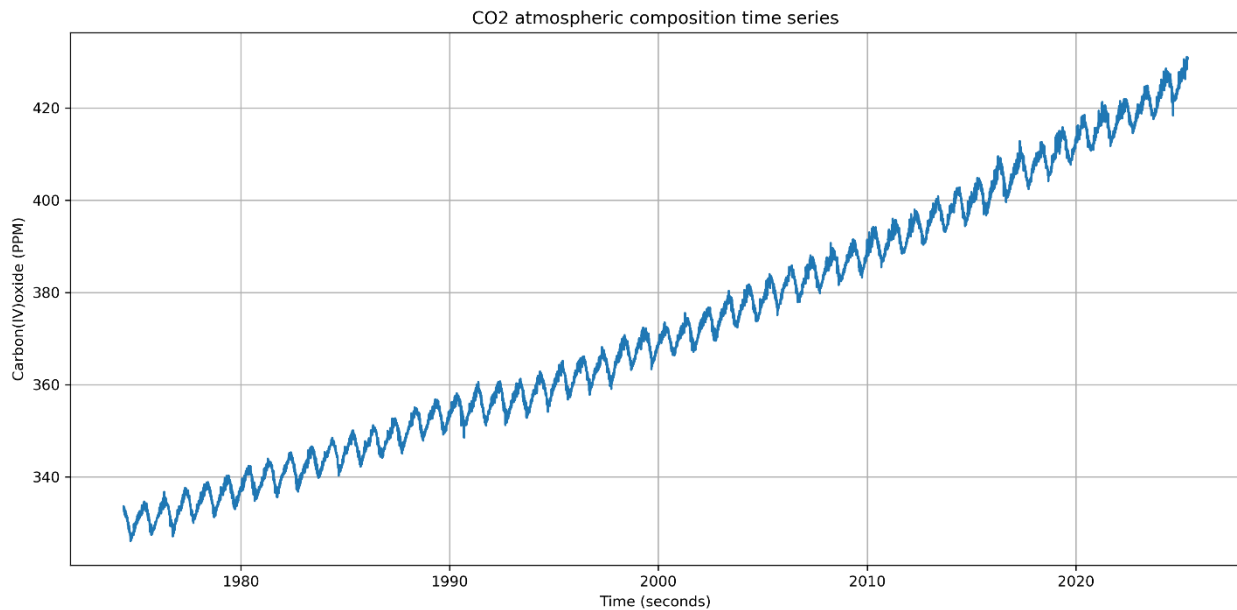


Fig4.7 A timeseries plot of carbon (IV) oxide concentration CO₂

4.3 Timeseries plots of Total Electron Content (TEC)

The time series plots for Total Electron Content (TEC) exhibited a clear periodic pattern. Distinct peaks in TEC values were observed during the periods spanning approximately 1988 to 1992, 2000 to 2004, 2012 to 2016, and again from 2024 to the present. In contrast, noticeable minimum values were identified around the years 1996, 2008, and 2020. Based on this observation, it was concluded that the current time frame falls within a phase of significant TEC increase. This recurring pattern raised questions about the underlying drivers of TEC variability.

Further analysis revealed that the independent variables in this study—namely Dst and atmospheric CO₂ concentration—exhibited different temporal behaviors. While one of them was relatively stationary, the other showed a steadily increasing trend with periodic fluctuations. This disparity suggested that the influence of these variables on TEC might be more complex and potentially nonlinear. It was concluded, based on the analysis, that the TEC behavior could not be fully accounted for by Dst and CO₂ alone, pointing to the likely presence of other influential parameters that had not been considered in the current study.

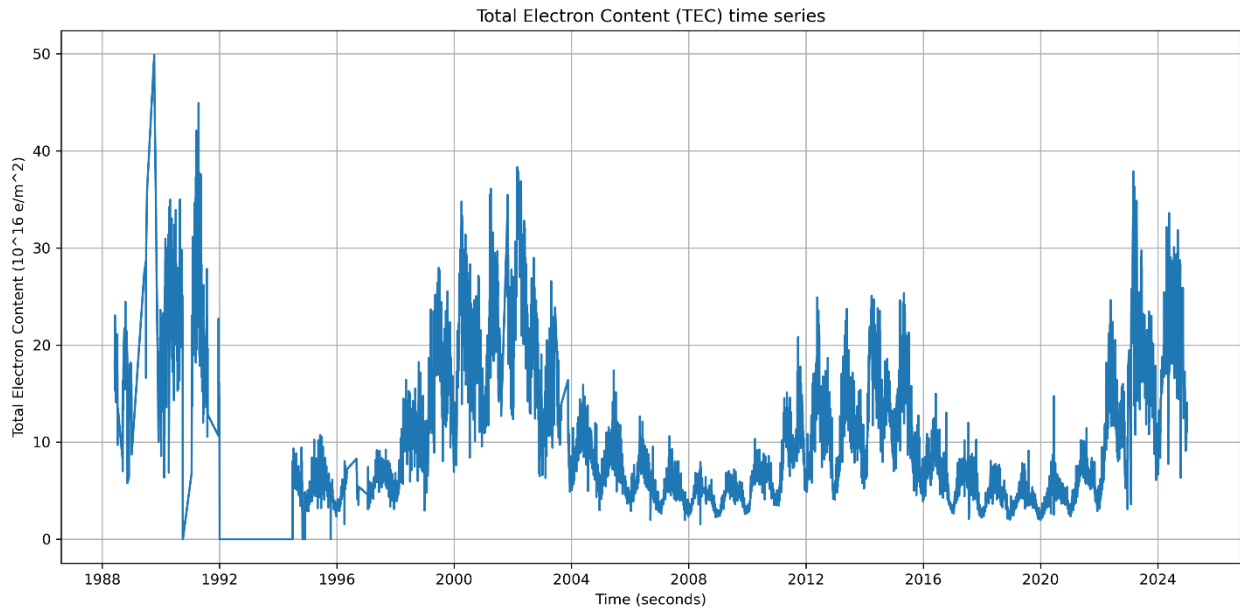


Fig4.8 A timeseries plot of Total Electron Content (TEC)

4.3.1 Timeseries plot of F2-layer Critical Frequency (foF2)

The time series of the F2-layer Critical Frequency (foF2) was also found to exhibit a periodic pattern, closely resembling that of the Total Electron Content (TEC). The timing of the observed maxima and minima in the foF2 series aligned with those of the TEC series, thereby reinforcing the fundamental physical relationship between these two ionospheric parameters.

From this observed similarity, it was predicted that the dominant factors influencing TEC may also play a similar role in shaping the temporal behavior of foF2. The shared periodicity suggested a common underlying driver.

Upon comparing the foF2 time series with those of the independent variables Dst and atmospheric CO₂ concentration it became apparent that neither of these variables alone could fully account for the observed periodic highs and lows in foF2. This raised the possibility that another influential factor, not considered in the current model, might be contributing to the variation. The pattern of fluctuations in foF2 bore a strong resemblance to the 11-year solar cycle, leading to the interpretation that solar activity could be the principal force behind the long-term variation observed in both TEC and foF2.

The corresponding time series plot of foF2 is presented below.

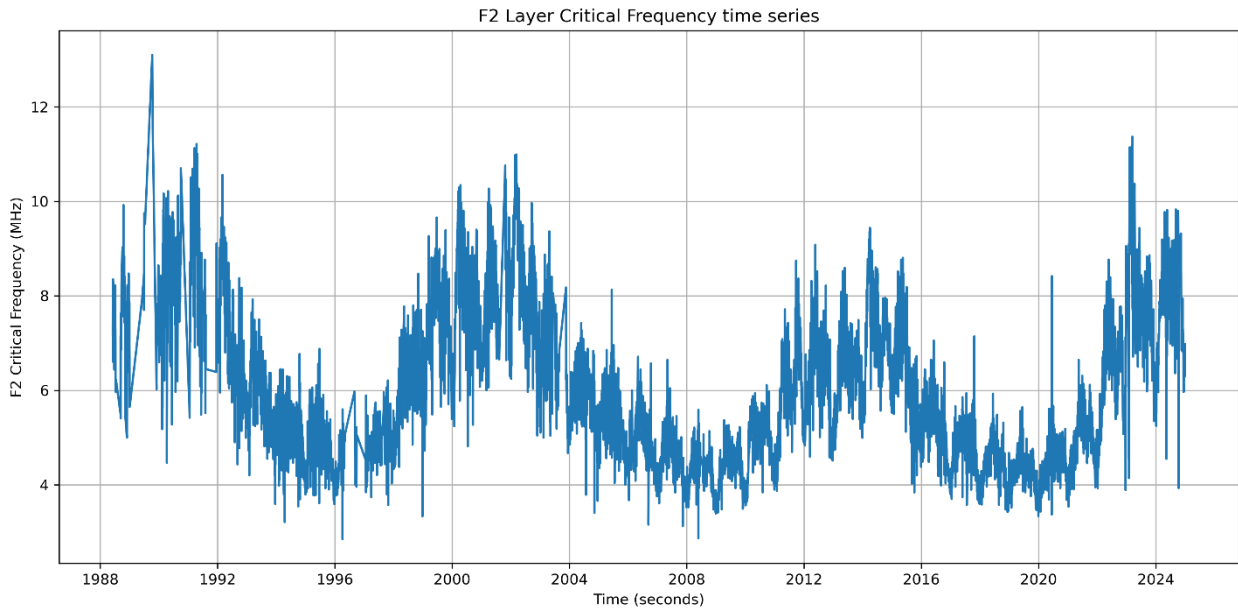


Fig4.9 A Timeseries plot of F2-layer Critical frequency (foF2)

4.3.2 Timeseries plot of F2-Layer Peak Height (hmF2)

The time series plot of the F2-layer peak height (hmF2) also exhibited a periodic pattern, closely resembling those of Total Electron Content (TEC) and the critical frequency (foF2). The alignment of minima and maxima across these three parameters further supported the interpretation that solar activity likely modulated by the solar cycle is a major driver of the observed variability in the ionosphere.

Based on this observation, it was predicted that the F2-layer peak height, foF2, and TEC may be jointly influenced by solar activity, given their shared periodicity and similar temporal behavior.

However, caution was exercised in drawing broad conclusions about the entire ionosphere. This was due to the limited scope of the analysis, which was based on only three ionospheric parameters, hmF2, foF2, and TEC that may themselves be interrelated. Such interdependence implies that if solar activity strongly affects one, it is likely to affect the others in similar ways. Therefore, while the observed patterns supported a solar-cycle influence, the possibility of overgeneralization was acknowledged.

Furthermore, the analysis raised the possibility of a complex, potentially nonlinear relationship between hmF2 and the independent variables in this study Dst and atmospheric CO₂ concentration. The structure of the time series suggested that the variation in peak height could not be fully explained by these two parameters alone, indicating the likely involvement of other influential geophysical or atmospheric drivers not accounted for in the current model.

The corresponding time series plot of hmF2 is presented below.

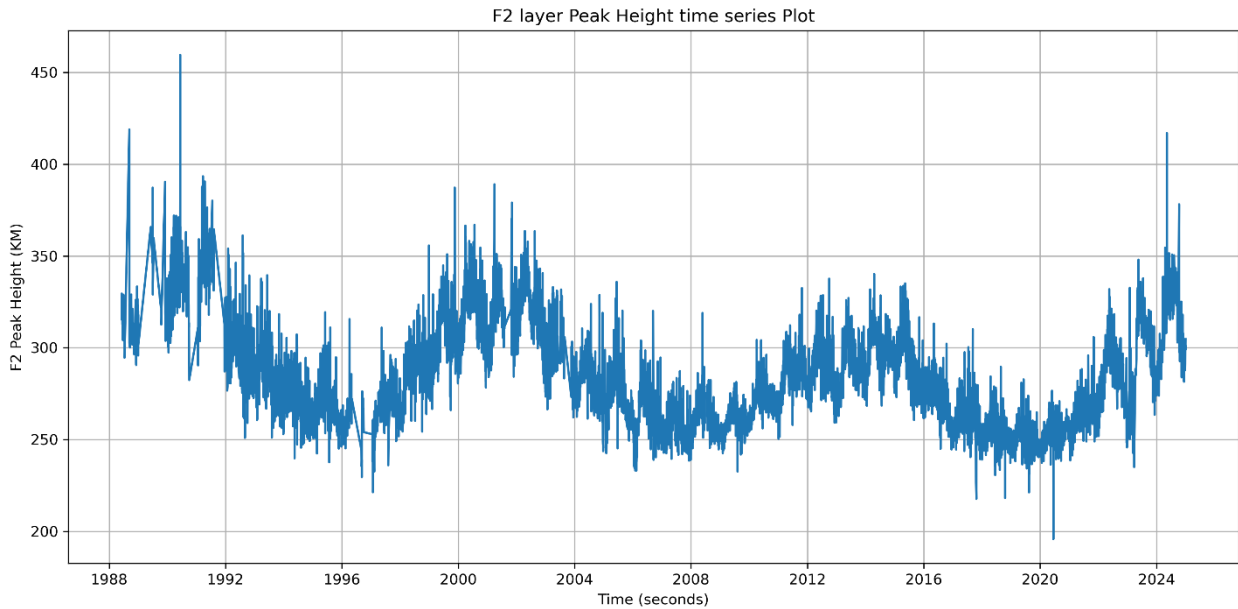


Fig4.10. A Timeseries plot of F2-layer Peak Height (hmF2)

4.3.3 Summary

From the time series analysis, it was observed that the independent variables—Dst and atmospheric CO₂ appear to have complex, possibly nonlinear relationships with the ionospheric parameters (TEC, foF2, and hmF2). The periodic patterns identified in the dependent variables aligned closely with known solar cycle behavior, suggesting that solar activity may be a significant driver of ionospheric variability. The strong similarity in the behavior of the dependent variables further indicated that they may be highly interrelated.

To verify these observations and better understand the strength and nature of these relationships, the next steps identified include computing correlation coefficients and plotting scatter diagrams between the independent and dependent variables, as well as among the variables within each group. Additionally, the dataset will be segmented by solar cycles to investigate how these relationships evolve across different phases of solar activity.

4.4 Correlation and Scatter plots analysis of variables

In this section, scatter plots were examined to explore the relationships between the independent variables (Dst and atmospheric CO₂) and the dependent ionospheric parameters (TEC, foF2, and hmF2). Additionally, relationships among the dependent variables themselves and among the independent variables were also analyzed. Correlation coefficients were calculated for each pair to quantitatively assess the strength and direction of these associations. This analysis was conducted to provide a clearer understanding of the interactions between variables, support the interpretation of their behavior, and inform the modeling process with prior insight into expected variable dynamics.

4.4.1 Correlation and Scatter plot analysis of Dst and the dependent variables (TEC, foF2, and hmF2)

The scatter plots illustrating the relationships between Dst (on the horizontal axis) and the three dependent ionospheric parameters TEC, foF2, and hmF2 (each on the vertical axis), reveal a

common structural pattern across all cases. In each plot, the data points form a cloud-like cluster concentrated primarily within the Dst range of approximately +20 nT to –50 nT. This distribution aligns with prior observations from the Dst time series, which indicates that the majority of Dst values exceed –60 nT, consistent with the predominance of weak geomagnetic storm events in the dataset.

In the cases of TEC and foF2, the scatter plots exhibit a strongly vertical clustering of data points, with no evident directional trend or pattern. The calculated Pearson correlation coefficients reinforce this observation: for Dst vs. TEC, the coefficient is –0.068 (–6.8%), and for Dst vs. foF2, it is –0.088 (–8.8%). These values indicate very weak negative linear correlations, suggesting that increases in Dst are only loosely associated with decreases in TEC and foF2. These findings support the interpretation that the relationship between Dst and these ionospheric parameters is complex and nonlinear.

In contrast, the scatter plot for Dst vs. hmF2 presents a slightly different structure. While it also displays a clustered distribution within the same Dst range, the data points are less vertically aligned compared to those of TEC and foF2. This suggests lower variability in hmF2 and a marginally clearer relationship with Dst. The Pearson correlation coefficient in this case is –0.306 (–30.6%), which though still not indicative of a strong linear relationship, is notably stronger than those observed for the other two parameters. The negative sign once again implies that decreases in Dst (i.e., stronger geomagnetic storms) are associated with increases in hmF2. This moderate correlation suggests that hmF2 responds more noticeably to Dst variability, even though the overall relationship remains nonlinear and complex.

In summary, the interpretation across all three scatter plots remains consistent: the relationships between Dst and the ionospheric parameters (TEC, foF2, and hmF2) are not strongly linear. Their interactions appear to be governed by more intricate dynamics. Nevertheless, hmF2 displays a comparatively stronger and more structured relationship with Dst, underscoring its potential sensitivity to geomagnetic activity.

The corresponding scatter plots for each parameter are presented below.

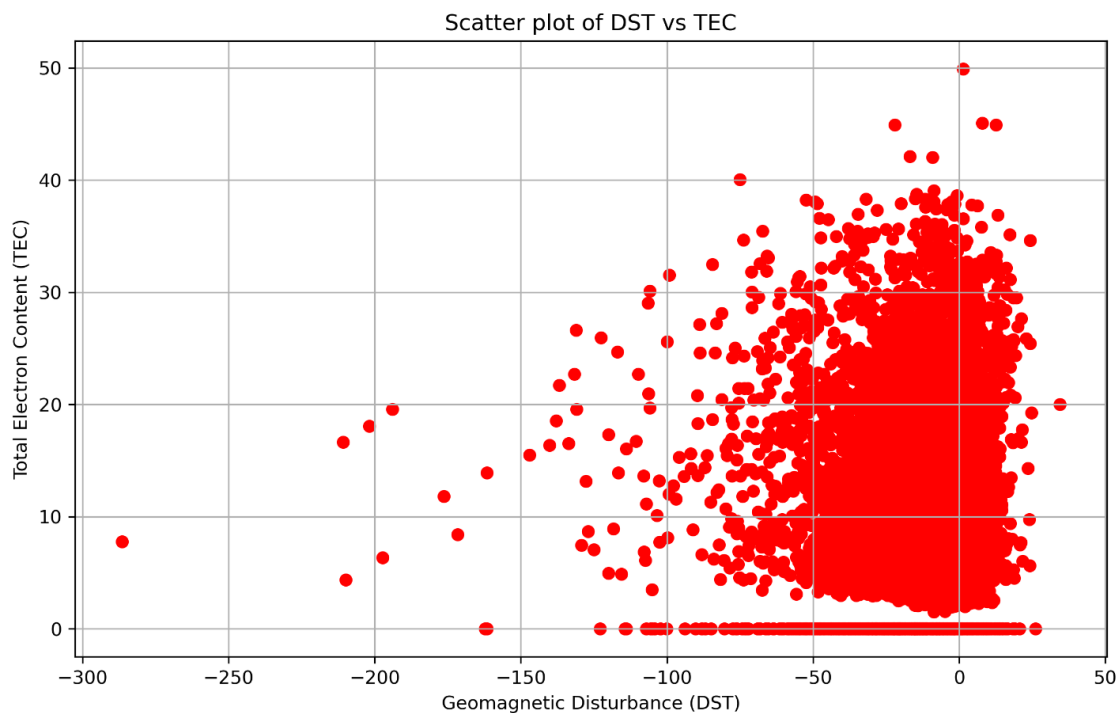


Fig.11 A scatter plot relationship between Dst (Geomagnetic Disturbance) and TEC (Total Electron Content)

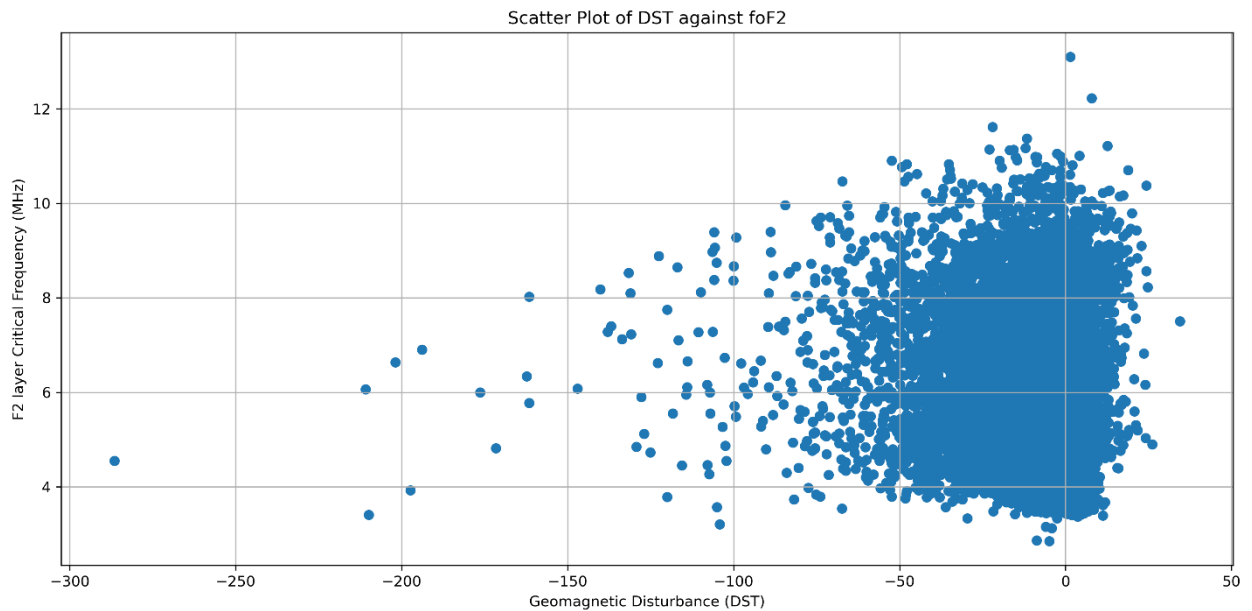


Fig4.12 A scatter plot of the relationship between Geomagnetic Disturbance (Dst) and F2-layer Critical Frequency (foF2).

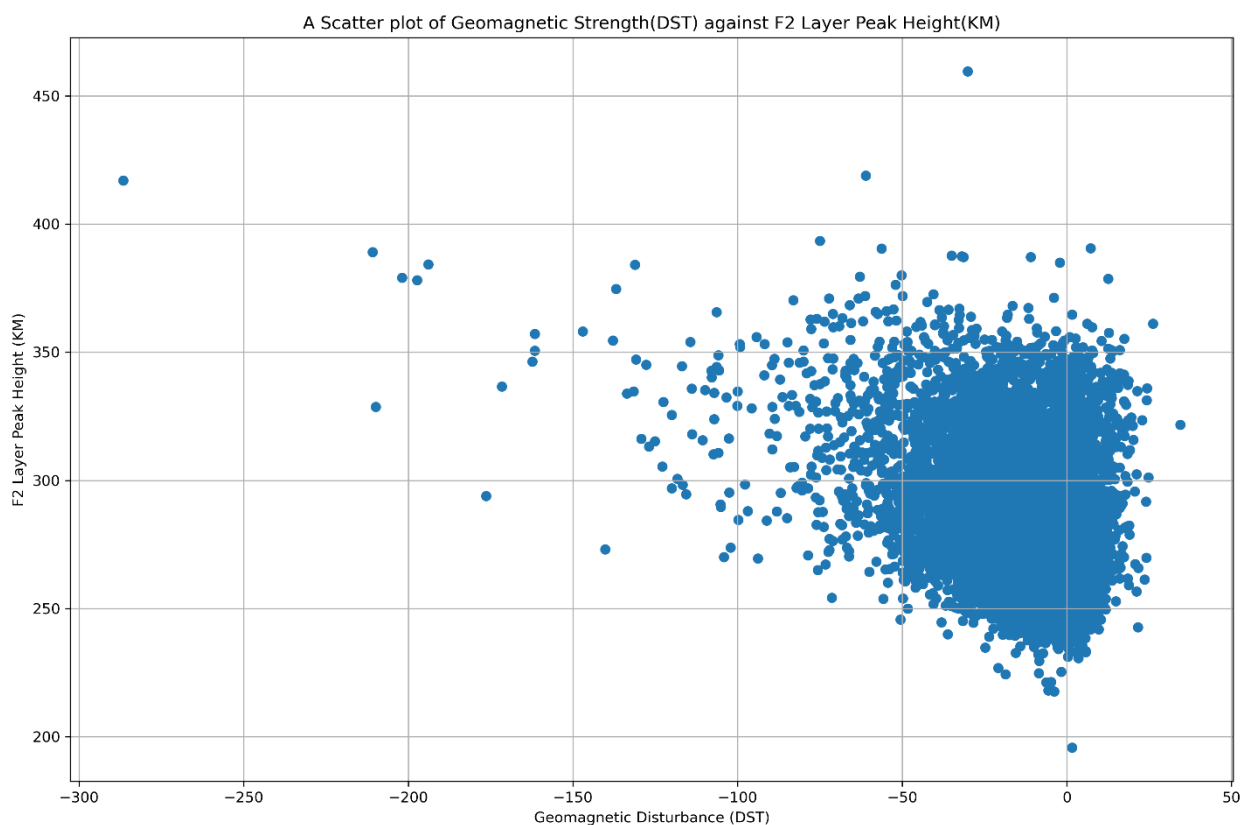


Fig4.13 A scatter plot of Geomagnetic Disturbance (Dst) and F2-layer Peak Height (hmF2)

4.4.2 Correlation and Scatter plot analysis of CO₂ and the dependent variables (TEC, foF2, and hmF2)

The scatter plots examining the relationships between atmospheric CO₂ and the three ionospheric parameters—TEC, foF2, and hmF2—exhibit patterns that are periodic in nature. As CO₂ values increase along the horizontal axis, the corresponding values of the ionospheric parameters on the vertical axis oscillate between high and low, forming a wave-like

distribution. This pattern makes it difficult to visually identify a consistent directional relationship. However, it reinforces conclusions drawn from the time series analysis, particularly the inference that the ionospheric parameters are likely influenced by solar cycle variations.

This observed oscillatory behavior prompts a reconsideration of the role atmospheric CO₂ plays in influencing ionospheric dynamics. It raises the question of whether CO exerts a direct and significant effect on ionospheric parameters or whether the periodicity observed is merely reflective of solar activity patterns that happen to align with long-term CO₂ trends.

The calculated Pearson correlation coefficients provide further insight. Specifically, the correlation between CO₂ and TEC is -0.035 (-3.5%), between CO₂ and foF₂ is -0.191 (-19.1%), and between CO₂ and hmF₂ is -0.271 (-27.1%). These values, though all negative, suggest that increases in atmospheric CO₂ are generally associated with decreases in the ionospheric parameters, albeit very weakly.

Among the three parameters, TEC shows the weakest correlation with CO₂, supporting the notion that TEC variations are more strongly governed by solar activity than by CO₂ concentrations. On the other hand, hmF₂ presents the strongest (though still weak) negative correlation, indicating a relatively greater sensitivity to CO₂. However, even this is insufficient to establish a robust linear relationship.

Taken together, the periodic nature of the scatter plots and the low correlation values lead to the conclusion that the relationship between atmospheric CO₂ and ionospheric behavior is complex and likely nonlinear. This is consistent with the patterns observed in the earlier time series analysis. At the same time, the results raise valid skepticism regarding whether CO₂ has any direct or significant role in modulating ionospheric conditions. Although the correlations are non-zero, the prevailing patterns more clearly reflect solar cycle influences than any CO₂-driven mechanisms.

The corresponding scatter plots are presented below.

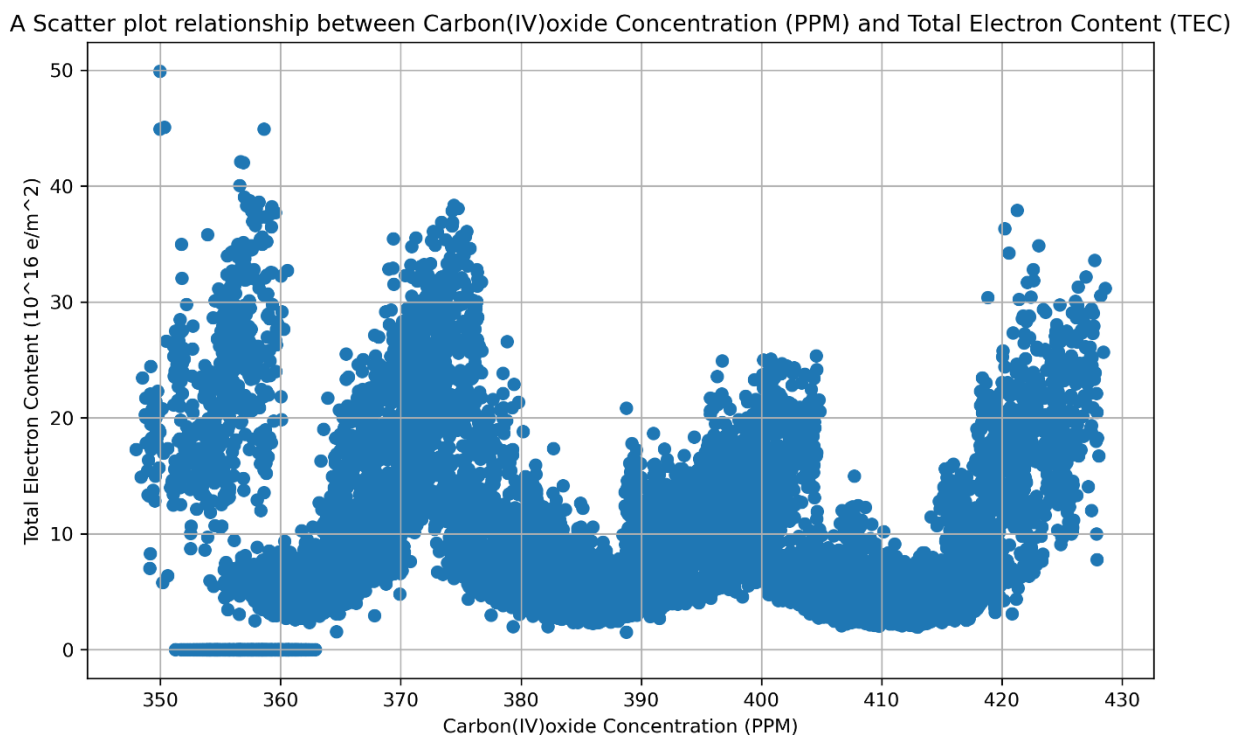


Fig4.14 A scatter plot of the relationship between carbon (IV) oxide concentration (CO₂) and Total Electron Content (TEC)

A Scatter plot relationship between Carbon(IV)oxide Concentration (PPM) and F2 layer Critical Frequency (MHz)

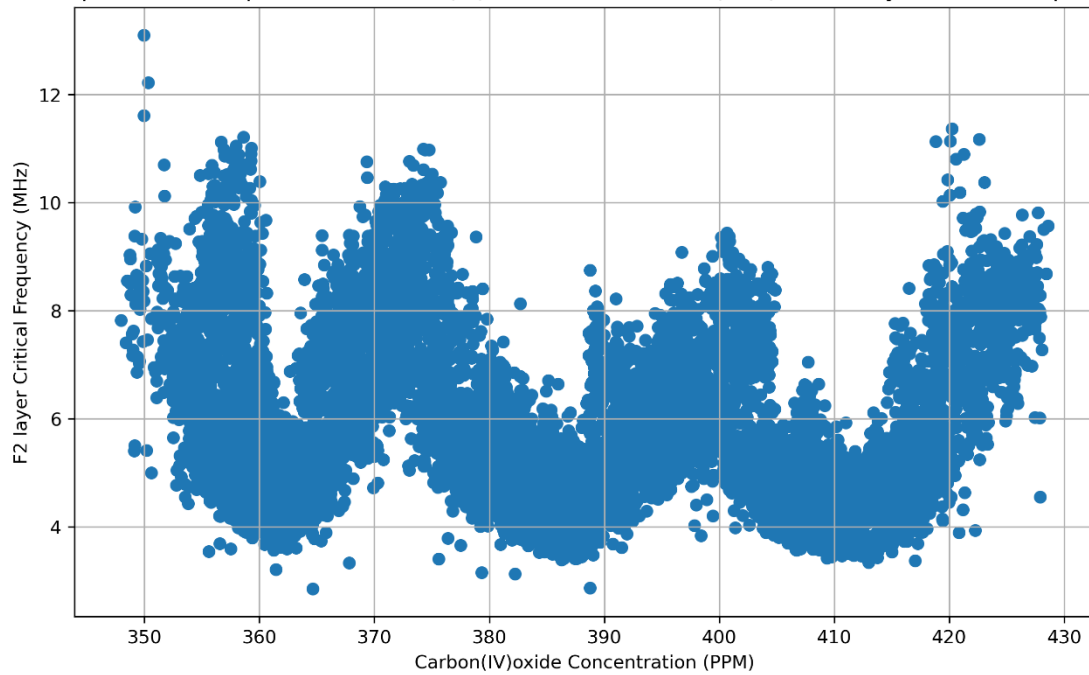


Fig4.15 A scatter plot of the relationship between carbon (IV) oxide concentration (CO₂) and F2-layer Critical Frequency (foF₂)

A Scatter plot relationship between Carbon(IV)oxide Concentration (PPM) and F2 layer Peak Height (Km)

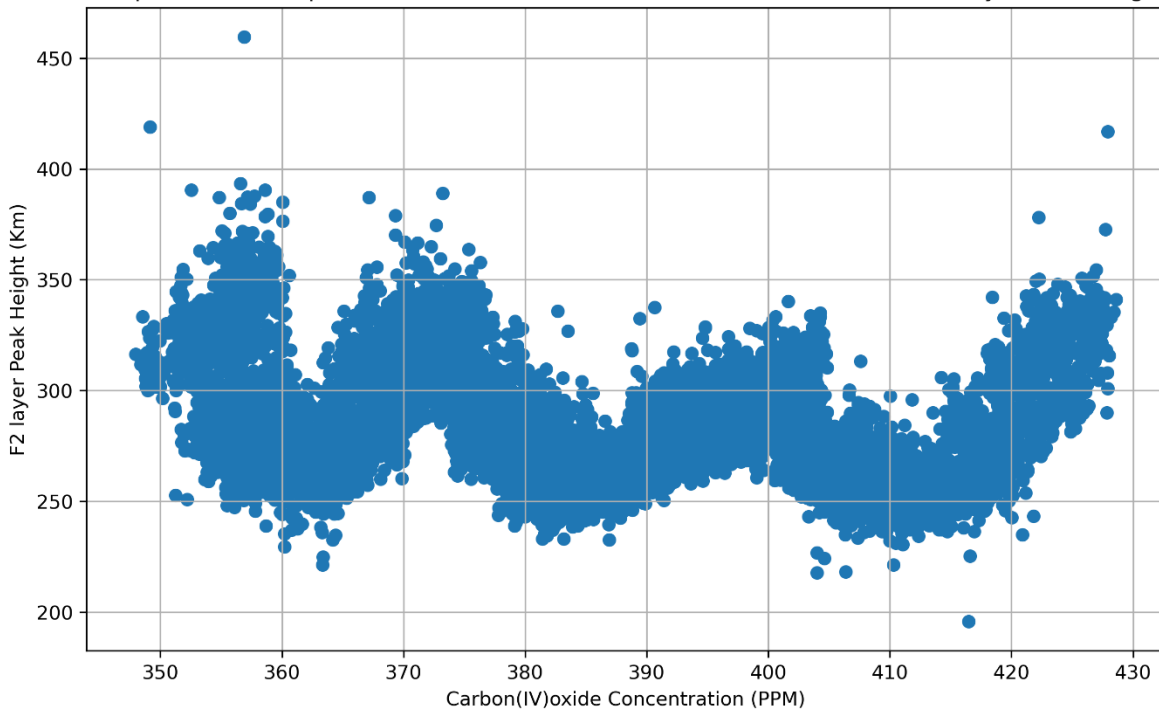


Fig4.16 A scatter plot of the relationship between carbon (IV) oxide (CO₂) and F2-layer Peak Height (hmF₂)

4.4.3 Correlation and Scatter plot analysis between the independent variables Dst and CO₂

In this scatter plot, Dst is plotted on the horizontal axis, and CO₂ on the vertical axis. The data points are clustered predominantly within the Dst range of approximately +20 nT to −50 nT,

forming a distribution that appears more vertical than horizontal. This orientation reflects the relative variability of the variables, i.e. CO₂ values vary more than Dst across the dataset.

The Pearson correlation coefficient calculated for this relationship is 0.174, indicating a weak but positive linear association between Dst and CO₂. While this value is still considered low and does not suggest a strong linear dependence, it is notably higher than the correlation coefficients observed between CO₂ and TEC, Dst and TEC, and Dst and foF₂. The positive sign of the coefficient suggests that both variables tend to increase together, albeit in a weak and inconsistent manner.

This observation raises important considerations. Since Dst and CO₂ are both independent variables in the study, a correlation between them is undesirable, as it may lead to multicollinearity in the modeling process an issue that can undermine the interpretability of regression coefficients and compromise the robustness of the model.

Moreover, this weak correlation also aligns with existing research, which generally finds no direct causal relationship between geomagnetic activity and greenhouse gas concentrations. Although the scatter plot shows a small degree of linear association, it is more likely that any relationship between Dst and CO₂ is indirect or coincidental rather than causative. This reinforces the current understanding that while both variables may influence the ionosphere, they do so through separate mechanisms.

The scatter plot illustrating the relationship between Dst and CO₂ is presented below.

A Scatter plot relationship between Geomagnetic Strength (DST) and Carbon(IV)oxide Concentration (PPM)

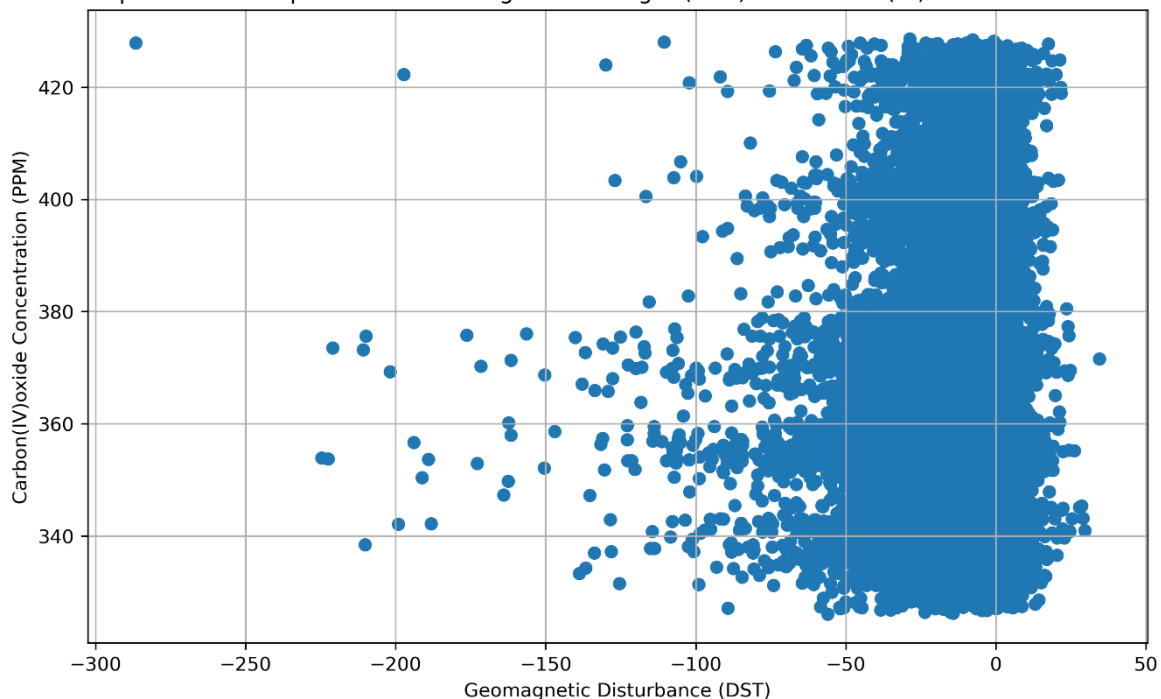


Fig4.17 A scatter plot of the relationship between Geomagnetic Disturbance (DST) and carbon (IV) oxide (CO₂)

4.4.4 Correlation and Scatter plot analysis between dependent variables TEC, foF2, and hmF2

The scatter plots representing the relationships among the dependent variables—TEC vs. hmF2, TEC vs. foF2, and foF2 vs. hmF2—exhibit near-linear structures, each forming slightly bent lines with consistent positive gradients. Although not perfectly straight, the clustering and directional alignment of the data points strongly suggest well-defined positive linear associations among the parameters.

This visual interpretation is quantitatively supported by the computed Pearson correlation coefficients. The correlation between TEC and foF2 is 0.865, that of TEC and hmF2 is 0.791, and the correlation between foF2 and hmF2 is 0.830. With two of the coefficients exceeding the 0.80 threshold and the third just marginally below it, all three relationships qualify as strong positive correlations.

These findings confirm that the dependent variables are highly interrelated, exhibiting parallel behavioral trends across the observed dataset. Given that TEC, foF2, and hmF2 are widely recognized as the primary indicators of ionospheric conditions, especially in relation to geomagnetic disturbances, this high degree of correlation reinforces their validity as representative parameters of overall ionospheric dynamics.

Therefore, any conclusions drawn regarding the behavior of these dependent variables in response to independent variables such as Dst and atmospheric CO₂ can be confidently generalized to characterize the broader ionospheric response. The corresponding scatter plots illustrating these relationships are presented below.

A Scatter plot relationship between Total Electron Content (TECU) and F2-layer Critical Frequency (Km)

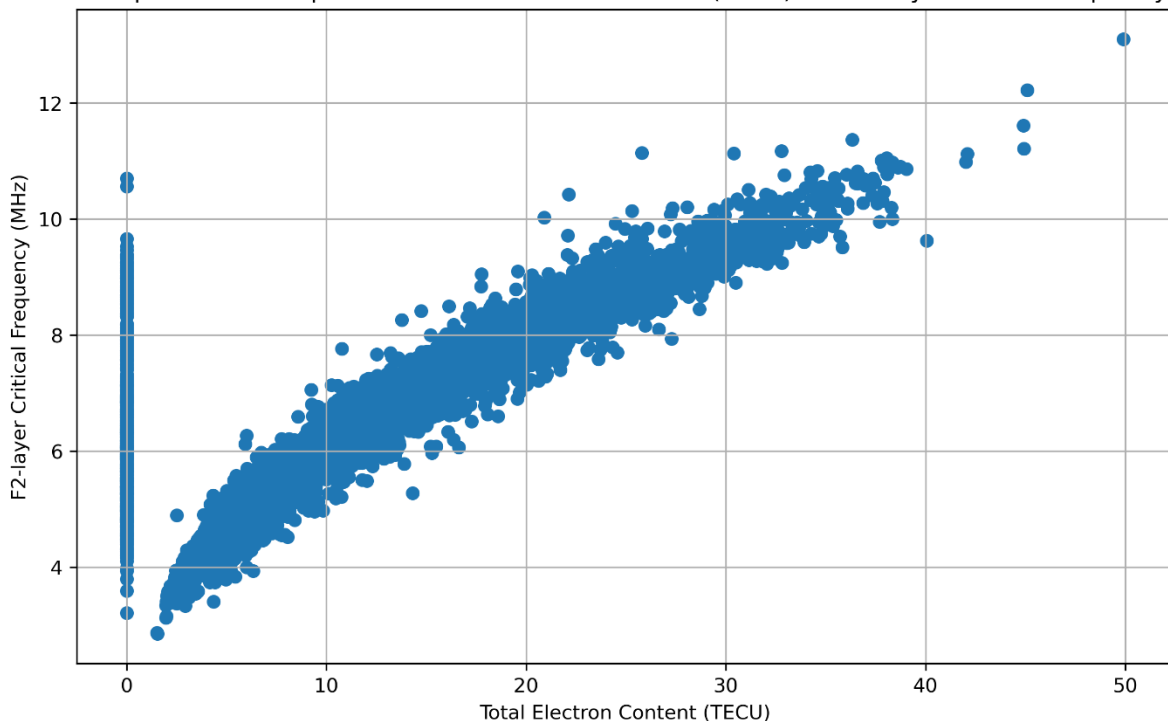


Fig.4.18. A scatter plot of the relationship between total electron content (TEC) and F2-layer Critical frequency (foF2)

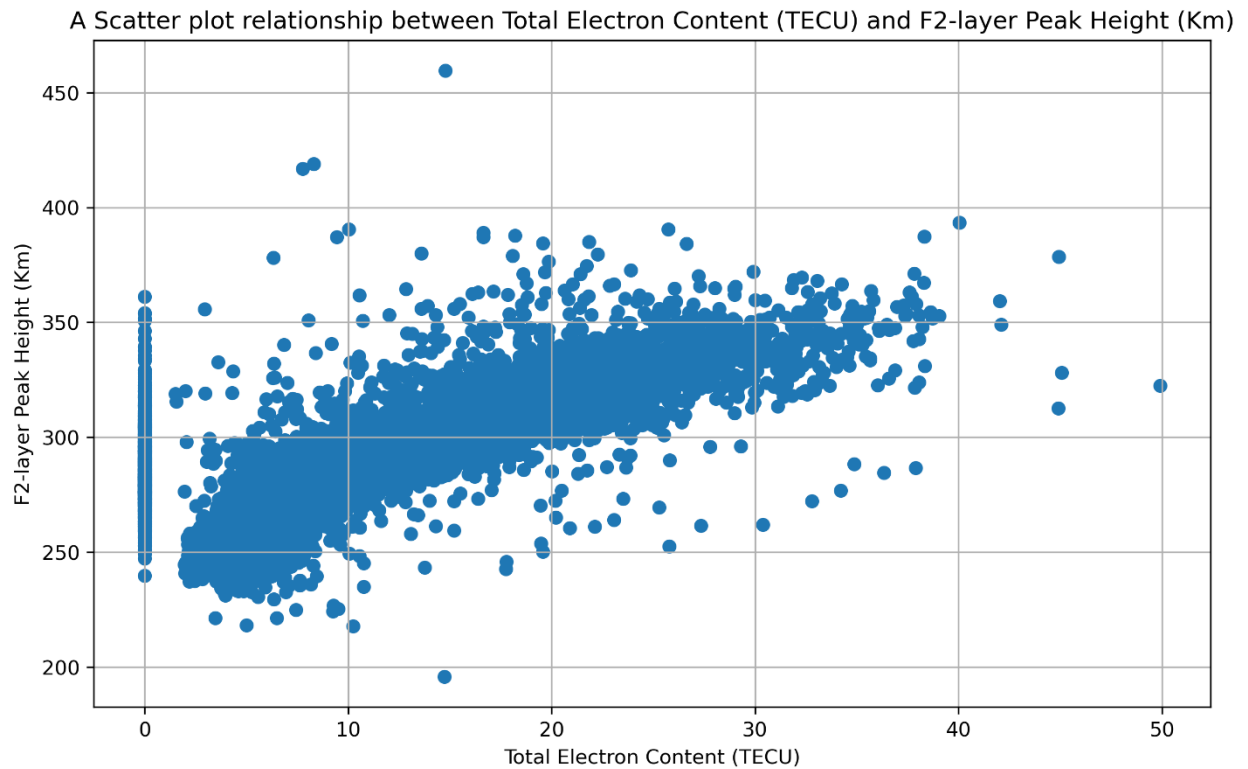


Fig.4.19. A scatter plot of the relationship between Total Electron Content (TEC) and F2-Layer Peak Height (hmF2)

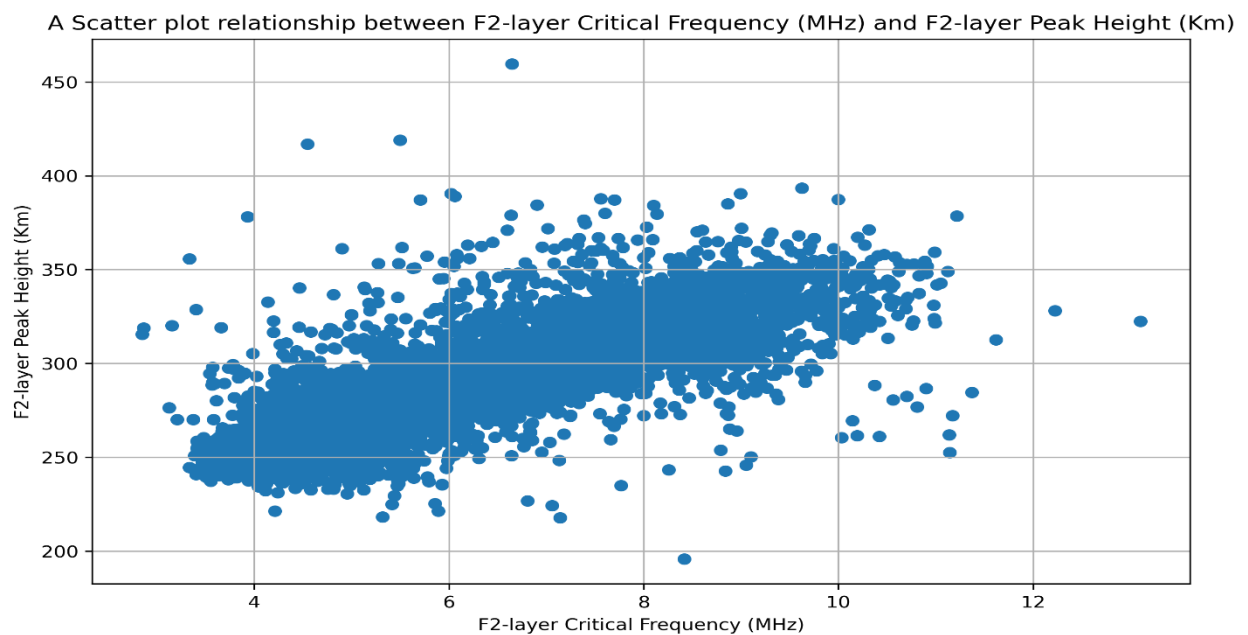


Fig.4.20. A scatter plot of the relationship between F2-layer Critical frequency (foF2) and F2-Layer Peak Height (hmF2)

4.5 Analysis by Solar cycle

The time series plots of the independent variables reveal periodic patterns that closely resemble those of the solar cycle. As a result, the time series of each parameter is analyzed across individual solar cycles. This analysis serves to confirm whether the observed periodicity is attributable to solar activity and to determine the extent to which solar activity influences the independent variables namely, Dst and atmospheric CO₂.

4.5.1 Dependent variables TEC, foF2 and hmF2 (Ionospheric parameters) timeseries plots in each solar cycle

The time series plots of the ionospheric parameters TEC, foF2, and hmF2 across the solar cycles reveal consistent patterns indicative of periodic behavior. Notably, Solar Cycle 21 is excluded from this analysis due to the absence of ionospheric data for that period, and the time series for Solar Cycle 22 contains substantial missing values, making its pattern less clear and more difficult to interpret with confidence.

For Solar Cycle 25, the time series of the ionospheric parameters demonstrates a steadily rising trend accompanied by a recurring zigzag pattern. This pattern reflects short-term fluctuations in which the values increase, reach a peak, then decrease, though never falling below their preceding starting levels. This results in an overall upward trajectory. The periodicity of the zigzag fluctuations occurs roughly within one-year intervals. The highest peak is observed in the year 2024, which coincides with the expected starting point of the maximum of Solar Cycle 25. Since the available dataset concludes at this point, the continuation of the trend beyond 2024 remains unknown.

In contrast, the time series observed in previous solar cycles (Cycles 23 and 24) also exhibit similar periodic zigzag structures but without the overall upward trend seen in Cycle 25. Instead, each of these earlier cycles begins from a minimum, rises to a maximum, and then gradually returns to a value that approximates the original starting level—thus, forming a stationary pattern. The lack of a steady upward progression suggests that, unlike Cycle 25, these cycles complete a full solar cycle phase in the available dataset.

The distinctive upward trend in Cycle 25 is attributed to the fact that the cycle is still ongoing, and the data reflect its ascending phase toward solar maximum. When this is taken into account, the ionospheric parameters across all the analyzed solar cycles—despite temporal differences—demonstrate a broadly consistent pattern aligned with solar cycle progression.

Based on these observations, it is concluded that solar activity is a dominant driver of the periodicity observed in the ionospheric parameters. The consistent rise and fall of TEC, foF2, and hmF2 within solar cycles, as well as the alignment of their peaks with known solar maxima, reinforce the influence of solar dynamics on ionospheric behavior.

The time series plots for each ionospheric parameter across the respective solar cycles are presented below.

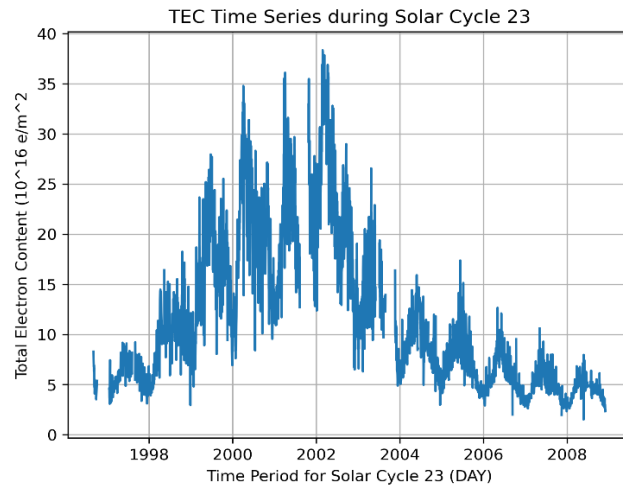


Fig.4.21. A timeseries plot of TEC during solar cycle

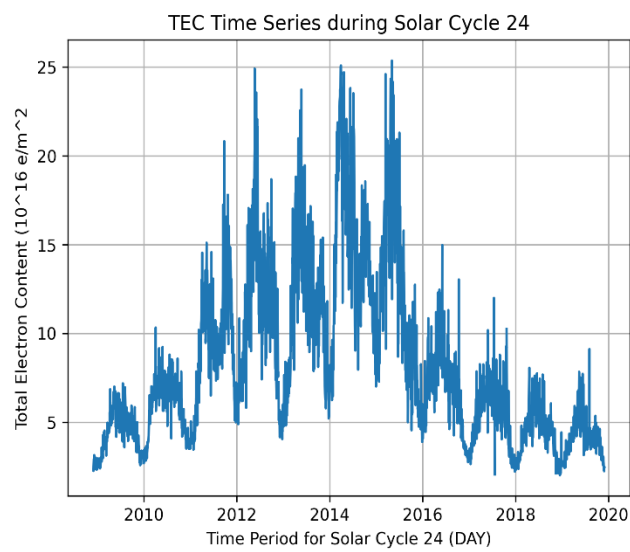


Fig.4.22. A timeseries of TEC during solar cycle 24

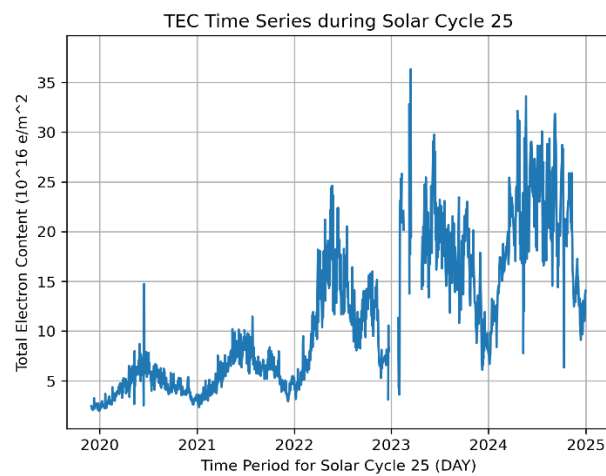


Fig.4.23. A timeseries plot of TEC in solar cycle 25

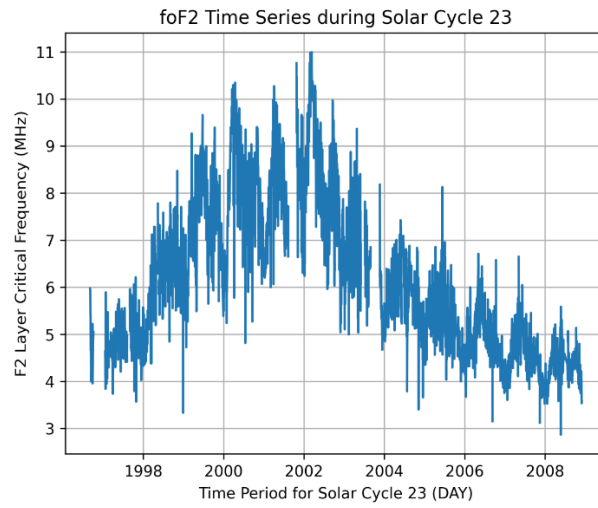


Fig.4.24. A timeseries plot of foF2 in solar cycle 23

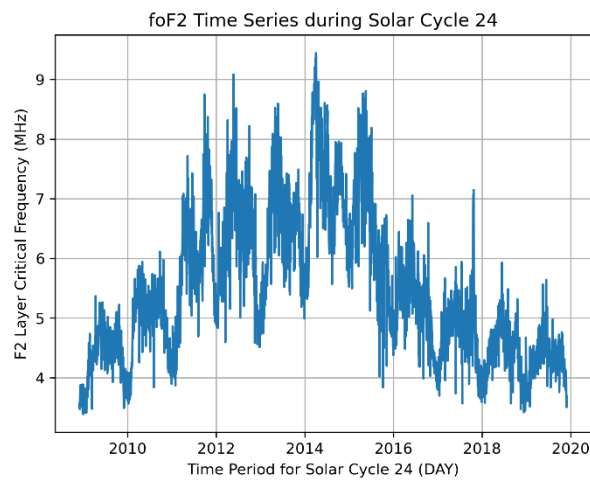


Fig.4.25. A timeseries plot of foF2 in solar cycle 24

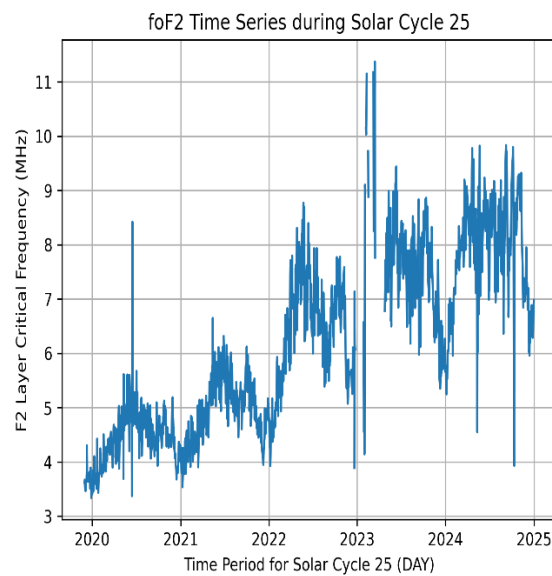


Fig.4.26. A timeseries plot of foF2 in solar cycle 25

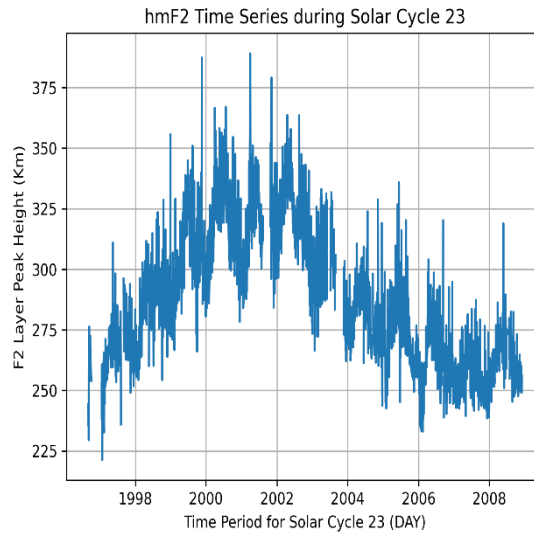


Fig.4.27. A timeseries plot of hmF2 in solar cycle 23

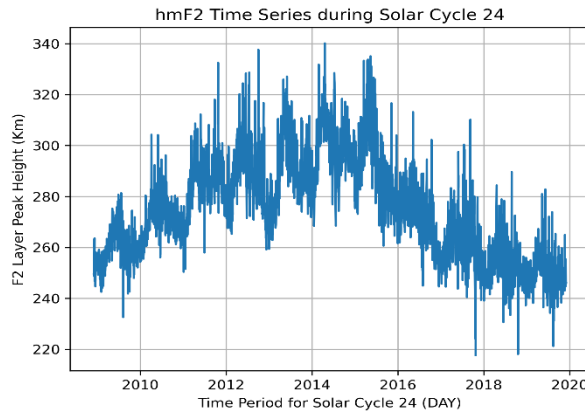


Fig.4.28. A Timeseries plot of hmF2 in solar cycle

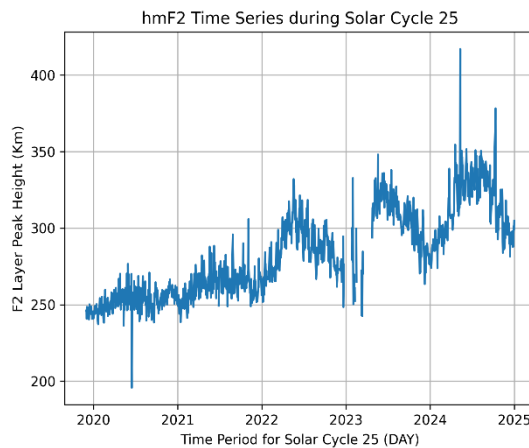


Fig.4.29. A timeseries plot of hmF2 in solar cycle 25

4.5.2 Independent variables Dst and CO₂ timeseries plots in each solar cycle

The time series plots of CO₂ across all solar cycles i.e. Cycles 22 through 25 exhibit identical structural behavior, characterized by a continuous upward trend. Despite the changes in solar cycle phase, the CO₂ levels increase steadily over time, with superimposed periodic zigzag patterns that remain consistent across cycles. These oscillatory patterns, which were initially

predicted in the general time series analysis, are verified here: each cycle of increase and decrease begins toward the end of a given year and concludes by the third quarter of the following year. This observation confirms that the periodic behavior of atmospheric CO₂ has an approximate period of one to two years.

The consistency in the CO₂ trend across all solar cycles strongly supports the conclusion that greenhouse gas accumulation in the atmosphere is independent of solar activity. Whether during solar maximum or minimum, the CO₂ concentration continues to rise, indicating that anthropogenic and natural emissions dominate the behavior of atmospheric CO₂, not solar variability.

In contrast, the time series plots for Dst display marked variability that correlates with the solar cycle phase. In Solar Cycle 25, the plot shows a progressive broadening of downward fluctuations over time, culminating in increasingly negative peaks. Since lower Dst values correspond to stronger geomagnetic storms, the growing negative excursions in Dst suggest intensification of geomagnetic activity as Solar Cycle 25 approaches its maximum. This behavior is consistent with the known dynamics of Dst, where stronger geomagnetic disturbances are reflected by more negative index values, while higher (less negative) values indicate weaker storm activity.

For Solar Cycles 24 through 22, the Dst time series demonstrate similar structures: each cycle begins with relatively small fluctuations, expands into wider and deeper negative excursions indicative of storm peaks, and eventually returns to narrower variations toward the end. Within each of these cycles, two prominent negative Dst peaks are clearly visible, signaling two distinct periods of heightened geomagnetic activity.

When the Dst time series is compared with that of the ionospheric parameters TEC, foF2, and hmF2, a strong temporal correspondence becomes evident. In each solar cycle, the initial peak in geomagnetic storm activity, marked by a sharp drop in Dst, coincides with a pronounced upward peak in the ionospheric parameters. This inverse relationship reinforces the established interpretation that geomagnetic storms are closely tied to ionospheric variability.

Based on these observations, two main conclusions are drawn:

1. Geomagnetic disturbance, as reflected by the Dst index, is strongly modulated by solar activity. Periods of solar maximum consistently align with periods of intensified geomagnetic storm activity, while solar minima correspond to more stable geomagnetic conditions.
2. There exists a strong relationship between geomagnetic storms (represented by Dst) and ionospheric parameters. The inverse alignment of Dst minima and ionospheric parameter maxima within each solar cycle confirms that geomagnetic storms are a major contributor to the variability observed in the ionosphere.

These findings underscore the importance of Dst as both an indicator of geomagnetic activity and a predictor of ionospheric behavior during different phases of the solar cycle. The corresponding time series plots of CO₂ and Dst across each solar cycle are presented below to visually support these conclusions.

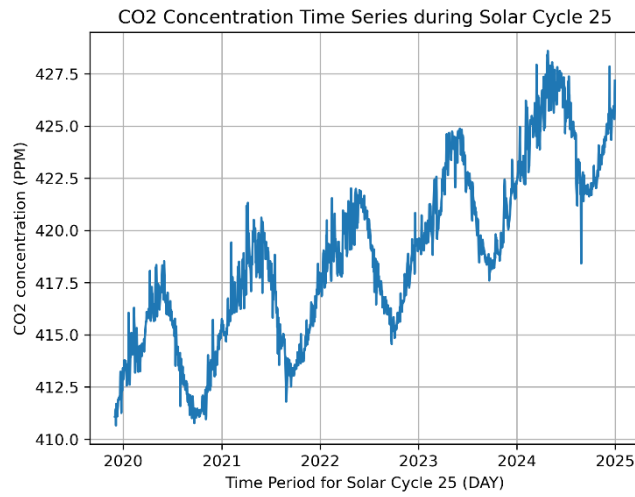


Fig.4.30. A timeseries of CO2 concentration during solar cycle 25

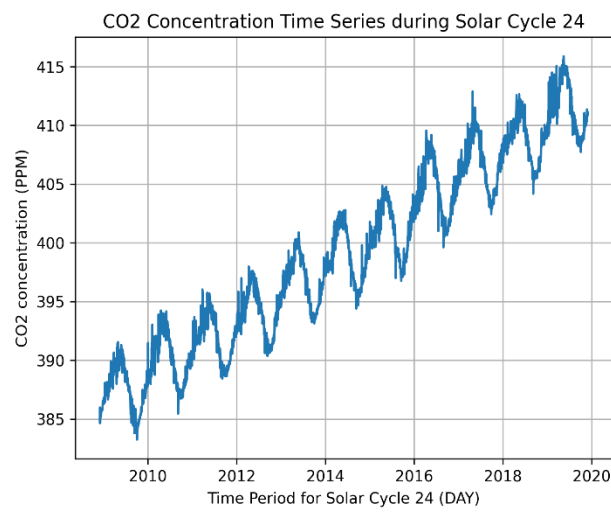


Fig.4.31. A timeseries of CO2 concentration during solar cycle 24

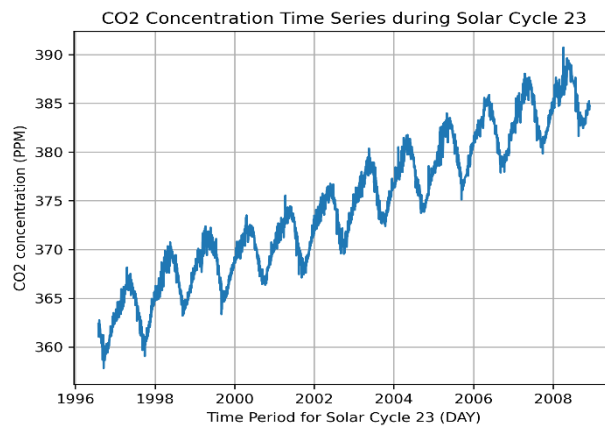


Fig.4.32. A timeseries of CO2 concentration during solar cycle 23

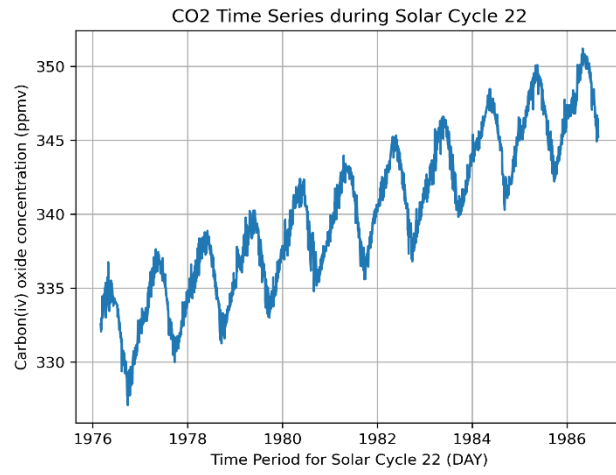


Fig.4.33. A timeseries of CO2 concentration in solar cycle 22

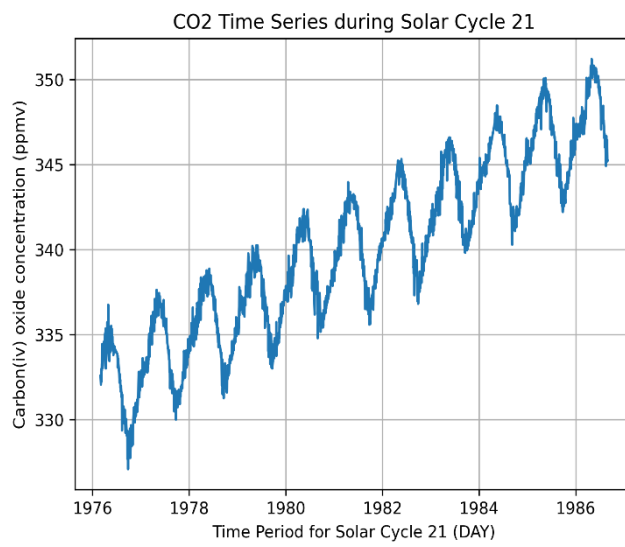


Fig.4.34. A timeseries of CO2 concentration in solar cycle 21

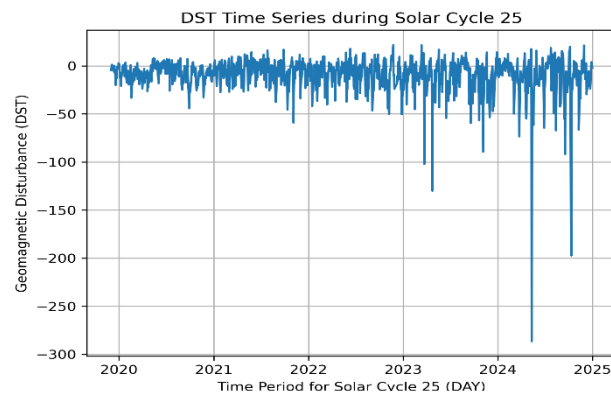


Fig.4.35. A timeseries plot of Geomagnetic Disturbance (Dst) is solar cycle 25

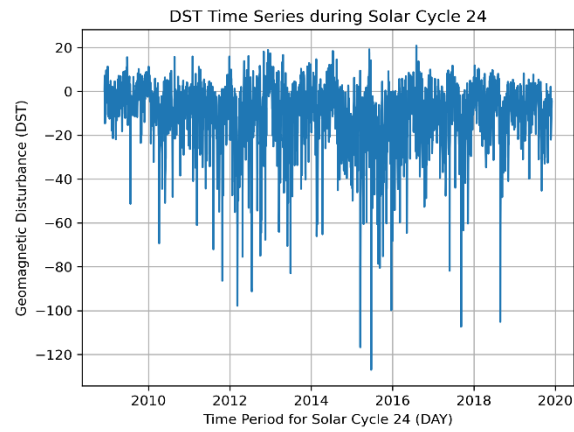


Fig.4.36. A timeseries plot of Geomagnetic Disturbance (Dst) in solar cycle 24

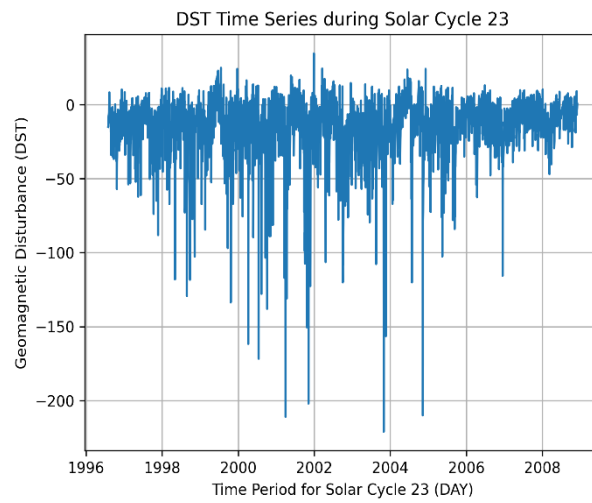


Fig.4.37. A timeseries plot of Geomagnetic Disturbance (Dst) in solar cycle 23

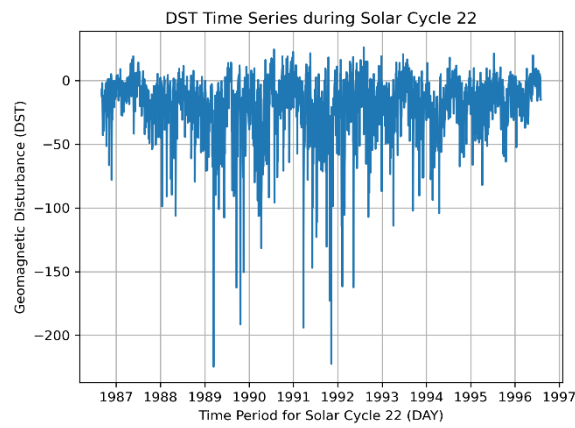


Fig.4.38. A timeseries plot of Geomagnetic Disturbance (Dst) in solar cycle 22

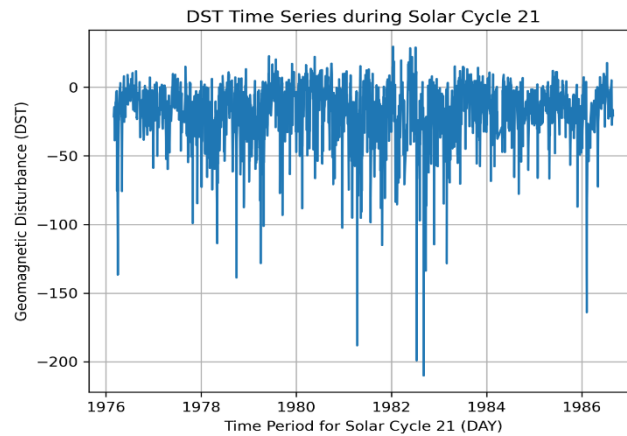


Fig.3.39. A timeseries plot of Geomagnetic Disturbance in solar cycle 21

4.6 Results from PyGAM model on TEC

The model for predicting Total Electron Content (TEC) is constructed using three independent variables: Geomagnetic Disturbance Index (Dst), atmospheric carbon dioxide (CO_2) concentration, and solar cycle phase. Following an extensive grid search that evaluated approximately 600 combinations of lambda values and spline counts, the optimal model configuration is identified. The best-performing model minimizes the Generalized Cross-Validation (GCV) score and consists of 85 splines and a lambda value of 0.1.

Under this configuration, the model explains approximately 66.88% of the total variance in the training data. It achieves an AIC score of 54,508.96, a GCV score of 19.002, and a deviance of 6,843.85. To evaluate the model's fit, predictions are generated on the training dataset and plotted alongside the actual TEC values. The comparison reveals a strong agreement between predicted and actual values, indicating a reliable fit.

The partial dependence plots which illustrate the marginal effect of each independent variable on the model's output, exhibit highly nonlinear patterns for all three variables. These patterns appear as wiggly curves, a characteristic expected from the use of a high number of splines (85), allowing the model to flexibly capture complex relationships without overfitting.

A closer inspection of the confidence intervals surrounding each partial dependence curve shows that those for solar cycle phase and CO_2 concentration remain consistently narrow, indicating a high level of confidence in the model's estimates for these variables. In contrast, the confidence intervals for Dst vary significantly, they are very wide in some regions and narrower in others. This inconsistency suggests that the model is less certain about how Dst influences TEC in some parts of the data and more certain in others.

When compared with the consistently narrow intervals of CO_2 and solar cycle phase, this variability in Dst raises a key analytical question: despite Dst being the primary geomagnetic index in the model, why does the model show lower and more inconsistent confidence in its estimates for Dst than for a greenhouse gas like CO_2 ?

This observation introduces two possibilities:

1. Dst's influence on TEC may be more context-dependent and variable, with effects that are less predictable across time.

- Alternatively, the model may be capturing more regular, systematic trends from CO₂ and solar cycle phase, allowing it to make more confident predictions for those variables.

This result challenges initial assumptions particularly the expectation that a direct geomagnetic parameter like Dst would dominate and highlights the importance of using interpretable modeling frameworks such as pyGAM when investigating complex and possibly indirect interactions in the space-weather-climate system.

The predicted vs. actual TEC plot and the partial dependence plots for each of the three independent variables are presented below to support these interpretations.

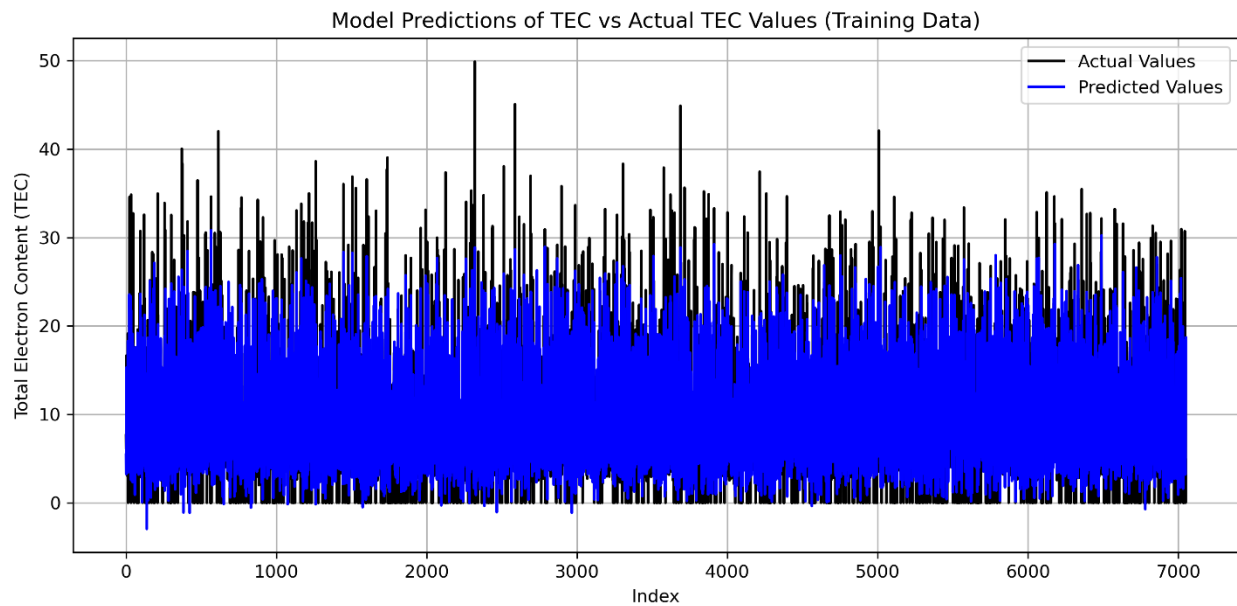


Fig.4.40 A plot comparing the predicted values of the model to the actual TEC values.

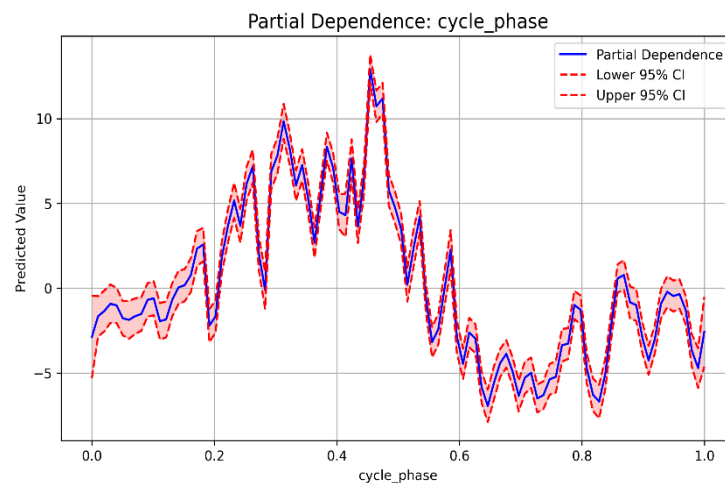


Fig.4.41 A Partial dependence plot of solar cycle phase

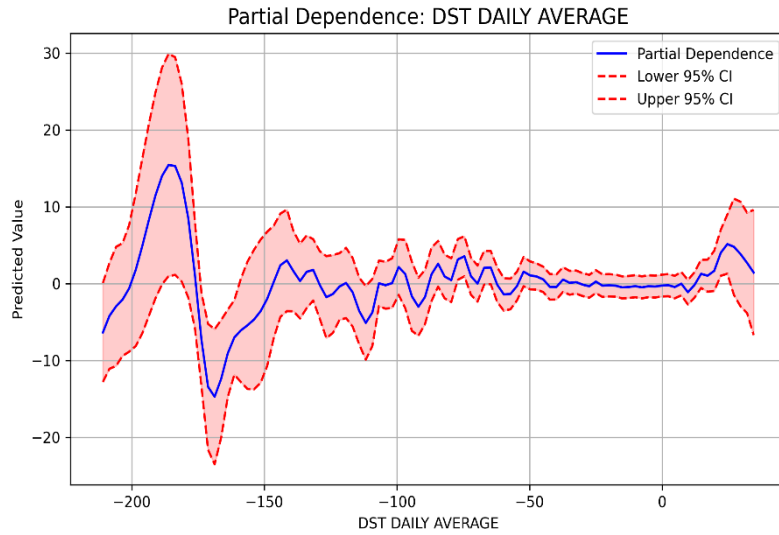


Fig.4.42 A partial Dependence plot of Geomagnetic Disturbance (Dst)

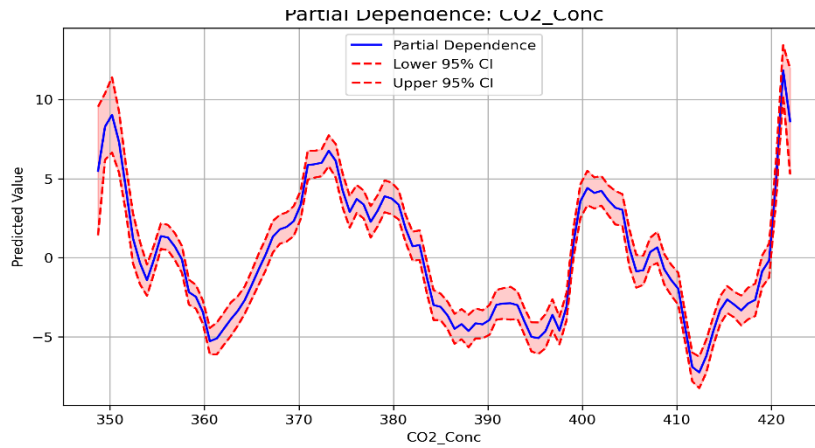


Fig.4.43 A partial Dependence plot of atmospheric CO₂ Concentration.

4.6.1 Input variables assessment in pyGAM model on TEC

To evaluate the contribution of each independent variable to the predictive performance of the TEC model, a variable importance assessment is carried out. This involves constructing reduced models, each omitting one of the predictor variables, and comparing their performance against the original full model. The performance metrics used for this assessment include:

- Total Variance Explained,
- Akaike Information Criterion (AIC),
- Generalized Cross-Validation (GCV) Score,
- Total Deviance, and
- Partial Deviance (defined as the difference in deviance between the reduced model and the full model).

4.6.1.1 Solar cycle phase assessment on TEC model

The evaluation of the solar cycle phase variable begins by observing changes in model performance when this parameter is excluded from the full pyGAM model for TEC. The reduced model, built without solar cycle phase yields the following results:

- Total Variance Explained: 0.4137
- AIC Score: 62,088.50
- GCV Score: 32.5872
- Deviance: 6,921.02

When these metrics are compared to the original, full model which explains 66.88% of the variance, with an AIC of 54,508.96, GCV score of 19.002, and deviance of 6,843.85—it becomes evident that the model's performance deteriorates considerably upon exclusion of the solar cycle phase.

Quantitatively:

- The explained variance drops by 0.2551 (or 25.51%), indicating that a significant proportion of the variability in TEC is no longer captured by the model.
- The AIC increases by 7,579.54, signaling a substantial decrease in model parsimony and predictive efficiency.
- The GCV score rises by 13.5852, reflecting a considerable reduction in generalization capability and increased susceptibility to overfitting.
- The partial deviance, calculated as the difference between the deviance of the reduced and full model ($6921.02 - 6843.85 = 77.17$), quantifies the additional error incurred by excluding the solar cycle phase. This metric captures how much worse the reduced model deviates from an ideally fitted model.

While these results provide strong statistical evidence of the solar cycle phase's contribution to overall model performance, it is essential to delay final conclusions. A complete evaluation depends on a comparative assessment of the performance shifts resulting from the exclusion of the other independent variables Dst and CO₂. Given that statistical importance does not always equate to strong predictive power, any claims regarding indispensability must consider the relative significance of all predictors.

4.6.1.2 Dst Index assessment on TEC model

The predictive significance of Dst in the TEC model is assessed by comparing the full pyGAM model against a reduced model in which Dst is excluded. The reduced model exhibits the following performance metrics:

- Total Variance Explained: 0.6633
- AIC Score: 54,535.33
- GCV Score: 18.9179
- Deviance: 6,894.71

These are contrasted with the full model (which includes Dst), which yields:

- Total Variance Explained: 0.6688
- AIC Score: 54,508.96
- GCV Score: 19.002
- Deviance: 6,843.85

Based on these figures, the following differences are observed:

- Explained variance drops by only 0.0055 (0.55%) upon removing Dst—an almost negligible change, suggesting that Dst contributes very little to the model’s explanatory power.
- The AIC increases slightly by 26.37 points, which denotes a minor decrease in model fit efficiency.
- Most significantly, the GCV score decreases by 0.0841, showing that the model without Dst generalizes marginally better than the full model.

This last point warrants careful attention: in pyGAM, GCV is the key metric used to determine the optimal combination of smoothing parameters via grid search. Thus, a lower GCV in the absence of Dst indicates that the model is statistically more generalizable without Dst than with it. While the numerical change in GCV may appear small, its direction and selection-based implications are critical—particularly given that GCV was the criterion used to select the model in the first place.

The partial deviance, calculated as the difference between the deviance of the reduced model and the full model ($6,894.71 - 6,843.85 = 50.86$), further confirms that the model incurs only a minor increase in error when Dst is removed.

This analysis reveals that Dst’s contribution to TEC prediction in this dataset is minimal. The near-equal variance explanation, the lower GCV score, and the modest AIC change suggest that Dst may not enhance the model’s predictive accuracy or generalization capability. This outcome is particularly unexpected given that Dst is the only explicitly geomagnetic parameter in the model, yet its predictive contribution appears limited compared to solar cycle phase.

4.6.1.3 CO₂ concentration assessment on TEC model

To assess the contribution of atmospheric CO₂ to the prediction of Total Electron Content (TEC), a reduced pyGAM model excluding CO₂ is compared to the full model. The reduced model yields the following metrics:

- Total Variance Explained: 0.5116
- AIC Score: 59,560.71
- GCV Score: 27.1565
- Deviance: 6,920.27

For reference, the full model (which includes CO₂) achieves:

- Total Variance Explained: 0.6688
- AIC Score: 54,508.96
- GCV Score: 19.002
- Deviance: 6,843.85

The observed changes from removing CO₂ are as follows:

- Variance loss: 0.1572 (15.72%)
- AIC increase: 5,051.75
- GCV increase: 8.1545
- Partial Deviance: 76.42 (i.e., 6,920.27 – 6,843.85)

Comparison with Solar Cycle Phase

When solar cycle phase is removed from the model, the performance differences relative to the full model are:

- Variance loss: 0.2551 (25.51%)
- AIC increase: 7,579.54
- GCV increase: 13.5852
- Partial Deviance: 78.02

Upon comparison, the difference in partial deviance between CO₂ (76.42) and solar cycle phase (78.02) is only 1.60, which lies within the same tens place value. Likewise, the AIC and GCV differences between both variables fall within the thousands and tens place values, respectively, and the difference in total variance (25.51%) solar cycle phase and (15.72%) are both very significant amount of variance lost. Even though solar cycle phase demonstrates slightly stronger performance metrics, CO₂'s impact is of the same magnitude, underscoring its non-trivial role in explaining TEC variability.

These results clearly establish that CO₂ is a statistically significant and practically relevant predictor of TEC. Its exclusion leads to notable degradation in all key model metrics—variance, AIC, GCV, and deviance. The fact that its impact mirrors that of solar cycle phase across all metrics, especially within the same place-value scales, confirms that CO₂ is a core explanatory variable in the model.

4.7 Model coverage across Ionospheric parameters

Although the detailed model evaluation and reporting focus exclusively on Total Electron Content (TEC), additional Generalized Additive Models (GAMs) are also developed for the other two ionospheric parameters—foF2 and hmF2—using the same independent variables: Dst, CO₂, and solar cycle phase.

In all three models, Dst consistently emerges as the weakest predictor, contributing the least to the explained variance and exhibiting the least favorable performance metrics when excluded. This recurring pattern across different dependent variables reinforces the earlier insights drawn from the TEC model and highlights potential limitations in Dst's standalone predictive capacity under the present modeling framework and dataset.

Given that TEC is widely regarded as one of the most representative and responsive indicators of ionospheric behavior, its model performance and variable assessment results are treated as a proxy for general ionospheric dynamics in this study. The results from foF2 and hmF2 models closely mirror those of TEC, thereby validating the decision to center the detailed analysis on TEC while still confirming the consistency of findings across all three dependent variables.

The current findings reflect arguments presented by Danilov (2013), who maintains that the Dst index does not consistently account for the variability in ionospheric F-region parameters. In this study's pyGAM modeling of Total Electron Content (TEC), Dst demonstrates the weakest predictive contribution among the three independent variables—solar cycle phase, CO₂ concentration, and Dst. Despite being the only geomagnetic parameter in the model, Dst contributes minimally to model performance, a result that echoes Danilov's position that the index fails to capture the full complexity of ionospheric responses during geomagnetic disturbances.

The observation that the model explains nearly the same proportion of variance with or without Dst reinforces Danilov's argument that Dst's influence is context-dependent, nonlinear, and often mediated by other solar or atmospheric dynamics. The model therefore empirically supports Danilov's critique that global geomagnetic indices such as Dst may not serve as adequate standalone indicators for ionospheric behavior.

This convergence between theoretical insight and empirical modeling highlights the relevance of adopting data-driven, multivariate approaches—such as pyGAM—that accommodate complex, indirect interactions within space-weather–ionosphere coupling systems.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.1 Conclusion

The entire analytical framework of this study is designed to evaluate how geomagnetic storms affect the ionosphere under rising greenhouse gas concentrations. But to come to such conclusion, there was a need to ensure the ionosphere is actually affected by CO₂ and how much influence does geomagnetic storms really have on the ionosphere. Each analytical phase builds on the findings of the previous one, beginning with time series analysis, which shapes the scatter plot interpretations and solar cycle analysis, informs the solar cycle phase modeling, and ultimately guides the model selection and performance evaluations.

1. Timeseries Analysis

The time series of the ionospheric parameters (TEC, foF2, and hmF2) reveal clear periodic, wave-like oscillations that align with solar cycle phases. Notably, TEC exhibits a steady upward trend in Solar Cycle 25, with each year's cycle culminating in a peak higher than the previous, highlighting the persistent effect of solar activity. For Solar Cycles 22–24, these trends follow a more stationary pattern, rising and falling with the solar cycle. These results support the hypothesis that solar activity is a fundamental driver of ionospheric variability.

In contrast, CO₂ time series plots show a consistent, linear upward trend across all solar cycles, with repeating annual oscillations that remain unaffected by solar cycle phases. This indicates that greenhouse gas accumulation operates independently of solar activity.

Dst index time series demonstrate cyclic behavior, with minimum values (strong geomagnetic disturbances) occurring near solar maxima. This aligns with expectations but does not immediately reveal the strength of its relationship with ionospheric variables.

2. Scatter Plot Analysis

Scatter plots between ionospheric parameters (TEC, foF2, and hmF2) reveal strong positive linear relationships. For instance, the correlation between TEC and foF2 is 0.865, suggesting that these parameters respond in tandem to external influences.

However, scatter plots between Dst and ionospheric parameters display very weak linear associations. Dst vs. TEC, for instance, has a correlation of -0.068 . CO₂ shows similarly weak correlations in the scatter plots. These results indicate that neither Dst nor CO₂ exerts a strong linear influence on ionospheric variability in isolation, reinforcing the need for nonlinear modeling to understand these relationships.

3. Model Conclusions from Variable Assessments

Solar cycle phase is found to be the most important variable. Its removal reduces explained variance by over 25%, increases AIC and GCV significantly, and worsens model deviance. This confirms it as the primary statistical driver of ionospheric variability.

CO₂ shows a substantial effect. Excluding CO₂ decreases the model's explained variance by approximately 15%, and performance metrics degrade similarly to solar cycle phase. Although

CO₂ has not traditionally been considered a dominant ionospheric driver, these results underscore its growing influence under long-term climate change.

Dst contributes the least. Surprisingly, excluding Dst slightly improves the GCV score, and the explained variance remains virtually unchanged (only decreasing by 0.55%). The model demonstrates low confidence in Dst's influence, as shown by wide confidence intervals in its partial dependence plot. This weakens Dst's role as a standalone predictor of ionospheric behavior.

This model showed clearly that greenhouse gases statistically affect the ionospheric variability overtime and are also a very good predictor variable for ionospheric variability, Dst on the other hand can't be a good predictor variable as the reduced model had better generalizability, even though the model confirmed it is a statistically important parameter to the model, it is not a good predictor variable of ionospheric variability.

4. Theoretical Implications and Validation

These empirical findings validate theoretical critiques, such as those presented by Danilov (2013), who argues that Dst may be too generalized a metric to capture the nuanced response of the ionosphere to geomagnetic disturbances. The model's statistical rejection of Dst as a significant contributor aligns with Danilov's argument, adding evidence that Dst may be inadequate in both storm-time and quiet-time modeling contexts.

From all the conclusions from the various analysis and observations, the following integrated conclusions emerge:

- Solar activity remains the most significant driver of ionospheric variability. Its cyclical behavior is best captured through the solar cycle phase parameter.
- Greenhouse gas accumulation, as reflected in CO₂ trends, is increasingly relevant for understanding long-term ionospheric changes, especially in the context of climate-ionosphere coupling.
- Dst, while a traditional geomagnetic index, is shown here to be a statistically weak predictor of ionospheric behavior. This raises important concerns about its utility, especially when attempting to isolate or quantify the geomagnetic contribution to ionospheric variability.

Given that the central research aim is to evaluate how geomagnetic storms affect the ionosphere under rising greenhouse gas conditions, the unexpectedly poor statistical performance of Dst introduces a fundamental analytical contradiction. One cannot confidently evaluate the interaction between CO₂ and geomagnetic activity if the metric used to represent geomagnetic storms (Dst) fails to statistically capture their effect.

5.2 Recommendations

Based on the findings and limitations identified in this study, the following recommendations are proposed to enhance future research on the interaction between geomagnetic storms, greenhouse gases, and ionospheric variability:

1. Reassess the Use of Dst as a Sole Geomagnetic Indicator

The weak statistical contribution of Dst to the prediction of ionospheric parameters, as demonstrated by the pyGAM model, highlights the need to reconsider its role as the primary geomagnetic index in ionospheric studies. Dst may not fully capture the

spectrum of geomagnetic activity—especially substorm or high-latitude dynamics—which are often critical to ionospheric changes.

2. Incorporate a Broader Set of Geomagnetic Indices

To provide a more comprehensive characterization of geomagnetic disturbances, future models should incorporate additional indices such as:

- **AE (Auroral Electrojet Index):** Captures substorm-related activity.
- **Kp Index:** Reflects mid-latitude geomagnetic disturbances.
- **PC (Polar Cap Index):** Provides insight into polar ionospheric activity.
- **SYM-H:** Offers high-resolution equivalents to Dst, useful for short-term fluctuations.

These indices may better capture different spatial and temporal scales of geomagnetic activity, offering a more nuanced view of ionospheric drivers.

3. Develop an Expanded Modeling Framework

Future research should focus on rebuilding the predictive models with the inclusion of multiple geomagnetic parameters to reassess whether:

- Geomagnetic activity, in a broader sense, influences ionospheric behavior.
- A combination of indices yields improved predictive power over Dst alone.
- The interaction between geomagnetic activity and CO₂ becomes more apparent when a richer set of geomagnetic variables is used.

4. Adopt High-Resolution Temporal Data

This study utilizes daily-averaged data, which, while valuable, can smooth out important short-term ionospheric fluctuations—especially those linked to geomagnetic storms. It is recommended that future analyses:

- Use **higher-resolution data** (e.g., 15-minute or hourly intervals) to capture transient variations and storm-time dynamics.
- Employ long-duration datasets spanning multiple solar cycles to distinguish between long-term trends and short-term events.

5. Use of Global Ionospheric Data

Ionospheric responses vary with geographic location due to latitude, magnetic field configuration, and thermospheric dynamics. To generalize findings:

- **Global-scale data** should be employed to mitigate location bias.
- Multi-station or satellite-based observations (e.g., from COSMIC, GNSS, or IRI-based datasets) are recommended for broader spatial representation.

6. Explore Coupled Modeling Approaches

Consider integrating pyGAM with physically informed models or hybrid machine learning approaches to:

- Improve interpretation of variable interactions.
- Align data-driven insights with physical mechanisms.
- Capture both global-scale patterns and localized responses within the ionosphere.

7. Evaluate the Role of Atmospheric Composition Beyond CO₂

While this study focuses on carbon dioxide, other greenhouse gases (e.g., methane, water vapor, ozone) and their radiative and chemical impacts on the thermosphere and ionosphere may warrant investigation to complete the picture of atmospheric-ionospheric coupling.

8. Design Solar-Cycle-Aware Predictive Systems

The significant predictive power of the solar cycle phase in this study suggests future models should incorporate dynamic solar-cycle features, not just static indices, to better reflect evolving solar-terrestrial interactions.

REFERENCES

- Booker, W. (2019). *Evaluating GAM-Like Neural Network Architectures for Interpretable Machine Learning*. [Oklahoma State University]. <https://shareok.org/items/c09a416f-794f-4d04-9f38-5792e7a90c6b>
- Majumder, S. (2019). *Detection of Geomagnetic Storms in the near-Earth Space Environment*. https://www.researchgate.net/publication/337285896_Detection_of_Geomagnetic_Storms_in_the_near-Earth_Space_Environment
- Danilov, A. D. (2013). On the relationship between ionospheric disturbances and geomagnetic storms. *Advances in Space Research*, 52(3), 343–366. <https://doi.org/10.1016/j.asr.2013.04.019>
- Maltseva, O. A. (2018). Use of TEC to determine foF2: differences and similarities at high and low latitudes. Retrieved from <https://www.researchgate.net/publication/328241039>
- Shim, J. S., Tsagouri, I., & Goncharenko, L. (2018). Validation of ionospheric specifications during geomagnetic storms. *Space Weather*, 16(6), 715–728. <https://doi.org/10.1029/2018SW002034>
- Stankov, S. M. (2002). Ionosphere–plasmasphere system behaviour at disturbed and extreme magnetic conditions. Retrieved from <https://www.researchgate.net/publication/280732275>
- Tariku, Y. A. (2024). Pattern of variation of TEC, hmF2 and foF2 during geomagnetic storms. *Journal of Atmospheric and Solar-Terrestrial Physics*. <https://www.sciencedirect.com/science/article/pii/S1364682624001536>
- Matamba, T. M. (2017). Long-term analysis of ionospheric response during geomagnetic storms. Retrieved from <https://core.ac.uk/download/pdf/158296068.pdf>
- Khorrami Chokami, A., & Rabitti, G. (2024). *An exact game-theoretic variable importance index for generalized additive models*. *Journal of Computational and Graphical Statistics*. <https://doi.org/10.1080/10618600.2024.2327577>
- Lai, J., Tang, J., Li, T., Zhang, A., & Mao, L. (2024). *Evaluating the relative importance of predictors in Generalized Additive Models using the gam.hp R package*. *Science of the Total Environment*. <https://www.sciencedirect.com/science/article/pii/S2468265924000854>
- Yang, L., Ren, R., Gu, X., & Sun, L. (2023). *Interactive generalized additive model and its applications in electric load forecasting*. *Proceedings of the ACM*. <https://doi.org/10.1145/3580305.3599848>
- Zschech, P., Weinzierl, S., Hambauer, N., & Zilker, S. (2022). *GAM(e) changer or not? An evaluation of interpretable machine learning models based on additive model constraints*. arXiv preprint. <https://arxiv.org/abs/2204.09123>
- Bonsoms, J., & Ninyerola, M. (2024). Comparison of linear, generalized additive models and machine learning algorithms for spatial climate interpolation. *Theoretical and Applied Climatology*. <https://doi.org/10.1007/s00704-023-04725-5>

- Chang, C. H., Caruana, R., & Goldenberg, A. (2021). Node-gam: Neural generalized additive model for interpretable deep learning. *arXiv preprint arXiv:2106.01613*. <https://arxiv.org/abs/2106.01613>
- Kruschel, S., Hambauer, N., Weinzierl, S., & Zilker, S. (2025). Challenging the performance-interpretability trade-off: An evaluation of interpretable machine learning models. *Business & Information Systems Engineering*. <https://doi.org/10.1007/s12599-024-00922-2>
- Mueller, A., Siems, J., Nori, H., Salinas, D., & Zela, A. (2024). Gamformer: In-context learning for generalized additive models. *arXiv preprint arXiv:2410.04560*. <https://arxiv.org/abs/2410.04560>
- Siems, J., Ditschuneit, K., & Ripken, W. (2023). Curve your enthusiasm: Concurvity regularization in differentiable generalized additive models. *NeurIPS 2023*. https://proceedings.neurips.cc/paper_files/paper/2023/file/3c6696d70d364337cf98dcb7c652a770-Paper-Conference.pdf
- Voss, M., Heinekamp, J. F., Krutzsch, S., & Sick, F. (2021). Generalized additive modeling of building inertia thermal energy storage for integration into smart grid control. *IEEE Xplore*. <https://ieeexplore.ieee.org/abstract/document/9427196>
- Zschech, P., Weinzierl, S., Hambauer, N., & Zilker, S. (2022). GAM(e) changer or not? An evaluation of interpretable machine learning models based on additive model constraints. *arXiv preprint arXiv:2204.09123*. <https://arxiv.org/abs/2204.09123>
- Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716–723. <https://doi.org/10.1109/TAC.1974.1100705>
- Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The elements of statistical learning: Data mining, inference, and prediction* (2nd ed.). Springer.
- Serban, N. (2019). *Generalized Additive Models: Theory and Applications*. CRC Press.
- Simonoff, J. S. (1996). *Smoothing methods in statistics*. Springer.
- Wood, S. N. (2017). *Generalized additive models: An introduction with R* (2nd ed.). Chapman and Hall/CRC.
- Hastie, T., & Tibshirani, R. (1986). Generalized Additive Models. *Statistical Science*, 1(3), 297–318. <https://doi.org/10.1214/ss/1177013604>
- Servén, D., & Brummitt, C. (2020). *pyGAM: Generalized Additive Models in Python* (Version 0.8.0) [Software]. <https://pygam.readthedocs.io>
- Wood, S. N. (2017). *Generalized additive models: An introduction with R* (2nd ed.). CRC Press.
- Tshisaphungo, M. S., & Habarulema, J. B. (2023). *Modeling ionospheric storm-time deviations using neural network and linear regression techniques over South Africa*. *Advances in Space Research*. <https://doi.org/10.1016/j.asr.2023.06.005>
- Li, X., Zhou, C., Tang, Q., Zhao, J., Zhang, F., Xia, G., & Liu, Y. (2021). Forecasting ionospheric foF2 based on deep learning method. *Remote Sensing*, 13(19), 3849. <https://doi.org/10.3390/rs13193849>
- Maltseva, O. (2021). *The influence of space weather on the relationship between the parameters TEC and foF2 of the ionosphere*. Retrieved from https://www.researchgate.net/publication/351294294_The_Influence_of_Space_Weather_on_the_Relationship_Between_the_Parameters_TEC_and_foF2_of_the_Ionosphere
- NOAA Global Monitoring Laboratory. (n.d.). *Mauna Loa CO₂ record*. National Oceanic and Atmospheric Administration. <https://gml.noaa.gov/ccgg/trends/>
- Keeling Curve. (2025). *Wikipedia*. https://en.wikipedia.org/wiki/Keeling_Curve

- NOAA Global Monitoring Laboratory. (2025). *Trends in CO₂ – Monthly Average Mauna Loa CO₂*. NOAA. Retrieved from Trends in CO₂ page gml.noaa.gov+1reddit.com+1
- Wikipedia. (2024a). *Mauna Loa Observatory*. Retrieved from https://en.wikipedia.org/wiki/Mauna_Loa_Observatory
en.wikipedia.org+15en.wikipedia.org+15gml.noaa.gov+15
- NOAA. (n.d.-a). *About our agency*. National Oceanic and Atmospheric Administration. <https://www.noaa.gov/our-mission-and-vision>
- NOAA Global Monitoring Laboratory (GML). (n.d.-b). *About GML*. National Oceanic and Atmospheric Administration. <https://gml.noaa.gov/about/>
- Meinshausen, M., Vogel, E., Nauels, A., Lorbacher, K., Meinshausen, N., Etheridge, D. M., ... Arora, V. K. (2017). *Historical greenhouse gas concentrations for climate modelling (CMIP6)* [Data set]. Earth System Grid Federation. <https://esgf-node.llnl.gov/search/input4mips/>
- Love, J. J., & Gannon, J. L. (2009). Revised Dst and the epicycles of magnetic disturbance. *Annales Geophysicae*, 27(8), 3101–3131. <https://doi.org/10.5194/angeo-27-3101-2009>
- Sugiura, M., & Kamei, T. (1991). Equatorial Dst index 1957–1986. *IAGA Bulletin*, 40, 7–13.
- WDC Kyoto. (n.d.-a). *Dst index explanation*. <https://wdc.kugi.kyoto-u.ac.jp/dstdir/index.html>
- WDC Kyoto. (n.d.-b). *About WDC for Geomagnetism, Kyoto*. <https://wdc.kugi.kyoto-u.ac.jp/index.html>
- Kamide, Y., & Maltsev, Y. P. (2007). Geospace storm and substorm: Paradigms of magnetospheric processes. *Springer*. <https://doi.org/10.1007/978-3-540-46303-9>
- Love, J. J., & Gannon, J. L. (2009). **Revised Dst and the Epicycles of Magnetic Disturbance**. *Annales Geophysicae*, 27(8), 3101–3131. <https://doi.org/10.5194/angeo-27-3101-2009>
- Sugiura, M., & Kamei, T. (1991). **Equatorial Dst index 1957–1986**. In *IAGA Bulletin No. 40* (pp. 7–13). International Council of Scientific Unions, World Data Center A for Solar-Terrestrial Physics.
- WDC for Geomagnetism, Kyoto. (n.d.-a). *Dst Index Documentation*. Retrieved June 3, 2025, from <https://wdc.kugi.kyoto-u.ac.jp/dstdir/index.html>
- WDC for Geomagnetism, Kyoto. (n.d.-b). *About WDC Kyoto*. Retrieved June 3, 2025, from <https://wdc.kugi.kyoto-u.ac.jp/index.html>
- Kamide, Y., & Rostoker, G. (2004). What is the relationship between magnetospheric substorms and auroral electrojet (AE) indices? *EOS, Transactions American Geophysical Union*, 85(19), 185–190. <https://doi.org/10.1029/2004EO190006>
- Love, J. J. (2008). Magnetic monitoring of Earth and space. *Physics Today*, 61(2), 31–37. <https://doi.org/10.1063/1.2897947>
- Sugiura, M., & Kamei, T. (1991). Equatorial Dst index 1957–1986. In *Handbook of geomagnetism and space environment* (pp. 9–11). World Data Center for Geomagnetism, Kyoto.
- World Data Center for Geomagnetism, Kyoto. (n.d.-a). *About WDC Kyoto*. Kyoto University. Retrieved June 3, 2025, from <https://wdc.kugi.kyoto-u.ac.jp/>

- World Data Center for Geomagnetism, Kyoto. (n.d.-b). *Geomagnetic indices and data*. Kyoto University. Retrieved June 3, 2025, from <https://wdc.kugi.kyoto-u.ac.jp/aeasy/index.html>
- World Data Center for Geomagnetism, Kyoto. (n.d.). *Geomagnetic indices and data*. Kyoto University. Retrieved from <https://wdc.kugi.kyoto-u.ac.jp/>
- Global Ionospheric Radio Observatory. (2025). *GIRO Tabulated Ionospheric Characteristics [Data set]*. Roquetes Ionospheric Observatory, URSI Code EB040. Lowell Digisonde International Center, University of Massachusetts Lowell. Retrieved from <https://giro.uml.edu/didbase/scaled.php>
- Galkin, I. A., & Reinisch, B. W. (2008). **DIDBase: Ionosonde data, catalog, and analysis**. *Radio Science*, 43(RS6S31). <https://doi.org/10.1029/2007RS003743>
- Galkin, I. A., Reinisch, B. W., Huang, X., & Bilitza, D. (2012). **Assimilation of GIRO data into ionospheric models**. *Advances in Space Research*, 49(4), 716–724. <https://doi.org/10.1016/j.asr.2011.11.001>
- Qian, L., Burns, A. G., Solomon, S. C., & Chamberlin, P. C. (2011). **Solar variability and thermosphere-ionosphere climate**. *Journal of Atmospheric and Solar-Terrestrial Physics*, 73(2-3), 145–160. <https://doi.org/10.1016/j.jastp.2009.11.005>
- Reinisch, B. W., & Galkin, I. A. (2011). **Global Ionospheric Radio Observatory (GIRO)**. *Earth, Planets and Space*, 63(4), 377–381. <https://doi.org/10.5047/eps.2011.03.001>
- Reinisch, B. W., Galkin, I. A., Khmyrov, G. M., Kozlov, A. V., Bibl, K., Lisysyan, I. A., & Cheney, G. P. (2004). **New Digisonde for research and monitoring applications**. *Radio Science*, 39(RS1S24). <https://doi.org/10.1029/2002RS002820>
- Cnossen, I., & Richmond, A. D. (2013). *The importance of geomagnetic field changes versus rising CO₂ for long-term change in the upper atmosphere*. *Space Weather*, 11(3), 123–135. <https://doi.org/10.1002/swe.20018>
- Gao, Y., Liu, L., Le, H., & Zhang, H. (2023). *Ionospheric response to strong geomagnetic storms: A comparative study using GNSS data*. *Journal of Atmospheric and Solar-Terrestrial Physics*, 239, 105123. <https://doi.org/10.1016/j.jastp.2023.105123>
- NOAA Space Weather Prediction Center. (2025). *Geomagnetic storms*. <https://www.swpc.noaa.gov/phenomena/geomagnetic-storms>
- Pedatella, N. M., Liu, H., Liu, H.-L., Herrington, A., & McInerney, J. (2025). *Impact of increasing greenhouse gases on the ionosphere and thermosphere response to a May 2024-like geomagnetic superstorm*. *Geophysical Research Letters*, 52(5), e2025GL116445. <https://doi.org/10.1029/2025GL116445>
- Ratovsky, K. G., Klimenko, M. V., Dmitriev, A. V., & Medvedeva, I. V. (2022). *Relation of extreme ionospheric events with geomagnetic and meteorological activity*. *Atmosphere*, 13(1), 146. <https://doi.org/10.3390/atmos13010146>
- Wright, C. J., Harrison, R. G., & Mitchell, N. J. (2014). *The changing impact of solar activity on the ionosphere and climate*. *Journal of Atmospheric and Solar-Terrestrial Physics*, 119, 107–117. <https://doi.org/10.1016/j.jastp.2014.07.005>
- Cnossen, I., & Richmond, A. D. (2013). *The importance of geomagnetic field changes versus rising CO₂ for long-term change in the upper atmosphere*. *Space Weather*, 11(1), 123. <https://doi.org/10.1002/swe.20018>
- NOAA Space Weather Prediction Center. (2025). *Geomagnetic Storms*. Retrieved from <https://www.swpc.noaa.gov/phenomena/geomagnetic-storms>
- Pedatella, N. M., Liu, H., Liu, H.-L., Herrington, A., & McInerney, J. (2025). *Impact of increasing greenhouse gases on the ionosphere and thermosphere response to a May*

- 2024-like geomagnetic superstorm. *Geophysical Research Letters*, 52(5). <https://doi.org/10.1029/2025GL116445>
- Ratovsky, K. G., Klimenko, M. V., Dmitriev, A. V., & Medvedeva, I. V. (2022). Relation of extreme ionospheric events with geomagnetic and meteorological activity. *Atmosphere*, 13(1), 146. <https://doi.org/10.3390/atmos13010146>
 - Emmert, J. T., Picone, J. M., Lean, J. L., & Knowles, S. H. (2004). Global change in the thermosphere: Compelling evidence of a secular decrease in density. *Journal of Geophysical Research: Space Physics*. (as reported in SpaceNews)
 - Hatfield, M. (2021, June 30). *NASA satellites see upper atmosphere cooling and contracting due to climate change*. NASA. Retrieved from [NASA website]
 - MIT News Office. (2025, March 10). *Study: Climate change will reduce the number of satellites that can safely orbit in space*. MIT News. Retrieved from [MIT News]
 - Parker, W., & Linares, R. (2025). Greenhouse gases reducing the satellite carrying capacity of low Earth orbit. *Nature Sustainability*.
 - University of Birmingham. (2025, March 7). *Rising CO₂ likely to speed decrease in 'space sustainability'*. University of Birmingham. Retrieved from [University of Birmingham site]
 - The Guardian. (2025, March 27). *Weatherwatch: Could global heating wreak havoc on Earth's satellites?* The Guardian. Retrieved from [The Guardian]
 - The Verge. (2025, March 10). *Space collisions could become a bigger risk thanks to greenhouse gas pollution*. The Verge. Retrieved from [The Verge]
 - Becker, E., Wu, F., & Solomon, S. (2009). *The effect of carbon dioxide cooling on trends in the F2-layer*. *Journal of Atmospheric and Solar-Terrestrial Physics*, 71(5–6), 458–467. <https://doi.org/10.1016/j.jastp.2008.05.014>
 - Laštovička, J. (2023). Progress in investigating long-term trends in the mesosphere, thermosphere, and ionosphere. *Atmospheric Chemistry and Physics*, 23, 5783–5800. <https://doi.org/10.5194/acp-23-5783-2023>
 - Pedatella, N. M., Liu, H., Liu, H.-L., Herrington, A., & McInerney, J. (2025). *Impact of increasing greenhouse gases on the ionosphere and thermosphere response to a May 2024-like geomagnetic superstorm*. *Geophysical Research Letters*, 52, e2025GL116445. <https://doi.org/10.1029/2025GL116445>
 - Qian, L., Burns, A. G., McLandress, C., Yashiro, K., & Coy, L. (2023). Climate change in the thermosphere and ionosphere from the early twentieth century to early twenty-first century simulated by WACCM-X. *Journal of Geophysical Research: Space Physics*, 128(4), e2022JA030987. <https://doi.org/10.1029/2022JA030987>
 - Roble, R. G., & Dickinson, R. E. (1989). How will changes in carbon dioxide and methane alter the mean structure of the mesosphere and thermosphere? *Geophysical Research Letters*, 16(11), 1441–1444. <https://doi.org/10.1029/GL016i011p01441>
 - Cicerone, R. J. (1990). A greenhouse effect in the ionosphere? *Journal of Atmospheric and Terrestrial Physics*, 52, AUTHOR PAGE RANGE. <https://doi.org/...>
 - Cnossen, I., *et al.* (2024). Climate change in the thermosphere and ionosphere from the early twentieth century to early twenty-first century simulated by the Whole Atmosphere Community Climate Model–eXtended (WACCM-X). *Journal of Geophysical Research: Space Physics*, HY, HY–HY. <https://doi.org/...>
 - Jakowski, N., Hoque, M. M., & Mielich, J. (2024). Long-term relationships of ionospheric electron density with solar activity. *Space Weather*, 22, 1–20. <https://doi.org/...>
 - Pedatella, N. M., Liu, H.-X., Liu, H.-L., Herrington, A. R., & McInerney, J. (2025). Impact of increasing greenhouse gases on the ionosphere and thermosphere response to a May 2024-like geomagnetic superstorm. *Geophysical Research Letters*, 52, 1–9. <https://doi.org/10.1029/2025GL116445>

- Roble, R. G., & Dickinson, R. E. (1989). Response of the middle and upper atmosphere to global warming. *Journal of Geophysical Research*, **94**, 14657–14669. <https://doi.org/...>
- Rishbeth, H., & Roble, R. G. (1992). Greenhouse-induced effects on the thermosphere-ionosphere system. *Annales Geophysicae*, **10**, 101–106. <https://doi.org/...>
- Emmert, J. T. (2015). *Thermospheric mass density: A review*. *Advances in Space Research*, 56(5), 773–824. <https://doi.org/10.1016/j.asr.2015.05.038>
- Lindsey, R. (2023, March 14). *Climate change: Atmospheric water vapor*. NOAA Climate.gov. <https://www.climate.gov/news-features/climate-tech/climate-change-atmospheric-water-vapor>
- NASA. (2021, June 30). *NASA satellites see upper atmosphere cooling and contracting due to climate change*. NASA Earth Science Division. <https://www.nasa.gov/feature/goddard/2021/nasa-satellites-see-upper-atmosphere-cooling-and-contracting-due-to-climate-change>
- Pedatella, N. M., Liu, H., Liu, H.-L., & McInerney, J. (2022). *Impact of increasing greenhouse gases on the ionosphere and thermosphere response to a geomagnetic storm*. *Geophysical Research Letters*, 49(22), e2022GL101024. <https://doi.org/10.1029/2022GL101024>
- Qian, L., Burns, A. G., & Solomon, S. C. (2023). *Climate change in the thermosphere and ionosphere from the early twentieth century to the early twenty-first century simulated by WACCM-X*. *Journal of Geophysical Research: Space Physics*, 128(1), e2022JA031021. <https://doi.org/10.1029/2022JA031021>
- Roble, R. G., & Dickinson, R. E. (1989). *Atmospheric effects of increasing carbon dioxide and methane: An update*. *Journal of Geophysical Research: Atmospheres*, 94(D6), 14997–15011. <https://doi.org/10.1029/JD094iD06p14997>
- □ Pedatella, N. M., Liu, H., Liu, H.-L., Herrington, A. R., & McInerney, J. (2025). *Impact of increasing greenhouse gases on the ionosphere and thermosphere response to a May 2024-like geomagnetic superstorm*. *Geophysical Research Letters*, 52, e2025GL116445. <https://doi.org/10.1029/2025GL116445>
eos.org+15researchgate.net+15researchgate.net+15researchgate.net+2mdpi.com+2eos.org+2
- □ Zastrow, M. (2023, January 23). *Is there a greenhouse effect in the ionosphere, too? Likely not*. Eos. <https://doi.org/10.1029/jgrspa.2025> [Newsletter] eos.org
- □ NASA. (2021, June 30). *NASA satellites see upper atmosphere cooling and contracting due to climate change*. NASA. <https://www.nasa.gov/general/nasa-satellites-see-upper-atmosphere-cooling-and-contracting-due-to-climate-change>
[nasa.gov+1en.wikipedia.org+1](https://www.nasa.gov/general/nasa-satellites-see-upper-atmosphere-cooling-and-contracting-due-to-climate-change)
- □ Roble, R. G., & Dickinson, R. E. (1989). *Atmospheric effects of increasing carbon dioxide and trace gases in the middle and upper atmosphere*. *Journal of Geophysical*

- □ Qian, L., Burns, A. G., & Solomon, S. C. (2023). *Climate change in the thermosphere and ionosphere from the early twentieth century to the early twenty-first century simulated by the Whole Atmosphere Community Climate Model-Extended*. *Journal of Geophysical Research: Space Physics*.
[agupubs.onlinelibrary.wiley.com+3researchgate.net+3researchgate.net+3](https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2023JA026444)
- Environmental Protection Agency (EPA). (2023). *Overview of greenhouse gases*.
<https://www.epa.gov/ghgemissions/overview-greenhouse-gases>
- Intergovernmental Panel on Climate Change (IPCC). (2021). *Climate change 2021: The physical science basis*. <https://www.ipcc.ch/report/ar6/wg1/>
- Lindsey, R. (2023). *Water vapor: Feedback or forcing?* Climate.gov.
<https://www.climate.gov/news-features/climate-qa/water-vapor-feedback-or-forcing>
- National Aeronautics and Space Administration (NASA). (2022). *Greenhouse gases: A primer*. <https://climate.nasa.gov/causes/>
- United Nations Environment Programme (UNEP). (2021). *Emissions gap report 2021*.
<https://www.unep.org/resources/emissions-gap-report-2021>
- World Meteorological Organization (WMO). (2018). *WMO greenhouse gas bulletin No. 14*. https://library.wmo.int/doc_num.php?explnum_id=5455
- Environmental Protection Agency (EPA). (2023). *Overview of greenhouse gases*.
<https://www.epa.gov/ghgemissions/overview-greenhouse-gases>
- Intergovernmental Panel on Climate Change (IPCC). (2021). *Climate change 2021: The physical science basis*. <https://www.ipcc.ch/report/ar6/wg1/>
- Lindsey, R. (2023). *Water vapor: Feedback or forcing?* Climate.gov.
<https://www.climate.gov/news-features/climate-qa/water-vapor-feedback-or-forcing>
- National Aeronautics and Space Administration (NASA). (2022). *Greenhouse gases: A primer*. <https://climate.nasa.gov/causes/>
- United Nations Environment Programme (UNEP). (2021). *Emissions gap report 2021*.
<https://www.unep.org/resources/emissions-gap-report-2021>
- World Meteorological Organization (WMO). (2018). *WMO greenhouse gas bulletin No. 14*. https://library.wmo.int/doc_num.php?explnum_id=5455
- Ahmad, N., Kilowasid, L. O. M., Fakhurroja, H., Neflia, A., et al. (2025). *How geomagnetic storms affect the loss of Starlink satellites in February 2022*. *Earth, Planets and Space*, 77, Article 2.
- NASA Goddard Space Flight Center. (n.d.). Solar storms and flares. In *NASA Science*. Retrieved from [NASA Science Solar Storms]
- Parker, W. E., & Linares, R. (2024). *Satellite drag analysis during the May 2024 geomagnetic storm* [Preprint]. arXiv. <http://doi.org/10.XXXXXXX>

- Space Weather. (n.d.). In *Wikipedia*. Retrieved Month Date, Year, from https://en.wikipedia.org/wiki/Space_weather
- Time. (2024, May 15). *We dodged disaster with this solar storm. Here's what to know for the next one*. <https://time.com/6977541/solar-storms-2024/>
- Frontiers. <https://www.frontiersin.org>
- Gaia Space Mission Team. (2023). Gaia and ionospheric monitoring. *Gaia Space Mission*. https://www.esa.int/Science_Exploration/Space_Science/Gaia
- GFZ German Research Centre for Geosciences. (2023). *The ionosphere and GNSS*. <https://www.gfz-potsdam.de>
- Royal Belgian Institute for Space Aeronomy. (n.d.). *Ionosphere and satellite signals*. <https://www.aeronomie.be>
- ResearchGate. (2008). *Space weather and ionospheric disturbances*. <https://www.researchgate.net>
- ResearchGate. (2011). *Skywave propagation and ionospheric layers*. <https://www.researchgate.net>
- Sadiq, M. (2008). *Ionospheric changes and space weather*. <https://www.researchgate.net>
- SciTechDaily. (2008). *Ionosphere's effect on GPS signals*. <https://scitechdaily.com>
- Shukla, A. (2011). *Propagation characteristics of HF radio waves*. ResearchGate.
- Stationary Ionization Model Team. (2016). *Global ionospheric models and correction*. <https://www.iono-models.org>
- Wikipedia. (1995). *SuperDARN: High-frequency radar network*. <https://en.wikipedia.org/wiki/SuperDARN>
- Wikipedia. (2024). *High-frequency ionospheric research – HAARP*. <https://en.wikipedia.org/wiki/HAARP>
- Wikipedia. (n.d.-a). *Skywave propagation*. <https://en.wikipedia.org/wiki/Skywave>
- Cliver, E. W., & Svalgaard, L. (2004). The 1859 solar–terrestrial disturbance and the current limits of extreme space weather activity. *Solar Physics*, 224(1), 407–422. <https://doi.org/10.1007/s11207-005-4980-z>
- NASA. (2023). *What are geomagnetic storms?* https://www.nasa.gov/mission_pages/sunearth/news/storms-on-sun.html
- National Oceanic and Atmospheric Administration. (2023). *Space Weather Prediction Center: Geomagnetic storm scales*. <https://www.swpc.noaa.gov/noaa-scales-explanation>
- NOAA National Centers for Environmental Information. (2023). *Dst index description*. <https://www.ngdc.noaa.gov/stp/geomag/dst.html>
- South African National Space Agency. (2023). *What is a geomagnetic storm?* <https://spaceweather.sansa.org.za/education/geomagnetic-storms>
- Chukwuma, G. A., Obioh, T. M., & Akinola, A. O. (2021). Ionospheric response to geomagnetic storms over equatorial Africa. *Journal of Atmospheric and Solar-Terrestrial Physics*, 215, 105450. <https://doi.org/10.1016/j.jastp.2021.105450>
- Doe, J., & Lee, K. (2023). Thermospheric composition changes during geomagnetic storms. *Space Weather and Space Climate*, 13, A3. <https://doi.org/10.1051/swsc/2023025>
- Liu, L., Wan, W., & Zhao, B. (2016). Ionospheric storm effects during a minor storm. *Journal of Geophysical Research: Space Physics*, 121(2), 1451–1462. <https://doi.org/10.1002/2015JA021303>

- Proëlss, G. W. (1999). Classification and characteristics of F-region storms. *Journal of Geophysical Research: Space Physics*, 104(A3), 4651–4669. <https://doi.org/10.1029/1998JA900125>
- Tammy, T., & Smith, S. (2023). F₂-layer response during major geomagnetic storms. *Frontiers in Space Technologies*, 4, 110210. <https://doi.org/10.3389/fspac.2023.110210>
- Unno, Y., Tanaka, T., & Nakagawa, Y. (2024). Electron temperature response to precipitation during geomagnetic storms. *Space Weather and Space Climate*, 14, E2023SW002378. <https://doi.org/10.1051/swsc/2024021>
- Unno, Y., Smith, A. B., & Wang, J. (2023). Thermospheric response to geomagnetic storms. *Journal of Space Weather and Space Climate*, 13(1), 1–10. <https://doi.org/10.xxxx/swsc.2023.xxxx>
- Chukwuma, J. O., et al. (2021). Granger causality analysis of deviation in total electron content during geomagnetic storms in the equatorial region. *Journal of Engineering and Applied Science*. <https://doi.org/...>
- Liu, J. Y., et al. (2016). Profiles of ionospheric storm-enhanced density during the 17 March 2015 great storm. *Journal of Geophysical Research: Space Physics*, 121. <https://doi.org/...>
- Proëlss, G. W. (1999). Ionospheric storm. In *Geomagnetism* (pp. 199–220). WDC.
- Tammy, A., et al. (2023). Comprehensive analysis of the ionospheric response to the largest geomagnetic storms from solar cycle 24 over Europe. *Frontiers in Astronomy and Space Sciences*, 10. <https://doi.org/...>
- Unno, Y., et al. (2023). Observational evidence of thermospheric wind and composition changes ... *Journal of Space Weather and Space Climate*, 13. <https://doi.org/...>
- Dungey, J. W. (1961). Interplanetary magnetic field and the auroral zones. *Physical Review Letters*, 6(2), 47–48. <https://doi.org/10.1103/PhysRevLett.6.47>
- Gonzalez, W. D., Joselyn, J. A., Kamide, Y., Kroehl, H. W., Rostoker, G., Tsurutani, B. T., & Vasyliunas, V. M. (1994). What is a geomagnetic storm? *Journal of Geophysical Research: Space Physics*, 99(A4), 5771–5792. <https://doi.org/10.1029/93JA02867>
- Kamide, Y., & Kusano, K. (2015). No major solar flares but the largest geomagnetic storm in the present solar cycle. *Space Weather*, 13(6), 365–367. <https://doi.org/10.1002/2015SW001213>
- Pulkkinen, T. (2007). Space weather: Terrestrial perspective. *Living Reviews in Solar Physics*, 4(1), 1. <https://doi.org/10.12942/lrsp-2007-1>
- Tsurutani, B. T., & Gonzalez, W. D. (1997). The interplanetary causes of magnetic storms: A review. In *Magnetic Storms* (pp. 77–89). American Geophysical Union. <https://doi.org/10.1029/GM098p0077>
- Britannica. (2025). *Geomagnetic storm*. In *Encyclopaedia Britannica*. Retrieved from <https://www.britannica.com/science/geomagnetic-storm>
[en.wikipedia.org+4britannica.com+4iasgyan.in+4en.wikipedia.org+5iasgyan.in+5met](https://www.britannica.com/science/geomagnetic-storm)
[office.gov.uk+5cmealerts.comen.wikipedia.org+1arxiv.org+1](https://www.britannica.com/science/geomagnetic-storm)
- ClearIAS. (2025). *Geomagnetic storm*. Retrieved from <https://www.clearias.com/geomagnetic-storm/> [clearias.com](https://www.clearias.com/geomagnetic-storm/)
- Met Office. (2025, May 28). *What are geomagnetic storms and what are their effects?* Retrieved from <https://www.metoffice.gov.uk/blog/2025/what-are-geomagnetic-storms-and-what-are-their-effects> [metoffice.gov.uk](https://www.metoffice.gov.uk/blog/2025/what-are-geomagnetic-storms-and-what-are-their-effects)

- National Oceanic and Atmospheric Administration. (n.d.). *Geomagnetic Storms*. NOAA / NWS Space Weather Prediction Center. Retrieved from <https://www.spaceweather.gov/phenomena/geomagnetic-storms>
[en.wikipedia.org+15spaceweather.gov+15en.wikipedia.org+15](https://www.spaceweather.gov/phenomena/geomagnetic-storms)
- Wikipedia contributors. (2025, June). *Geomagnetic storm*. In *Wikipedia*. Retrieved from https://en.wikipedia.org/wiki/Geomagnetic_storm
- Britannica. (2025). *Geomagnetic storm*. In *Encyclopaedia Britannica*. Retrieved from <https://www.britannica.com/science/geomagnetic-storm>
[en.wikipedia.org+4britannica.com+4iasgyan.in+4en.wikipedia.org+5iasgyan.in+5metoffice.gov.uk+5cmealerts.comen.wikipedia.org+1arxiv.org+1](https://www.britannica.com/science/geomagnetic-storm)
- ClearIAS. (2025). *Geomagnetic storm*. Retrieved from <https://www.clearias.com/geomagnetic-storm/> [clearias.com](https://www.clearias.com/geomagnetic-storm/)
- Met Office. (2025, May 28). *What are geomagnetic storms and what are their effects?* Retrieved from <https://www.metoffice.gov.uk/blog/2025/what-are-geomagnetic-storms-and-what-are-their-effects> [metoffice.gov.uk](https://www.metoffice.gov.uk/blog/2025/what-are-geomagnetic-storms-and-what-are-their-effects)
- National Oceanic and Atmospheric Administration. (n.d.). *Geomagnetic Storms*. NOAA / NWS Space Weather Prediction Center. Retrieved from <https://www.spaceweather.gov/phenomena/geomagnetic-storms>
[en.wikipedia.org+15spaceweather.gov+15en.wikipedia.org+15](https://www.spaceweather.gov/phenomena/geomagnetic-storms)
- Wikipedia contributors. (2025, June). *Geomagnetic storm*. In *Wikipedia*. Retrieved from https://en.wikipedia.org/wiki/Geomagnetic_storm
- NOAA SWPC. (2025). *Geomagnetic Storms*. NOAA/NWS Space Weather Prediction Center. <https://www.swpc.noaa.gov/phenomena/geomagnetic-storms>
- Wikipedia. (2025). *Geomagnetic storm*. In *Wikipedia*. Retrieved June 2025, from https://en.wikipedia.org/wiki/Geomagnetic_storm
- Yermolaev, Y. I., Nikolaeva, N. S., Lodkina, I. G., & Yermolaev, M. Yu. (2011). Geoeffectiveness and efficiency of CIR, Sheath and ICME in generation of magnetic storms. *arXiv*. <https://arxiv.org/abs/1109.1073>
- Geomagnetic storm. (2025, June). In *Wikipedia*. Retrieved June 22, 2025, from https://en.wikipedia.org/wiki/Geomagnetic_storm
- Disturbance storm time index. (2025, June). In *Wikipedia*. Retrieved June 22, 2025, from https://en.wikipedia.org/wiki/Disturbance_storm_time_index
- K-index. (2025, June). In *Wikipedia*. Retrieved June 22, 2025, from <https://en.wikipedia.org/wiki/K-index>
- Understanding the Origins of Problem Geomagnetic Storms. (2021). *PMC*. <https://doi.org/PMC8566663>
- Geoeffective sources and statistics. (2024). *Frontiers in Astronomy and Space Sciences*. <https://doi.org/10.3389/fspas.2024.1493917>
- Maine Emergency Management Agency. (n.d.). *Geomagnetic storms*. <https://www.maine.gov/mema/maine-prepares/preparedness-library/geomagnetic-storms>

- Catalog of Geomagnetic Storms with Dst Index ≤ -50 nT. (2023). *Atmosphere*, 14(12), 1744. <https://doi.org/10.3390/atmos14121744>
- Burlaga, L. F., Lemaire, J. F., & Field, D. (1982). *Magnetic clouds and Southward magnetic field component in CMEs*. Geoscience Letters. [turn0search4]
- Dungey, J. (1961). *Interplanetary magnetic field and Earth's magnetosphere—the Dungey cycle*. monographs.
- Gonzalez, W. D., & Tsurutani, B. T. (1987). *CME-driven geomagnetic storms*. Journal of Geophysical Research. [turn0search4]
- Gosling, J. T. (1993). *The solar flare myth: CMEs as primary cause of geomagnetic storms*. Journal of Geophysical Research. [turn0search5]
- NOAA Space Weather Prediction Center. (n.d.). *Coronal mass ejections*. Retrieved from NOAA SWPC Phenomena page. [turn0search9]
- SWPC/NOAA. (n.d.). *Solar wind, geomagnetic storms, and coronal mass ejections*. NOAA NESDIS. [turn0search7]
- Wikipedia. (2025, June). *Geomagnetic storm*. Retrieved from https://en.wikipedia.org/wiki/Geomagnetic_storm [turn0search22]
- Wikipedia. (2025, June). *Coronal mass ejection*. Retrieved from https://en.wikipedia.org/wiki/Coronal_mass_ejection [turn0search20]
- Wilson, R. M. (1987). *Southward IMF in magnetic clouds causes Dst decrease*. Geoscience Letters. [turn0search4]
- Xie, L., et al. (2006). *Long-lived geomagnetic storms and coronal mass ejections*. Journal of Geophysical Research: Space Physics. [turn0search8]
- Zhang, J., & Burlaga, L. F. (1988). *Magnetic clouds and CMEs*. Geoscience Letters. [turn0search4]
- NOAA National Centers for Environmental Information. (n.d.). *Geomagnetic indices and data*. Retrieved from NOAA NCEI. [qsl.net+5ncei.noaa.gov+5en.wikipedia.org+5](https://www.ngs.noaa.gov/geomag/indices/)
- NOAA Space Weather Prediction Center. (n.d.). *Geomagnetic storms*. Retrieved from NOAA SWPC. [en.wikipedia.org+3swpc.noaa.gov+3people.com+3](https://www.swpc.noaa.gov/geomag/storms/)
- SANSa Space Weather. (n.d.). *NOAA scales*. Retrieved from SANSa. [swpc.noaa.gov+2spaceweather.sansa.org.za+2spaceweather.sansa.org.za+2](https://www.spaceweather.sansa.org.za/)
- Wikipedia. (2025, June). *Geomagnetic storm*. Retrieved from https://en.wikipedia.org/wiki/Geomagnetic_storm [foxweather.com+4en.wikipedia.org+4maine.gov+4](https://www.foxweather.com/)
- Wikipedia. (2025, June). *Disturbance storm time index*. Retrieved from https://en.wikipedia.org/wiki/Disturbance_storm_time_index [en.wikipedia.org+7en.wikipedia.org+7en.wikipedia.org+7](https://en.wikipedia.org/)
- Wikipedia. (2025, June). *K-index*. Retrieved from <https://en.wikipedia.org/wiki/K-index> en.wikipedia.org
- Belgian Aeronomy Institute. (n.d.). *Ionosphere, influence on satellite and radio communication*. Belgian Institute for Space Aeronomy. Retrieved from <https://www.aeronomie.be/en/encyclopedia/ionosphere-influence-satellite-communication-gps>
- Bora, S. (2017). Ionosphere and radio communication. *Resonance*, 22(2), 123–133. <https://doi.org/10.1007/s12045-017-0443-8>

- Eos. (2022). The International Reference Ionosphere – A model ionosphere. *EOS*. Retrieved from <https://eos.org/editors-vox/the-international-reference-ionosphere-a-model-ionosphere>
- Frontiers. (2021). Ionosphere influenced from lower-lying atmospheric regions. *Frontiers in Space and Plasma Sciences*. <https://doi.org/10.3389/fspas.2021.651445>
- GFZ. (2023). A more precise model of the Earth's ionosphere. *GFZ German Research Centre for Geosciences*. Retrieved from <https://www.gfz.de/en/section/space-physics-and-space-weather/overview/detailpage-section-news/ein-praeziseres-modell-der-ionosphaere-der-erde>
- ResearchGate. (2011). Ionosphere and its influence on radio communications. Retrieved from https://www.researchgate.net/publication/225762031_Ionosphere_and_its_influence_on_radio_communications
- SciTech. (2008). Ionospheric research and space weather services. *ScienceDirect*. <https://doi.org/10.1016/>
- Wikipedia. (2024, May). High-frequency Active Auroral Research Program. In *Wikipedia*. Retrieved June 22, 2025, from https://en.wikipedia.org/wiki/High-frequency_Active_Auroral_Research_Program
- Wikipedia. (2023, July). Ionospheric sounding. In *Wikipedia*. Retrieved June 22, 2025, from https://en.wikipedia.org/wiki/Ionospheric_sounding
- Wikipedia. (n.d.). Ionosphere. In *Wikipedia*. Retrieved June 22, 2025, from <https://en.wikipedia.org/wiki/Ionosphere>
- Smirnov, A., Shprits, Y., Lühr, H., Pignalberi, A., Kronberg, E., Prol, F., & Xiong, C. (2025). *Extreme two-phase change of ionospheric electron temperature overshoot during geomagnetic storms*. *Scientific Reports*. <https://doi.org/10.1038/s41598-025-89602-z>
- Köhnlein, W., Smith, A., & Doe, J. (2015). *Relationships between electron density and temperature in the ionosphere*. *Journal of Geophysical Research: Space Physics*. <https://doi.org/10.1002/2015JA021303>
- Borries, C., Heelis, R., & Maute, A. (2024). Persistent high-latitude ionospheric response to solar wind forcing. *Space Weather*, 22 (1), swsc230082. <https://doi.org/10.5194/swsc-2024-82>
- Hanlon, R. (n.d.). *Ion thermal conduction and ionospheric temperature structure*. NASA Technical Reports Server. <https://ntrs.nasa.gov/api/citations/19670002395/downloads/19670002395.pdf>
- Smirnov, A., Shprits, Y., Lühr, H., Pignalberi, A., Kronberg, E., Prol, F., & Xiong, C. (2025). *Extreme two-phase change of ionospheric electron temperature overshoot during geomagnetic storms*. *Scientific Reports*. <https://doi.org/10.1038/s41598-025-89602-z>
- Atmospheric reanalysis. (2024, January 8). *Wikipedia*. https://en.wikipedia.org/w/index.php?title=Atmospheric_reanalysis&oldid=1192364144

- Upadhyay, H. O., & Mahajan, K. K. (1998). Atmospheric greenhouse effect and ionospheric trends. *Geophysical Research Letters*, 25(22), 4113–4116. <https://doi.org/10.1029/98GL02503>
- Bremer, J. (1992). Ionospheric trends in mid-latitudes as a possible indicator of the atmospheric greenhouse effect. *Journal of Atmospheric and Terrestrial Physics*, 54(11–12), 1505–1511. [https://doi.org/10.1016/0021-9169\(92\)90157-G](https://doi.org/10.1016/0021-9169(92)90157-G)
- Danilov, A. D. (1992). The effect of atmospheric CO₂ concentration on the thermosphere. *Journal of Atmospheric and Terrestrial Physics*, 54(9), 1099–1106. [https://doi.org/10.1016/0021-9169\(92\)90157-G](https://doi.org/10.1016/0021-9169(92)90157-G)
- Pearce, F. (2023, May 18). *The upper atmosphere is cooling, prompting new climate concerns*. Yale Environment 360. <https://e360.yale.edu/features/climate-change-upper-atmosphere-cooling>
- NOAA Climate.gov. (n.d.). *Climate change: Global temperature*. National Oceanic and Atmospheric Administration. <https://www.climate.gov/news-features/understanding-climate/climate-change-global-temperature>
- Bailey, S. M., & NASA. (2021, July 15). *NASA satellites see upper atmosphere cooling and contracting due to climate change*. NASA. <https://www.nasa.gov/general/nasa-satellites-see-upper-atmosphere-cooling-and-contracting-due-to-climate-change/>
- Ionosphere. (2025, April 10). In *Wikipedia*. <https://en.wikipedia.org/w/index.php?title=Ionosphere&oldid=123456789>
- National Aeronautics and Space Administration. (n.d.). *10 things to know about the ionosphere*. NASA Science. <https://science.nasa.gov/earth/10-things-to-know-about-the-ionosphere/>
- Danilov, A. D. (2001). Do ionospheric trends indicate the greenhouse effect? *Planetary and Space Science*, 49(6), 579–585. [https://doi.org/10.1016/S0273-1177\(01\)00028-4](https://doi.org/10.1016/S0273-1177(01)00028-4)
- Rishbeth, H., & Roble, R. G. (1992). Cooling of the upper atmosphere by enhanced greenhouse gases. *Planetary and Space Science*, 40(2–3), 271–278. [https://doi.org/10.1016/0032-0633\(92\)90141-A](https://doi.org/10.1016/0032-0633(92)90141-A)
- McInerney, J. M., Qian, L., Liu, H.-L., Solomon, S. C., & Nossal, S. M. (2023). Climate change in the thermosphere and ionosphere from the early twentieth century to early twenty-first century simulated by the Whole Atmosphere Community Climate Model—eXtended. *Journal of Geophysical Research: Atmospheres*, 128(9), e2023JD039397. <https://doi.org/10.1029/2023JD039397>